AI Benchmark Half-Life in Recursive Corpora:

A Theory of Validity Decay under Semantic Leakage and Regeneration

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Abstract

Modern AI benchmarks no longer live outside the training and evaluation ecosystem. They live inside recursive corpora, where benchmark items, paraphrases, explanations, model outputs, benchmark-aware curation rules, and judge conventions re-enter the public knowledge stream and can later affect both models and evaluators. Existing literatures on recursive training, contamination detection, dynamic benchmark construction, construct validity, benchmark saturation, and human–AI co-produced knowledge each capture part of this problem, but they do not yet provide a unified lifetime theory for benchmarks as measurement instruments. This paper develops an origin-free theory of *benchmark half-life*: the first time a benchmark lineage loses half of its initial validity under a declared operational evaluation specification. We reject nominal data origin ("human" versus "synthetic") as a primitive because origin labels are often latent, unstable under mixed production, and neither necessary nor sufficient for benchmark quality. Instead, we model benchmark decay through observable or estimable state variables: semantic leakage, verified novelty inflow, lineage observability, verification fidelity, support diversity, evaluator coupling, and construct drift. Validity is defined relative to an explicit operational class and decomposed into discriminative power and a construct-validity component that combines resolved-pair mass with thresholded order fidelity. We separate weak passive-envelope assumptions, which suffice for intervalwise retirement bounds and cross-ecology ordering, from stronger state-reduced laws, which suffice for statewise comparison, proxy-based identification, factor-targeted experimentation, and control once declared regime-time jumps are recorded explicitly. The framework is jump aware: passive hazard statements apply only between declared benchmark interventions, while validity envelopes and identification formulas carry separate multiplicative jump factors. This keeps first-hitting-time objects, regeneration, and safe control logically compatible. The paper then develops anchor-based construct-validity identification under scalar, directional, abstention, and nondifferential class-conditional error models; portfolio design and monitoring under common-mode decay and uncertain coverage; and safe sequential monitoring and control under partial observability, transition-kernel ambiguity, frontier-pruned portfolio actions, and a joint estimation-to-control certificate. The practical implication is that benchmark reports should carry operational scope, identification status, validity intervals, dependence budgets, exclusion logic, jump logs, coverage uncertainty budgets, coverage-frontier certificates, ambiguity sets for control, estimation-to-control validity budgets, and life-cycle status, not raw scores alone.

Machine-Readable YAML

```yaml
title: AI Benchmark Half-Life in Recursive Corpora
core_question: >
  How should benchmark validity be modeled, identified, monitored, and controlled
  when benchmarks live inside recursive corpora rather than outside them?
operational_specification:
  Omega_t:
    components: [M_t, Pi_t, delta_t, eps_t]
    meaning:
      M_t: operational model class
      Pi_t: pair distribution over M_t x M_t
      delta_t: effective score-resolution threshold
      eps_t: target-resolution threshold
validity_objects:
  D_B_t: pairwise separation probability above delta_t
  R_B_t: target-resolved pair mass above eps_t
  O_B_t: thresholded order fidelity conditional on target-resolved pairs
  C_B_t: construct-validity component = R_B_t * O_B_t
  V_B_t: benchmark validity = D_B_t * C_B_t
  tau_half: first time on the full cadlag lineage path that V_B_t <= 0.5 * V_B_0
ecological_state:
  X_eco_t:
    coordinates:
      S_t: semantic leakage mass
      N_t: verified novelty inflow
      F_t: verification fidelity
      U_t: support diversity / tail coverage
      J_t: evaluator coupling
      K_t: construct drift
observability_state:
  A_obs_t:
    coordinates:
      L_t: lineage observability
      H_t: audit and calibration metadata
      Lambda_jump_t: declared jump log and intervention metadata
full_governance_state:
  Z_t: [X_eco_t, A_obs_t]
dynamics:
  path_class: cadlag_with_locally_finite_regime_times
  passive_intervals: >
    Between declared regime times, the benchmark artifact and operational
    specification are held fixed and validity evolves under passive decay.
  intervention_times: >
```

```
    Regeneration, scope changes, judge changes, or benchmark revisions may create
    jumps. Passive hazard results apply intervalwise between jumps.
  jump_accounting: >
    Declared regime-time jump multipliers are tracked separately from passive
    hazards. Identification and envelope formulas therefore use cumulative jump
    factors times passive exponential terms.
core_assumptions:
  weak:
    - passive envelope-valid benchmark dynamics
  strong:
    - state-reduced passive hazard law
    - coordinatewise monotonicity
    - factor-specific passive hazard law
main_result_blocks:
  - Decay dynamics and benchmark lifetime
  - Identification, proxies, and factor attribution
  - Benchmark portfolios and coverage uncertainty
  - Sequential monitoring and safe control
identification_status:
  score_trace_only: >
    composite hazard identified;
    latent ecological coordinates not identified
  calibrated_injective_proxy: state path point identified
  support_dense_design: hazard map identified on visited support
  off_support: only set-valued extrapolation under added shape restrictions
  aggregate_validity_only: factor hazards not point identified
  anchors: >
    order-fidelity component partially identified under
    scalar error budgets, directional and nondifferential class-conditional
    anchor error, dependence, abstention, and directional resolved-set mismatch budgets
governance_outputs:
  - half-life and threshold-invalidity summaries
  - validity intervals and identification labels
  - dependence and exclusion budgets
  - coverage uncertainty sets
  - transition / observation ambiguity sets
  - coverage-frontier certificates for action pruning
  - estimation-to-control validity budgets
  - retirement / regeneration / scope-restriction rules
theorem_registry:
  decay:
    - thm:env_compare
    - thm:comparison
```

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  - thm:finite
  - thm:avg_budget
identification:
  - thm:score_nonid
  - thm:interval
  - thm:hazard_support
  - thm:lipschitz_extrap
  - thm:factor_support
  - thm:shared_iv
  - thm:imperfect_exclusion_sharp
  - thm:shared_iv_sensitivity
  - thm:structured_sharp
anchors:
  - thm:anchor_validation
  - thm:anchor
  - thm:anchor_directional
  - thm:anchor_direrr
  - thm:anchor_nd_sensitivity
  - thm:anchor_qualified
  - thm:shared_blind
  - thm:anchor_latent
portfolio:
  - thm:portfolio_common
  - thm:portfolio_design
  - thm:coverage_frontier
  - thm:frontier_prune
  - thm:coverage_budget
  - thm:coverage_learned
  - thm:coverage_learned_dep
  - thm:coverage_dep_construct
  - thm:coverage_regime_switch
control:
  - thm:seq_retire
  - thm:safe_policy
  - thm:bellman_safe
  - thm:robust_bellman
  - thm:pomdp_safe
  - thm:robust_pomdp
  - thm:rect_hull
  - thm:safe_explore
  - thm:approx_regret
  - thm:grid_safe
  - thm:e2c
```

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estimation_fail_closed:
  zero_resolved_mass: report status under_resolved and set C_B_t = 0
  zero_anchor_denominator: report anchor_status = insufficient_resolved_pairs
  zero_anchor_overlap: report anchor_status = insufficient_overlap and retain [0,1]
  qualified_anchor_only: >
    if anchor qualification is observed only on a subset of resolved pairs, report
    qualification mass lower bound and widen order-fidelity interval outside the
    qualified region rather than imputing missing accuracy
  score_resolved_direrr: >
    directional class-conditional anchor-error formulas apply only on the
    score-resolved and target-resolved subset; carry the score-resolved mass and
    positive-order prevalence as separate latent or calibrated inputs
  transition_model_uncertainty: >
    when control transitions are estimated, report an ambiguity set and solve the
    robust safe Bellman recursion rather than plugging in a single kernel
  nonrectangular_ambiguity: >
    if exact stagewise rectangularity is not defensible, report the nonrectangular
    family and its rectangular outer hull separately; use the hull only as a
    conservative lower-value planner, not as if it were an exact model
lifecycle_status:
  active: lower validity envelope > governance threshold
  watchlist: lower envelope <= threshold < upper envelope
  regenerate_or_replace: >
    upper envelope <= threshold but an admissible intervention exists
  retire_or_scope_restrict: >
    upper envelope <= threshold and no admissible repair exists for the current
    operational class
main_dependencies:
  thm:env_compare:
    - ass:envelope
    - matched_regime_times
    - order_preserving_jumps
  thm:comparison:
    - ass:state_v
    - ass:monotone_v
    - common_passive_hazard_map
    - matched_regime_times
    - order_preserving_jumps
  thm:finite:
    - ass:envelope
    - passive_window
    - positive_lower_rate_on_window
  cor:no_redesign:
```

```
  - thm:finite
  - certified_no_redesign_window
thm:avg_budget:
  - ass:state_v
  - linear_rate_upper_bound
  - declared_lower_jump_envelope
  - nonincreasing_lower_validity_envelope
thm:score_nonid:
  - positive_validity_trace
thm:interval:
  - ass:state_v
  - calibrated_hazard_map
  - declared_jump_factor_envelopes
thm:hazard_support:
  - ass:state_v
  - state_recovery_regime
  - support_dense_calibration
thm:lipschitz_extrap:
  - identified_support
  - known_lipschitz_modulus
thm:factor_support:
  - ass:factor_rate
  - factor_state_recovery
  - projected_support_richness
thm:shared_iv:
  - local_shared_coordinate_design
  - full_rank_design_matrix
  - exclusion_restriction
thm:imperfect_exclusion_sharp:
  - thm:shared_iv
  - attainable_direct_effect_class
thm:shared_iv_sensitivity:
  - local_shared_coordinate_design
  - full_rank_design_matrix
  - bounded_direct_effect_class
thm:structured_sharp:
  - thm:shared_iv_sensitivity
  - attainable_structural_violation_class
thm:anchor_validation:
  - declared_anchor_validation_design
  - validated_reference_or_identified_error_set
thm:anchor:
  - resolved_pair_event
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  - admissible_anchor_validation_design
thm:anchor_directional:
  - positive_overlap_or_fail_closed_status
cor:anchor_nested_sharp:
  - thm:anchor_directional
cor:anchor_direrr_nested:
  - thm:anchor_direrr
  - identified_or_calibrated_score_resolved_inputs
thm:anchor_qualified:
  - thm:anchor
  - observed_anchor_qualification_rule
thm:anchor_direrr:
  - def:anchor_direrr
  - score_resolved_target_resolved_pairs
  - identified_or_calibrated_score_resolved_inputs
thm:anchor_nd_sensitivity:
  - def:anchor_direrr
  - def:anchor_nd_budget
  - score_resolved_target_resolved_pairs
thm:shared_blind:
  - odd_anchor_panel
  - shared_blind_spot_budget
thm:anchor_latent:
  - latent_factor_anchor_model
  - conditional_independence_given_latent_factor
thm:portfolio_common:
  - ass:state_v
  - common_mode_hazard_decomposition
  - no_upward_component_jumps_on_certified_horizon
thm:portfolio_design:
  - declared_scenario_survival_profiles
thm:coverage_frontier:
  - def:coverage_design
  - feasible_coverage_constraints
thm:frontier_prune:
  - thm:coverage_frontier
  - portfolio_summaries_control_sufficient
  - measurable_frontier_selector
thm:coverage_budget:
  - budgeted_coverage_uncertainty_set
  - strong_duality_of_inner_lp
thm:coverage_learned:
  - learned_coverage_set_with_uniform_coverage
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thm:coverage_learned_dep:
  - thm:coverage_learned
  - stopping_time_uniform_set_coverage
thm:coverage_dep_construct:
  - declared_audit_dependence_regime
  - valid_confidence_sequences
thm:coverage_regime_switch:
  - declared_regime_switching_or_posterior_mixture_audit_model
  - regimewise_confidence_sequences
thm:seq_retire:
  - anytime_valid_envelope
thm:safe_policy:
  - def:safe_control
  - anytime_valid_lower_envelope
thm:bellman_safe:
  - def:safe_control
thm:robust_bellman:
  - def:robust_safe_control
thm:pomdp_safe:
  - def:pomdp_safe
  - exact_belief_or_control_sufficient_state
thm:robust_pomdp:
  - def:robust_pomdp_safe
  - rectangular_ambiguity
  - exact_belief_or_control_sufficient_state
thm:rect_hull:
  - def:rect_hull
  - declared_planning_domain_contains_original_reachable_states
thm:safe_explore:
  - def:safe_explore
  - confidence_event
  - declared_nonstationarity_budget
thm:approx_regret:
  - def:approx_safe
thm:grid_safe:
  - projected_grid_abstraction
  - safe_set_margin_or_projection_invariance
  - value_lipschitz_certificate
  - reward_and_kernel_lipschitzness
thm:e2c:
  - def:e2c
  - anytime_validity_event
  - kernel_coverage_event
```

Symbol Table

Symbol	Meaning
Ω_t	Operational evaluation specification at time t with components $(\mathcal{M}_t, \Pi_t, \delta_t, \varepsilon_t)$.
$Y_B(m,t)$	Benchmark score of model m on benchmark B at time t.
$G_t(m)$	Target-claim functional the benchmark is meant to support.
$\mathcal{D}_B(t)$	Discriminative power: pairwise score separation probability above δ_t.
$R_B(t)$	Resolved-pair mass: probability that target difference exceeds ε_t.
$\mathcal{O}_B(t)$	Fail-closed thresholded order fidelity conditional on target-resolved pairs.
$\mathcal{C}_B(t)$	Construct-validity component, defined as $R_B(t)\mathcal{O}_B(t)$.
$\mathcal{V}_B(t)$	Benchmark validity, defined as $\mathcal{D}_B(t)\mathcal{C}_B(t)$.
X_t^{eco}	Ecological state vector $(S_t, N_t, F_t, U_t, J_t, K_t)$ governing passive decay.
A_t^{obs}	Observability and audit state, including lineage observability L_t, calibration metadata, and declared jump logs.
$\Lambda_t^{\mathrm{jump}}$	Declared jump log and intervention metadata used to certify regime-time multiplier envelopes.
Z_t	Full governance state $(X_t^{\mathrm{eco}}, A_t^{\mathrm{obs}})$.
q_V, q_D, q_C	Passive hazard maps for benchmark validity, discriminative power, and the construct-validity component.
$\rho_{B,\delta}(t)$	Score-resolution mass on target-resolved pairs, $\mathbb{P}(\mathcal{G}_t^{\delta} \mid \mathcal{R}_t)$.
$\underline{V}_n, \overline{V}_n$	Anytime-valid lower and upper validity envelopes at monitoring epoch n.
$\kappa_A(t)$	Qualification mass of an abstaining anchor on the resolved-pair event.
$\eta_A(t)$	Symmetric resolved-set mismatch probability $\mathbb{P}(\mathcal{R}_t \triangle \mathcal{R}_t^{(A)})$.
$\eta_A^+(t), \eta_A^-(t)$	Directional resolved-set mismatch masses $\mathbb{P}(\mathcal{R}_t^{(A)} \setminus \mathcal{R}_t)$ and $\mathbb{P}(\mathcal{R}_t \setminus \mathcal{R}_t^{(A)})$.
$\Gamma_A^{\mathrm{obs}}(t)$	Observable benchmark-anchor agreement on the anchor-resolved set.
$e_A^+(t), e_A^-(t)$	Class-conditional anchor error rates $\mathbb{P}(A_t = -1 \mid T_t = +1, \mathcal{R}_t)$ and $\mathbb{P}(A_t = +1 \mid T_t = -1, \mathcal{R}_t)$ under a nondifferential anchor-error model.
$p_{B,+}(t)$	Benchmark positive-order prevalence $\mathbb{P}(B_t = +1 \mid \mathcal{R}_t)$ on target-resolved pairs.
$\mathfrak{P}_n(I, a)$	Transition-kernel ambiguity set for robust safe control at epoch n, information state I, action a.
$\mathfrak{Q}_n(b, a)$	Joint transition-observation ambiguity set for robust safe belief-state control at epoch n, belief state b, action a.
$\mathrm{Rect}(\widetilde{\mathfrak{Q}})$	Stagewise rectangular outer hull of a reported nonrectangular ambiguity family $\widetilde{\mathfrak{Q}}$.
$v(\rho; T)$	Coverage-frontier value: maximal certified worst-case survival at horizon T subject to coverage target ρ.
$\tau_{1/2}(B)$	First time benchmark validity falls to half of its initial value.

1 Introduction

Benchmarks are now treated as public instruments of scientific and engineering progress in AI. They anchor model comparisons, guide model release narratives, and increasingly influence procurement, deployment, and policy decisions. Yet the environment in which these instruments operate has changed. A benchmark

no longer sits cleanly outside the corpus ecology. Instead, benchmark items, paraphrases, solution traces, model-generated explanations, benchmark-aware filtering heuristics, and community discussions circulate through the same public channels that later model builders, evaluators, and downstream users consume. The benchmark has become one component of a recursive corpus.

Several adjacent literatures already illuminate parts of this landscape. Recursive training can induce model collapse when generated data recursively replace real data, especially in distribution tails [1, 2]. But collapse is not inevitable: when synthetic data accumulate alongside real data, degradation can be reduced or avoided [3]. Contamination studies have shown that benchmark overlap inflates apparent performance and that detection is harder than exact-match filtering suggests [4, 5, 6, 7, 8, 9]. Dynamic benchmarks such as LiveBench, AntiLeak-Bench, and LiveXiv respond by refreshing questions, relying on objective answers, and injecting temporally new content [10, 11, 12, 13]. Measurement-oriented work argues that even uncontaminated benchmarks may fail to measure the intended construct [14, 15, 16, 17]. More recent studies show both soft contamination by semantic duplicates and variant contamination beyond exact overlap [18, 19] and benchmark saturation as models converge toward empirical ceilings [20]. Human–AI collective-knowledge models add a further complication: the open web is increasingly co-produced by humans and LLMs, blurring any stable boundary between human-origin and AI-origin text [21].

These lines of work are individually useful but jointly incomplete. Recursive-training papers study training-distribution degradation, not the lifespan of benchmarks as instruments. Contamination papers study overlap and detection, not the joint evolution of discriminative power and construct validity. Dynamic benchmark papers give design recipes, not a general decay law. Construct-validity papers explain why score-based claims can fail, but not when benchmarks should be retired or regenerated. Saturation studies quantify plateauing, but do not integrate semantic leakage, evaluator coupling, construct drift, and regeneration into a single dynamical quantity.

This paper proposes that quantity. We define benchmark validity as the product of discriminative power and a construct-validity component. The first measures whether the benchmark still separates relevant systems above the declared score-noise floor. The second combines two logically distinct issues: whether the operational class still contains enough target-resolved pairs to matter, and whether the benchmark preserves the target ordering on those resolved pairs once score-resolution limits are respected. The benchmark half-life is then the first time this validity falls below half of its initial value. The resulting theory is deliberately origin-free. We do not use "human" and "synthetic" as primitive ontological categories because they are increasingly latent, unstable, and poorly aligned with what actually harms or preserves benchmark quality. A high-novelty synthetic stream can be instrumentally useful; a low-novelty human-origin stream can still erode a benchmark through exposure and derivative leakage. The relevant primitives are semantic leakage, verified novelty inflow, verification fidelity, support diversity, evaluator coupling, construct drift, and provenance observability.

The contributions are organized into four blocks. First, the paper gives an origin-free mathematical setup for recursive benchmark ecology, clarifies operational-scope dependence, and separates passive decay from explicit benchmark revisions or regeneration jumps. Second, it develops decay and identification theory: envelope and statewise comparison results, finite-lifetime and survival guarantees, non-identification theorems for score traces and aggregate validity, calibrated-support identification results, off-support extrapolation results under weak shape restrictions, and anchor-based partial-identification results for the construct-validity component under resolved-set mismatch, directional mismatch sharpening, anchor dependence, and qualification-abstention. Third, it extends the analysis from single benchmarks to

benchmark portfolios with common-mode hazards, uncertain coverage matrices, and learned uncertainty sets under dependent or nonstationary audits. Fourth, it connects lifetime estimation to governance by deriving anytime-valid monitoring rules, safe retirement and regeneration control, robust ambiguity-set control, robust belief-state control under partial observability, an explicit estimation-to-control interface, and approximate safe-control regret bounds.

The theory is intentionally modest. It does not claim that all benchmark decay is captured by a single scalar, nor that origin metadata are useless. Rather, it argues that benchmark life-cycle governance should be based on state variables and uncertainty bounds, not on raw scores or nominal source labels alone.

Reader roadmap. The paper is organized so that readers can stop at the weakest layer that answers their use case. Sections 3 and 5 introduce the operational objects and give intervalwise lifetime results that need only passive-envelope control. Section 6 adds stronger assumptions only where point identification, factor attribution, or anchor sharpening require them. Section 7 lifts the analysis from single benchmarks to benchmark families with uncertain coverage. Section 8 then adds sequential monitoring and safe control, including robust Bellman recursions for both information-state and belief-state ambiguity, conservative outer-hull reductions for nonrectangular ambiguity, and coverage-frontier action pruning. Readers interested mainly in implementation can jump from the setup directly to Sections 9 and 11.

2 Related Work and Positioning

2.1 Recursive training and model collapse

Shumailov et al. showed that recursively training on generated data can distort distributions and erase tails, producing what they called the curse of recursion [1, 2]. Gerstgrasser et al. then showed that collapse is not inevitable if real data are continually accumulated alongside synthetic data [3]. These papers are crucial but they address the health of training distributions, not the lifetime of benchmarks as evaluative instruments. Our object is benchmark validity, not corpus fidelity alone.

2.2 Benchmark contamination and detection

A growing literature surveys contamination definitions, detectors, and remediation strategies [4, 5, 6, 7]. Samuel et al. emphasize inconsistencies and oracle challenges in modern contamination detection [8]. Singh et al. propose ConTAM and argue that contamination metrics should be judged by whether the flagged examples actually benefit models downstream [9]. Liang et al. introduce DVD, a detector aimed specifically at variant contamination, where semantically equivalent items evade simpler matching rules [19]. This literature is highly relevant to our estimation protocol, but its center of gravity is detection quality rather than benchmark lifetime dynamics.

2.3 Dynamic benchmarks and regeneration

LiveBench, AntiLeak-Bench, and LiveXiv exemplify the move from static to dynamic evaluation [10, 11, 12]. Chen et al. survey this transition and make clear that the field is moving toward time-sensitive, contamination-aware benchmark maintenance rather than purely static test sets [13]. These works motivate our regeneration variable, but they do not yield a general state-variable theory of how much regeneration is enough to control benchmark half-life.

2.4 Adaptive reuse, dynamic creation, and psychometric benchmarking

Adaptive data analysis showed earlier that repeatedly reusing an evaluation set can itself destroy inferential validity, even without any ontological change in the data source [27, 28]. This is conceptually adjacent to benchmark half-life: repeated exposure and adaptation can make an apparently fixed instrument progressively less trustworthy. Dynabench operationalizes dynamic, human-and-model-in-the-loop benchmark creation, where annotators actively search for examples that current models fail on [29]. More recent psychometric and adaptive approaches such as Fluid Benchmarking use item-response structure and dynamic item selection to improve validity and reduce saturation [30]. These works strongly motivate our emphasis on reuse pressure, regeneration, and operational-class-relative validity, but they do not by themselves provide a general hazard-based lifetime theory.

2.5 Partial identification and interval inference

In settings with interval-valued or incompletely observed variables, econometrics treats many targets as *partially identified* rather than point identified [31, 32, 37]. That stance is directly relevant here. Recursive-corpus contamination, provenance, and benchmark-coupling states are often observed only through noisy detectors, incomplete logs, or heuristic proxies. Our interval results therefore adopt a deliberately conservative logic: absent an injective and calibrated observation model, benchmark validity and half-life should be reported as identification regions rather than faux-precision point estimates. For projected-support extrapolation, the relevant comparison class is not unrestricted smooth curve fitting but shape-restricted extension—for example Lipschitz or monotone continuation from calibrated support, in the spirit of classical extension results such as McShane's theorem [41]. Design-based exclusion and instrumental-variables logic are also relevant once one moves from passive monitoring to factor attribution: with the right exclusion restrictions, aggregate responses can identify otherwise hidden structural components [42]. When those exclusions are only approximate, Conley–Hansen–Rossi style plausibly exogenous logic becomes the right comparison class: one should report sensitivity regions indexed by declared violation budgets rather than pretend the instrument is pristine [47].

2.6 Latent-panel and annotator-error models

Anchor panels are measurement devices with their own error structure. Classical latent-annotator models such as Dawid–Skene estimate observer error rates from repeated noisy labels without direct access to ground truth [38]. That tradition motivates our insistence that anchor quality and dependence be modeled explicitly rather than smuggled in as an unexamined source of "expert" truth. Our latent-factor panel result is deliberately weaker than a full Dawid–Skene model: it does not try to identify every anchor-specific confusion matrix, only to upper-bound construct-validity error under common latent difficulty or blind-spot structure.

2.7 Portfolio design and robust allocation

Once benchmark half-life is extended from a single instrument to a benchmark family, allocation questions become unavoidable. Classical portfolio theory studies allocation under known or estimated return-risk trade-offs [39], while robust optimization studies allocation under uncertainty sets and worst-case guarantees [40, 45]. Tail-risk functionals such as conditional value-at-risk provide another lens for allocation when one cares about rare but severe failures rather than only mean performance [46]. Data-driven

robust optimization adds a further lesson that matters directly here: uncertainty sets can be learned from audit data and can carry finite-sample transfer guarantees when the true object lands inside the learned set with known confidence [48]. Our benchmark-portfolio design results are much simpler than those literatures in objective and constraints, but they inherit the same conceptual lesson: diversification is only rational once one acknowledges uncertainty, common-mode exposure, uncertain coverage accounting, learned uncertainty sets, or explicit breadth constraints. Under a single known deterministic hazard profile, the optimal horizon-survival design collapses to the best single benchmark.

2.8 Sequential monitoring and anytime-valid inference

Online benchmark governance is a sequential decision problem because scores, audits, and contamination evidence arrive over time while stakeholders keep peeking. Confidence-sequence and always-valid inference frameworks therefore provide the right comparison class for retirement and continuation rules under optional stopping: one wants intervals or envelopes that remain valid no matter when monitoring stops [49, 50]. Our sequential monitoring extension does not derive new confidence sequences from first principles; it shows how any declared anytime-valid validity envelope can be converted into safe retirement and continuation rules for benchmark governance and then into a constrained dynamic control problem for retirement versus regeneration.

2.9 Dependent auditing, mixing, and martingale concentration

Once audit rows are clustered by time, evaluator, family, or provenance chain, independence becomes a decorative fiction. Strong-mixing surveys such as Bradley's show both the usefulness and the limits of classical dependence regimes [51]. On the martingale side, Freedman-type inequalities and later time-uniform concentration tools show how predictable variation can replace independence as the relevant control variable for sequential inference [52, 49]. Our dependence-aware learned-set results borrow this viewpoint: the relevant question is not whether audits are independent in a metaphysical sense, but whether one has a valid sequential uncertainty construction under the actual dependence structure.

2.10 Nonstationary audits and regime switching

Dependence is only part of the story. Audit processes can also be nonstationary because benchmark taxonomies drift, evaluator pools change, or benchmark families move between different operating regimes. Classical regime-switching time-series models show how piecewise or latent changes in data-generating structure can be modeled without collapsing everything into one stationary law [53]. Our nonstationary audit extension adopts the same sober lesson: if audit regimes change, the uncertainty set should track the current or posterior-weighted regime rather than pretending that one fixed coverage matrix governs all times.

2.11 POMDP control and exploration–exploitation

The control side is naturally close to partially observed Markov decision processes, because the true benchmark-ecology state is rarely observed directly. Standard POMDP results show that optimal control can be written on the belief state rather than the latent state itself [58]. Robust MDP work shows that stagewise rectangular ambiguity over transition kernels can be propagated through a max–min Bellman

recursion rather than hidden behind a single plug-in estimate [54, 55]. Robust and distributionally robust POMDP work extends the same lesson to partially observed systems, where ambiguity can sit on the joint transition–observation law and still admit belief-state dynamic programming under suitable structure [56, 57]. Once learning and action are coupled, one also enters the exploration–exploitation territory studied in reinforcement learning, where confidence-based optimism can deliver finite regret guarantees under explicit uncertainty accounting [59]. Our extension borrows these lessons but keeps the benchmark-governance twist: the action set is safety-trimmed by anytime-valid validity bounds before any exploration is allowed, the ambiguity set is an explicit reported object, and the estimation layer is tied to control through a joint fail-closed certificate rather than a silent handoff.

2.12 Approximate dynamic programming and finite-horizon control

Once retirement and regeneration are treated as a sequential decision problem, exact dynamic programming quickly becomes expensive on realistic information states. Approximate dynamic programming and neuro-dynamic programming provide the natural comparison class [60]. Finite-sample and performance analyses for approximate value iteration and fitted value iteration show how value-function error propagates into policy loss [61]. Our safe-control extension is simpler than those general frameworks because the action space is explicitly safety-trimmed by the anytime-valid envelope, but the same moral applies: approximate control should come with explicit residual or regret certificates rather than hand-wavy optimization theater.

2.13 Construct validity, psychometrics, and model metrology

The construct-validity component used here sits inside a much older measurement tradition. Cronbach and Meehl framed construct validity as the disciplined relation between test behavior, theory, and the inferential claims a measurement instrument is meant to support [33]. Messick later unified validity under an evidential and consequential interpretation, emphasizing that score meaning cannot be detached from intended use [34]. Measurement-oriented critiques in AI now make a parallel point. Bean et al. show that many benchmark claims rest on weak construct-validity foundations [14]. Saxon et al. call for model metrology rather than scoreboard-centric benchmarking [15]. Salaudeen et al. provide a validity-centered framework linking evidence to evaluative claims [16]. Vendrow et al. show that even benchmark labels and adjudications can be systematically wrong enough to distort reliability claims, motivating more carefully curated "platinum" benchmarks [23]. Welty et al. earlier argued that AI evaluation should import metrological thinking about instrument resolution and uncertainty [17]. Our paper is squarely in this tradition, but adds explicit decay dynamics and a half-life object.

2.14 Item response theory and dynamic discrimination

Item-response theory studies how test items discriminate among latent ability levels and how item quality changes with item difficulty and population mix [35]. That perspective is conceptually adjacent to our discriminative-power term: both ask whether an instrument still separates the operational class it is meant to resolve. The difference is that we study the temporal decay of an entire benchmark lineage under recursive exposure, regeneration, and construct drift rather than the static calibration of individual items. Still, IRT provides an important antecedent for thinking about saturation, operational-class dependence, and the distinction between item difficulty and instrument validity.

2.15 Social choice and rank aggregation

Benchmark orderings are also related to social-choice and rank-aggregation problems. Arrow's classical work shows that there is no frictionless way to aggregate heterogeneous pairwise information into a universally satisfactory ordering rule [36]. That is not the same problem studied here, but it clarifies why construct-validity auditing cannot rely on one benchmark, one anchor, or one judge as if they revealed a uniquely privileged ranking. Our anchor-panel and mismatch results should therefore be read partly as metrological analogues of rank-aggregation robustness: the question is not only who wins under one ordering device, but how stable the inferred order is under noisy and partially overlapping comparison systems.

2.16 Design-based causal inference and matched intervention logic

The intervention-design part of the paper is also aligned with design-based causal inference. Matched designs, exclusion restrictions, and local instrumental-variables logic are standard ways to recover causal structure when direct observation is insufficient [42, 43, 44]. Our factor-targeted intervention results adapt that design logic to benchmark ecology: matched contrasts, exclusion budgets, and structural spillover classes are not rhetorical decoration but the objects that determine what can and cannot be identified from aggregate validity traces.

2.17 Benchmark-quality self-assessment and adversarial benchmarking

Some recent work evaluates benchmarks themselves rather than only the models scored on them. Qian et al. propose Benchmark^2, which measures ranking consistency, discriminability, and capability-alignment deviation across benchmark families [24]. Marro et al. reformulate frontier mathematical benchmarking as an adversarial generation–evaluation game and report stable rankings that correlate with external capability measures [25]. These papers are close in spirit to our aims, but they remain essentially cross-sectional: they diagnose benchmark quality or redesign benchmark construction at a time slice rather than treating validity as a dynamical state variable with a half-life.

2.18 Soft contamination, saturation, and recursive knowledge ecologies

Spiesberger et al. show that semantic duplicates can confound benchmark gains even when exact matching misses the overlap [18]. Akhtar et al. show that benchmark saturation is already widespread and that benchmark age and test scale matter more than several common folk remedies [20]. Nettasinghe and Zhao model the web as a human–AI collective-knowledge system with feedback loops, regime changes, and quality dilution risks [21]. Salaudeen et al. show, in a neighboring benchmark domain, that apparently robust benchmark behavior can be an artifact of misspecification rather than genuine measurement of the intended phenomenon [26]. These works motivate four core choices of our theory: semantic leakage must be broader than exact overlap, validity must be time-indexed, construct drift must be treated as a first-class failure mode, and nominal data origin is no longer a stable primitive.

Gap statement. To our knowledge, the literature still lacks a unified state-variable theory that treats benchmark validity as a dynamical quantity in recursive corpora, derives explicit lifetime bounds from semantic leakage, evaluator coupling, construct drift, support erosion, and regeneration, makes explicit which quantities are point identified, set identified, or not identified under noisy anchors, resolved-set

mismatch, incomplete provenance, imperfect contamination detectors, and aggregate-versus-factor hazard observability, sharpens imperfect-exclusion analysis into explicit identified sets, extends lifetime reasoning from single benchmarks to benchmark portfolios exposed to both idiosyncratic and common-mode decay together with explicit design choices under scenario uncertainty, and connects those lifetime summaries to safe sequential control of retirement and regeneration. This is the gap addressed here.

3 Mathematical Setup

3.1 Operational specification, scores, and target claims

Let $t \geq 0$ denote time. Let $\mathcal{M}_t$ be the operational class of models or systems that a benchmark is intended to compare at time t, and let Π_t be a probability distribution over ordered pairs $(m, m') \in \mathcal{M}_t \times \mathcal{M}_t$. The distribution Π_t is part of the benchmark specification: it determines whether the benchmark is meant to separate frontier systems, all publicly released systems, or some narrower deployment-relevant class.

Definition 1 (Operational evaluation specification)**.** An operational evaluation specification at time t is

$$\Omega_t := (\mathcal{M}_t, \Pi_t, \delta_t, \varepsilon_t), \tag{1}$$

where $\delta_t > 0$ is the effective score-resolution threshold used to declare practical score separation and $\varepsilon_t > 0$ is the target-resolution threshold used to decide which model pairs are resolved enough to count toward construct validity. All validity objects below are relative to a declared Ω_t. To keep notation readable, we suppress Ω_t whenever the specification is fixed or clear from context.

For a benchmark B, let $Y_B(m, t) \in [0, 1]$ denote the benchmark score assigned to model m at time t under the then-current evaluation protocol. Let $G_t(m) \in \mathbb{R}$ denote the target-claim functional, i.e., the latent or external capability quantity the benchmark is meant to support claims about. We do not assume G_t is fully observable. It is a conceptual target against which construct validity is defined.

Definition 2 (Discriminative power)**.** Relative to the declared operational specification, the discriminative power of benchmark B at time t is

$$\mathcal{D}_B(t) := \mathbb{P}_{(m,m') \sim \Pi_t}(|Y_B(m, t) - Y_B(m', t)| > \delta_t). \tag{2}$$

This definition absorbs score noise into the declared threshold δ_t: a benchmark is operationally discriminative only when it separates relevant systems by more than the effective noise floor associated with the stated use case.

Definition 3 (Construct validity, resolved mass, and resolution status)**.** Define the target-resolved-pair event

$$\mathcal{R}_t := \{(m, m') : |G_t(m) - G_t(m')| > \varepsilon_t\}, \tag{3}$$

the resolved-pair mass

$$R_B(t) := \mathbb{P}_{\Pi_t}(\mathcal{R}_t), \tag{4}$$

and the thresholded score-sign map

$$\operatorname{sgn}_{\delta_t}(z) := \begin{cases} +1, & z > \delta_t, \\ 0, & |z| \le \delta_t, \\ -1, & z < -\delta_t. \end{cases} \tag{5}$$

Define the resolution status

$$\Sigma_B^{\mathrm{res}}(t) := \begin{cases} \texttt{under-resolved}, & R_B(t) = 0, \\ \texttt{resolved}, & R_B(t) > 0. \end{cases} \tag{6}$$

If $R_B(t) = 0$, set $\mathcal{O}_B(t) := 0$ and $\mathcal{C}_B(t) := 0$. Otherwise define the order-fidelity component

$$\mathcal{O}_B(t) := \mathbb{P}\Big(\operatorname{sgn}_{\delta_t}\big(Y_B(m,t) - Y_B(m',t)\big) = \operatorname{sgn}(G_t(m) - G_t(m')) \;\Big|\; (m,m') \in \mathcal{R}_t\Big), \tag{7}$$

and the construct-validity component

$$\mathcal{C}_B(t) := R_B(t)\,\mathcal{O}_B(t). \tag{8}$$

This definition closes two boundary holes at once. First, an operational class with vanishingly few resolved target pairs should not look healthy merely because the benchmark preserves ordering on that tiny residue; the resolved-pair mass is therefore part of the construct-validity component. Second, order fidelity should respect the declared score-resolution threshold rather than pretend that infinitesimal score differences are meaningful. The status variable and the mass term separate three failure modes: too few resolved pairs, poor ordering on resolved pairs, and low discriminative power.

Proposition 4 (Fail-closed score-resolution factorization of order fidelity)**.** *Let*

$$\mathcal{G}_t^{\delta} := \{(m,m') : |Y_B(m,t) - Y_B(m',t)| > \delta_t\} \tag{9}$$

be the score-resolved event on the declared benchmark scale and define

$$\rho_{B,\delta}(t) := \mathbb{P}(\mathcal{G}_t^{\delta} \mid \mathcal{R}_t) \tag{10}$$

whenever $R_B(t) > 0$, with the convention $\rho_{B,\delta}(t) := 0$ when $R_B(t) = 0$. If

$$\mathcal{O}_B^{\delta}(t) := \mathbb{P}\big(\operatorname{sgn}_{\delta_t}\big(Y_B(m,t) - Y_B(m',t)\big) = \operatorname{sgn}(G_t(m) - G_t(m')) \mid \mathcal{R}_t \cap \mathcal{G}_t^{\delta}\big) \tag{11}$$

is interpreted only when $\mathbb{P}(\mathcal{R}_t \cap \mathcal{G}_t^{\delta}) > 0$, then

$$\mathcal{O}_B(t) = \rho_{B,\delta}(t)\,\mathcal{O}_B^{\delta}(t). \tag{12}$$

Proof. On $\mathcal{R}_t \setminus \mathcal{G}_t^{\delta}$, the thresholded score comparator is 0 while the target comparator is ± 1, so agreement is impossible under the fail-closed definition. The agreement event is therefore exactly the event of score resolution times correct sign on $\mathcal{R}_t$, and the displayed factorization follows from conditioning on $\mathcal{G}_t^{\delta}$ inside $\mathcal{R}_t$. □

Remark 5 (Fail-closed versus abstention-style order fidelity)**.** An alternative abstention-style object would condition only on $\mathcal{R}_t \cap \mathcal{G}_t^\delta$ and report $\mathcal{O}_B^\delta(t)$ as the primary order statistic, effectively excluding unresolved score pairs from the denominator. That object is useful diagnostically, and we use it later when sharper anchor models condition on score-resolved pairs. The benchmark-validity primitive in this paper is instead the fail-closed quantity $\mathcal{O}_B(t)$ because governance should treat unresolved score pairs as lost discriminatory evidence rather than as silent abstentions.

Definition 6 (Benchmark validity)**.** The benchmark validity is

$$\mathcal{V}_B(t) := \mathcal{D}_B(t)\,\mathcal{C}_B(t) \in [0,1]. \tag{13}$$

The multiplicative form is not unique, but it now has the intended fail-closed behavior: no discriminative power, no resolved mass, or no order fidelity each break the instrument.

Definition 7 (Benchmark half-life)**.** For $\mathcal{V}_B(0) > 0$, define the benchmark half-life relative to the declared operational specification by

$$\tau_{1/2}(B) := \inf\{t \geq 0 : \mathcal{V}_B(t) \leq \tfrac{1}{2}\mathcal{V}_B(0)\}. \tag{14}$$

The definition applies to the full right-continuous validity lineage, including any declared regime-change or regeneration jumps. Because the object is a first hitting time, a later regeneration jump cannot erase an earlier half-life event.

Operational ambiguity is therefore not a side issue. If Ω_t is changed, benchmark validity and half-life may change even when the underlying score process does not. The threshold $1/2$ is only a canonical reference level, not the only lifetime scalar of interest. Later sections work with the general threshold-invalidity time T_α for arbitrary $\alpha \in (0,1)$.

3.2 Recursive corpora and semantic leakage

Let $\mathcal{X}$ denote the space of publicly circulating artifacts that can influence future training, evaluation, or benchmark-aware behavior: benchmark items, paraphrases, worked solutions, tutorials, model-generated traces, discussion threads, retrieval targets, judge instructions, and so on.

Let μ_t be an exposure measure on $\mathcal{X}$ representing the artifact mass available in the recursive corpus environment at time t. For each benchmark item $b \in B$, let $\kappa_b : \mathcal{X} \to [0,1]$ be a semantic-neighborhood kernel that measures how much an artifact x leaks the evaluative content of b. Exact duplicates correspond to sharp kernels; semantic duplicates, paraphrases, and worked variants correspond to broader kernels.

Definition 8 (Semantic leakage mass)**.** The semantic leakage mass of benchmark B at time t is

$$S_B(t) := \frac{1}{|B|} \sum_{b \in B} \int_{\mathcal{X}} \kappa_b(x)\,\mu_t(\mathrm{d}x). \tag{15}$$

When the benchmark is clear from context we write S_t.

This definition is functional rather than ontological. Leakage depends on what an artifact reveals about benchmark content, not on whether that artifact is labeled human, synthetic, or hybrid.

3.3 State variables

We separate the latent state into an *ecological state*, which governs passive validity decay, and an *observability state*, which governs how sharply that ecology can be inferred and audited.

The ecological state is

$$X_t^{\mathrm{eco}} := (S_t, N_t, F_t, U_t, J_t, K_t). \tag{16}$$

Its coordinates are benchmark-specific and have the following meanings:

$$\begin{aligned} S_t &:= \text{semantic leakage mass}, \\ N_t &:= \text{verified novelty inflow}, \\ F_t &:= \text{verification fidelity}, \\ U_t &:= \text{support diversity / tail coverage}, \\ J_t &:= \text{evaluator coupling}, \\ K_t &:= \text{construct drift}. \end{aligned}$$

Their intended meanings are:

- S_t increases when the recursive corpus contains benchmark items or semantically nearby artifacts that make benchmark success easier for benchmark-specific rather than capability-general reasons.
- N_t measures the inflow of items that are genuinely new relative to plausible exposure and that remain verifiable enough to serve as evaluation material.
- F_t measures answer verifiability and evaluator fidelity: objective ground truth, validator robustness, and low judge fragility all increase F_t.
- U_t measures whether the benchmark still covers diverse or hard regions of the intended task distribution, including tails rather than only central modes.
- J_t measures coupling between the evaluated system and the evaluative apparatus, including shared judge families, benchmark-generation overlap, style-matching effects, or common curation pipelines.
- K_t measures drift between what the benchmark was designed to support claims about and what it now effectively measures.

The observability state is

$$A_t^{\mathrm{obs}} := (L_t, H_t, \Lambda_t^{\mathrm{jump}}), \tag{17}$$

where L_t is lineage observability, H_t denotes audit and calibration metadata such as proxy-quality summaries or validation-design records, and $\Lambda_t^{\mathrm{jump}}$ records the declared jump log. Its intended meanings are:

- L_t measures how well provenance, timestamps, revisions, or construction logs support auditing and partial identification.
- H_t stores calibration metadata, proxy-quality summaries, validation-study outputs, and other objects used to construct identified sets and uncertainty budgets.
- $\Lambda_t^{\mathrm{jump}}$ records the declared intervention log needed to certify regime-time jump multipliers and their envelopes.

The full governance state is

$$Z_t := (X_t^{\mathrm{eco}}, A_t^{\mathrm{obs}}). \tag{18}$$

Low F_t can directly harm validity because poor verification distorts both scores and claims. Low L_t mainly widens uncertainty because it obstructs provenance-aware auditing, weakens jump-factor certification, and restricts the action set that can be justified safely. We therefore let F_t enter the passive decay law directly, while L_t enters the identification, envelope-width, and control-admissibility results through the observability state.

Proposition 9 (Observability affects feasible governance without changing the passive hazard law)**.** *Suppose two benchmark lineages share the same ecological state path X_t^{eco} and declared operational specification but have different observability states A_t^{obs} and $(A_t^{\mathrm{obs}})'$. Under any passive hazard law depending only on X_t^{eco}, the two lineages have the same passive validity path between declared jumps. However, any identification region, jump-factor envelope, safe action set, or ambiguity set built from the reported audit objects may differ across the two observability states. Thus observability need not enter the passive hazard to matter operationally.*

Proof. If the passive hazard depends only on X_t^{eco}, identical ecological paths imply identical passive differential equations and therefore identical passive validity trajectories between declared jumps. All remaining differences arise through the inferential objects built from auditability, calibration, and jump certification, which are by construction functions of the observability state. □

3.4 Modeling assumptions

Operational specifications, judge protocols, or benchmark contents need not evolve smoothly. We therefore distinguish *passive decay intervals* from *declared regime times*. Let

$$0 = \tau_0 < \tau_1 < \tau_2 < \cdots$$

be a locally finite sequence containing all declared benchmark revisions, operational-scope changes, judge changes, or explicit regeneration actions. On each open interval (τ_k, τ_{k+1}), the benchmark artifact and operational specification are held fixed and validity evolves passively. At the regime times themselves, the observed validity path is assumed càdlàg and may jump because the instrument has been changed rather than because passive ecological pressure acted continuously. All envelope and hazard statements below apply intervalwise on passive intervals, while half-life is always defined on the full càdlàg path.

Assumption 10 (Envelope-valid passive benchmark dynamics)**.** On each passive interval (τ_k, τ_{k+1}) on which $\mathcal{V}_B(t) > 0$, the restriction of $\mathcal{V}_B$ to that interval is absolutely continuous and there exist measurable effective-rate envelopes $\underline{q}_B, \overline{q}_B : [0, \infty) \to [0, \infty)$ such that for almost every t in that interval,

$$-\overline{q}_B(t) \le \frac{\mathrm{d}}{\mathrm{d}t} \log \mathcal{V}_B(t) \le -\underline{q}_B(t). \tag{19}$$

This weak assumption is sufficient for intervalwise envelope bounds, finite half-life upper bounds, and fail-closed retirement rules. Jumps at declared regime times are handled separately by the càdlàg path convention rather than forced into the hazard itself.

Proposition 11 (Integral-envelope sufficiency for passive lifetime bounds)**.** *All conclusions in this paper that use theorem 10 only through integrated rate bounds remain valid under the following weaker condition: on each passive interval (a, b) with positive validity, there exist measurable envelopes $\underline{q}_B, \overline{q}_B \geq 0$ such that for every $a < s < t < b$,*

$$-\int_s^t \overline{q}_B(u)\,\mathrm{d}u \leq \log \mathcal{V}_B(t) - \log \mathcal{V}_B(s) \leq -\int_s^t \underline{q}_B(u)\,\mathrm{d}u. \tag{20}$$

In particular, theorems 23, 31 and 34 and every corollary or proposition derived only from those passive-interval envelope inequalities continue to hold without assuming pointwise differentiability of $\log \mathcal{V}_B$ on passive intervals.

Proof. Each listed result uses only ordered integrals of the log-validity path on passive intervals and concatenation across declared regime times. Whenever those integral inequalities are available directly, the same proofs go through with (20) in place of integrating (19). □

Definition 12 (Declared jump multipliers and cumulative jump factors)**.** For each declared regime time $\tau_k > 0$ with $\mathcal{V}_B(\tau_k^-) > 0$, define the benchmark-validity jump multiplier

$$\xi_B(\tau_k) := \frac{\mathcal{V}_B(\tau_k)}{\mathcal{V}_B(\tau_k^-)}. \tag{21}$$

If there is no jump at τ_k, set $\xi_B(\tau_k) := 1$. The cumulative jump factor up to time t is

$$J_B(t) := \prod_{k:\ 0<\tau_k\leq t} \xi_B(\tau_k), \tag{22}$$

with the empty product equal to 1. Because the regime schedule is locally finite, the product is well defined on every bounded horizon. When only interval information on jump multipliers is available, write

$$\underline{J}_B(t) := \prod_{k:\ 0<\tau_k\leq t} \underline{\xi}_B(\tau_k), \qquad \overline{J}_B(t) := \prod_{k:\ 0<\tau_k\leq t} \overline{\xi}_B(\tau_k) \tag{23}$$

for any declared multiplier envelopes $0 \leq \underline{\xi}_B(\tau_k) \leq \xi_B(\tau_k) \leq \overline{\xi}_B(\tau_k)$.

Let $\mathcal{S} \subseteq \mathbb{R}^6$ be a measurable ecological-state domain containing the range of X_t^{eco} on the horizon of interest. When delayed effects matter, one may augment X_t^{eco} with memory coordinates such as exponentially weighted leakage debt, judge-convention stock, public-exposure stock, or regeneration debt. Throughout the paper, the displayed six-dimensional ecological state is the minimal readable specification, not a claim that benchmark decay is intrinsically Markovian. The comparison and interval arguments below extend verbatim to any augmented sufficient ecological state once the relevant monotonicity assumptions are imposed coordinatewise.

Assumption 13 (State-reduced passive hazard law)**.** On each passive interval (τ_k, τ_{k+1}) during which $\mathcal{V}_B(t) > 0$, the restriction of $\mathcal{V}_B$ to that interval is absolutely continuous and there exists a measurable map

$$q_V : \mathcal{S} \to [0, \infty)$$

such that for almost every $t \geq 0$,

$$\frac{\mathrm{d}}{\mathrm{d}t} \log \mathcal{V}_B(t) = -q_V(X_t^{\mathrm{eco}}). \tag{24}$$

Assumption 14 (Coordinatewise monotonicity of the effective rate)**.** The map q_V is nondecreasing in S_t, J_t, K_t, $1 - F_t$, and $1 - U_t$, and nonincreasing in N_t.

Assumption 15 (Factor-specific passive hazard law)**.** On each passive interval (τ_k, τ_{k+1}) during which $\mathcal{D}_B(t) > 0$ and $\mathcal{C}_B(t) > 0$, the restrictions of $\mathcal{D}_B$ and $\mathcal{C}_B$ to that interval are absolutely continuous and there exist measurable maps

$$q_D : \mathcal{S} \to [0, \infty), \qquad q_C : \mathcal{S} \to [0, \infty),$$

such that for almost every $t \geq 0$,

$$\frac{\mathrm{d}}{\mathrm{d}t} \log \mathcal{D}_B(t) = -q_D(X_t^{\mathrm{eco}}), \tag{25}$$

$$\frac{\mathrm{d}}{\mathrm{d}t} \log \mathcal{C}_B(t) = -q_C(X_t^{\mathrm{eco}}). \tag{26}$$

If either $\mathcal{D}_B$ or $\mathcal{C}_B$ reaches zero inside a passive interval, the corresponding factor remains zero on the remainder of that passive interval. A later declared regime-time intervention may reset the factor to a positive value because the instrument itself has been changed.

Assumption 16 (Coordinatewise monotonicity of factor rates)**.** The factor-rate maps are coordinatewise monotone in the following sense:

- q_D is nondecreasing in S_t, J_t, K_t, $1 - F_t$, and $1 - U_t$, and nonincreasing in N_t.
- q_C is nondecreasing in S_t, J_t, K_t, and $1 - F_t$, nonincreasing in N_t, and weakly nonincreasing in U_t whenever broader support improves construct alignment.

Definition 17 (Exclusive and shared factor coordinates as optional design regimes)**.** The baseline factor law in theorem 15 allows every ecological coordinate to affect both factor hazards. An *exclusive-coordinate design regime* is an additional substantive restriction under which a declared coordinate block affects one factor hazard directly and the other factor hazard only through a known zero or bounded spillover channel. The exact-zero special case is used only in the cleanest intervention-identification results. Bounded-spillover variants are treated later as sensitivity classes rather than as background truths.

Definition 18 (Effective decay rate)**.** Under theorem 13, the effective benchmark decay rate is

$$q_B(t) := q_V(X_t^{\mathrm{eco}}). \tag{27}$$

Under theorem 15, one may additionally write

$$q_B(t) = q_D(X_t^{\mathrm{eco}}) + q_C(X_t^{\mathrm{eco}}) \tag{28}$$

on intervals where $\mathcal{V}_B(t) > 0$.

The assumption ladder is deliberate. The envelope assumption is weakest and is enough for fail-closed retirement bounds and, when rate envelopes separate across ecologies, for cross-ecology ordering. The state-reduced effective-rate law is stronger and is used for statewise comparison, proxy-based point identification, conditional interval propagation, and regeneration arguments. The factor-specific exact law is stronger still and is used only when one wants separate statements about discriminative power and construct validity. Exclusive-coordinate restrictions are *not* part of the baseline law; they are optional design-side assumptions that can be invoked, rejected, or weakened through explicit sensitivity budgets.

3.5 A useful linear proxy

For closed-form governance thresholds it is convenient to consider a linear proxy for the *passive* hazard,

$$q_B^{\mathrm{lin}}(t) := \Big[a_S S_t + a_J J_t + a_K K_t + a_F(1-F_t) + a_U(1-U_t) - a_N N_t\Big]_+, \tag{29}$$

with nonnegative coefficients $a_\bullet$ and $[x]_+ := \max\{x, 0\}$. The positive-part operator matters. Verified novelty inflow can offset passive decay pressure down to zero, but passive hazard itself should not become negative. Validity gains above the current level are modeled through declared regeneration or redesign jumps at regime times, not by pretending that a negative decay rate is the same thing as maintenance. The main comparison and interval results do not require linearity. The proxy is used only for explicit thresholds and phase sketches.

4 Why Origin Labels Fail as Primitives

The usual temptation is to classify data by nominal origin: human-origin artifacts are presumed safer, synthetic artifacts are presumed riskier. In recursive corpora this is not a stable primitive. A text may be human-written with AI assistance, AI-generated and lightly edited by a human, or produced by a human whose own cognitive state has been shaped by prior AI interaction. More importantly, even when provenance is known, origin does not coincide with instrumental quality. High-novelty synthetic content can improve downstream reasoning or help stabilize recursive training regimes, while low-novelty human-origin content can still leak benchmark structure and erode validity [3, 22, 21].

We formalize this in an observational and decision-theoretic way.

Definition 19 (Latent production history)**.** Let $\mathcal{Z}$ be a latent space of production histories. A history $z \in \mathcal{Z}$ includes the causal path by which an artifact entered the recursive corpus, including any mixture of human and AI intervention. Let $o(z)$ denote a nominal origin label, and let $\mathcal{O} : \mathcal{Z} \to \mathcal{Y}$ be the observation map that retains only publicly available information relevant to governance.

Proposition 20 (Observation-level origin non-identifiability)**.** *Suppose there exist $z, z' \in \mathcal{Z}$ such that*

$$\mathcal{O}(z) = \mathcal{O}(z') \qquad \textit{but} \qquad o(z) \neq o(z'). \tag{30}$$

Then no governance rule measurable with respect to the observed sigma-field can distinguish z from z' by nominal origin. If, in addition, z and z' induce the same declared operational specification, the same ecological state trajectory X_t^{eco}, and the same declared jump-multiplier path, then they are indistinguishable with respect to passive validity trajectories, jump-accounted validity paths, and any other validity objects in the present model that depend only on those quantities. They need not be indistinguishable with respect to identified sets, jump-factor envelopes, or safe action sets unless the observability states also agree.

Proof. Any governance rule based only on observed information factors through $\mathcal{O}$. If $\mathcal{O}(z) = \mathcal{O}(z')$, the rule must take the same value on both histories. For the second claim, theorem 13 implies that on passive intervals the validity path is determined by the declared operational specification and the ecological state trajectory alone. The jump-accounting convention then multiplies those passive factors by the same declared jump multipliers, so the resulting full càdlàg validity path agrees as well. The final sentence is exactly the distinction formalized in theorem 9: observability affects inferential and control objects even when it leaves the passive hazard law unchanged. □

Observation-level non-identifiability is only the first step. Even when an origin label is observed, it can still be an insufficient statistic for governance.

Proposition 21 (Origin labels are not sufficient statistics)**.** *Suppose there exist two feasible recursive-corpus histories $z, z' \in \mathcal{Z}$ with the same nominal origin label process $o(z) = o(z')$ but distinct observed state summaries. Let $\mathcal{A}$ be a feasible action set for benchmark governance, and suppose there exist actions $a, a' \in \mathcal{A}$ such that an origin-only policy must choose the same action on both histories, whereas a state-based policy may choose a on one history and a' on the other. Assume furthermore that the induced post-action ecological state trajectories satisfy a shared passive hazard law with the same monotone map q_V and satisfy the coordinatewise dominance inequalities later formalized in the statewise comparison principle. Then origin-only governance cannot, in general, attain the best feasible validity trajectory on both histories.*

Proof. An origin-only rule cannot distinguish the two histories by assumption, so it must choose the same feasible action on both. By construction, the state-based policy class can condition on the observed state summaries and choose different feasible actions. Under the stated shared-hazard and coordinatewise-dominance conditions, the usual monotone comparison argument implies that the state-based rule weakly dominates the origin-only path on both histories and is strictly better on at least one whenever the dominance is strict on a set of positive measure. Thus origin labels are not sufficient statistics for attainable benchmark-governance quality. □

Remark 22. These propositions are deliberately modest. They do not claim that provenance metadata are useless. They claim only that nominal origin is neither reliably identifiable from observations nor sufficient for validity governance once state-overlap and human–AI co-production are allowed.

5 Decay Dynamics and Benchmark Lifetime

5.1 Comparison principle and source-agnostic quality ordering

The weakest cross-ecology ordering result in the paper uses only rate envelopes. It does not require a common state map.

Theorem 23 (Envelope comparison under rate separation with order-preserving jumps)**.** *Consider two benchmark ecologies A and B for the same benchmark and declared operational specification, with identical initial validity $\mathcal{V}_A(0) = \mathcal{V}_B(0) > 0$. Suppose there is a common declared regime schedule*

$$0 = \tau_0 < \tau_1 < \cdots < \tau_K \leq T$$

that partitions $[0, T]$ into passive intervals (τ_k, τ_{k+1}). Suppose each ecology satisfies theorem 10 on every passive interval with envelope pairs $(\underline{q}_A, \overline{q}_A)$ and $(\underline{q}_B, \overline{q}_B)$, and that

$$\overline{q}_A(t) \leq \underline{q}_B(t) \qquad \text{for almost every } t \in (\tau_k, \tau_{k+1}),\ k = 0, \ldots, K-1. \tag{31}$$

Assume moreover that every declared regime-time jump is order preserving in the sense that for each $k = 1, \ldots, K$,

$$\mathcal{V}_A(\tau_k^-) \geq \mathcal{V}_B(\tau_k^-) \quad \Longrightarrow \quad \mathcal{V}_A(\tau_k) \geq \mathcal{V}_B(\tau_k). \tag{32}$$

Then

$$\mathcal{V}_A(t) \geq \mathcal{V}_B(t) \qquad \text{for all } t \in [0,T]. \tag{33}$$

Consequently, if the same passive-interval rate inequality and order-preserving jump condition hold for all $t \geq 0$, then $\tau_{1/2}^{(A)}(B) \geq \tau_{1/2}^{(B)}(B)$.

Proof. Proceed by induction over passive intervals. On (τ_0, τ_1), theorem 10 implies for almost every t,

$$\frac{\mathrm{d}}{\mathrm{d}t} \log \mathcal{V}_A(t) \geq -\overline{q}_A(t) \geq -\underline{q}_B(t) \geq \frac{\mathrm{d}}{\mathrm{d}t} \log \mathcal{V}_B(t).$$

Integrating from 0 to any $t < \tau_1$ yields $\mathcal{V}_A(t) \geq \mathcal{V}_B(t)$. The order-preserving jump condition then gives $\mathcal{V}_A(\tau_1) \geq \mathcal{V}_B(\tau_1)$. Repeating the same argument on each interval (τ_k, τ_{k+1}) with initial condition at τ_k proves the claim on $[0,T]$. The half-life ordering follows from the pointwise validity ordering. □

The next result sharpens theorem 23 by reducing cross-ecology ordering to coordinatewise state ordering under a *shared* monotone hazard map.

Theorem 24 (Statewise comparison principle with order-preserving jumps)**.** *Consider two benchmark ecologies A and B for the same benchmark and declared operational specification, with identical initial validity $\mathcal{V}_A(0) = \mathcal{V}_B(0) > 0$. Suppose there is a common declared regime schedule $0 = \tau_0 < \tau_1 < \cdots < \tau_K$ such that on every passive interval (τ_k, τ_{k+1}) and for almost every t,*

$$\begin{aligned} &S_t^{(A)} \leq S_t^{(B)}, && N_t^{(A)} \geq N_t^{(B)}, && F_t^{(A)} \geq F_t^{(B)}, \\ &U_t^{(A)} \geq U_t^{(B)}, && J_t^{(A)} \leq J_t^{(B)}, && K_t^{(A)} \leq K_t^{(B)}. \end{aligned}$$

Assume that both ecologies satisfy theorems 13 and 14 with the same *passive hazard map $q_V : \mathcal{S} \to [0,\infty)$, and that every regime-time jump is order preserving for validity:*

$$\mathcal{V}_A(\tau_k^-) \geq \mathcal{V}_B(\tau_k^-) \quad \Longrightarrow \quad \mathcal{V}_A(\tau_k) \geq \mathcal{V}_B(\tau_k), \qquad k \geq 1. \tag{34}$$

Then

$$\mathcal{V}_A(t) \geq \mathcal{V}_B(t) \qquad \text{for all } t \geq 0. \tag{35}$$

If on some passive interval at least one coordinate inequality is strict on a set of positive measure and q_V is strictly monotone in the corresponding coordinate there, then the inequality is strict for some positive time in that interval.

Proof. On any passive interval (τ_k, τ_{k+1}), coordinatewise monotonicity of q_V gives $q_A(t) \leq q_B(t)$ for almost every t. Using (24) for both ecologies and integrating from τ_k to $t \in (\tau_k, \tau_{k+1})$ yields

$$\log \frac{\mathcal{V}_A(t)}{\mathcal{V}_A(\tau_k)} = -\int_{\tau_k}^{t} q_A(u)\,\mathrm{d}u \geq -\int_{\tau_k}^{t} q_B(u)\,\mathrm{d}u = \log \frac{\mathcal{V}_B(t)}{\mathcal{V}_B(\tau_k)}. \tag{36}$$

If $\mathcal{V}_A(\tau_k) \geq \mathcal{V}_B(\tau_k)$, this implies $\mathcal{V}_A(t) \geq \mathcal{V}_B(t)$ throughout that passive interval. Starting from $k = 0$ and propagating across regime times using the order-preserving jump condition proves the claim. Strictness follows whenever the rate inequality is strict on a set of positive measure inside some passive interval. □

Corollary 25 (Ordered hazard-family comparison)**.** *Let ecologies A and B satisfy passive laws*

$$\frac{\mathrm{d}}{\mathrm{d}t}\log \mathcal{V}_A(t) = -q_V^{(A)}(X_t^{(A)}), \qquad \frac{\mathrm{d}}{\mathrm{d}t}\log \mathcal{V}_B(t) = -q_V^{(B)}(X_t^{(B)}),$$

with possibly different monotone hazard maps. If the same coordinatewise state ordering as in theorem 24 holds and

$$q_V^{(A)}(x) \le q_V^{(B)}(y) \qquad \text{whenever } x \preceq y \tag{37}$$

for the validity-improving order introduced later in theorem 52, then the same validity ordering conclusion as in theorem 24 follows. Thus a common hazard map is sufficient but not necessary; an ordered hazard family is enough.

Proof. The proof is identical to theorem 24 after replacing equality of the hazard map by the displayed cross-map order relation. □

Corollary 26 (Source-agnostic quality dominance)**.** *Suppose two candidate data-update policies or source streams induce state trajectories satisfying the ordering in theorem 24. Then the lower-rate stream weakly dominates the higher-rate stream in benchmark validity over time, regardless of nominal source label.*

This is the formal version of "human versus synthetic is the wrong contrast." The relevant contrast is low-rate versus high-rate contribution to the recursive corpus.

Proposition 27 (Matched-intervention sign identification)**.** *Fix a coordinate j of the state vector and a time window $[t_0, t_1]$. Consider two matched benchmark ecologies A and B with the same benchmark, the same declared operational specification, and equal validity at time t_0. Suppose that on $[t_0, t_1]$ all state coordinates coincide except coordinate j. If q_V is nondecreasing in coordinate j and*

$$X_{A,j}(t) \le X_{B,j}(t) \qquad \text{for almost every } t \in [t_0, t_1], \tag{38}$$

then

$$\frac{1}{t_1 - t_0}\log\frac{\mathcal{V}_A(t_1)}{\mathcal{V}_A(t_0)} \ge \frac{1}{t_1 - t_0}\log\frac{\mathcal{V}_B(t_1)}{\mathcal{V}_B(t_0)}. \tag{39}$$

If the coordinate inequality is strict on a set of positive measure and q_V is strictly monotone there, then the inequality is strict. If coordinate j is one in which q_V is nonincreasing, the conclusion reverses with the ordering of the state coordinate. Thus matched interventions can empirically identify the sign *of a coordinate effect without point-identifying the full hazard map.*

Proof. Under the stated matching, the only rate difference between the ecologies comes from coordinate j. Coordinatewise monotonicity therefore gives $q_A(t) \le q_B(t)$ for almost every $t \in [t_0, t_1]$. Integrating (24) on $[t_0, t_1]$ yields the average log-decay ordering. □

Definition 28 (Normalized validity survival)**.** For any benchmark ecology with $\mathcal{V}_B(0) > 0$, define its normalized validity survival curve by

$$\widetilde{V}_B(t) := \frac{\mathcal{V}_B(t)}{\mathcal{V}_B(0)}. \tag{40}$$

Lemma 29 (Half-life ordering from normalized survival)**.** *Consider two benchmark ecologies with positive initial validity. If*

$$\widetilde{V}_B^{(1)}(t) \le \widetilde{V}_B^{(2)}(t) \qquad \text{for all } t \ge 0, \tag{41}$$

then

$$\tau_{1/2}^{(1)}(B) \le \tau_{1/2}^{(2)}(B). \tag{42}$$

If, in addition, $\mathcal{V}_B^{(1)}(0) = \mathcal{V}_B^{(2)}(0)$*, then the same conclusion follows from the raw validity ordering* $\mathcal{V}_B^{(1)}(t) \le \mathcal{V}_B^{(2)}(t)$ *for all* t*.*

Proof. The event $\widetilde{V}_B^{(i)}(t) \le \frac{1}{2}$ is exactly the half-life event for ecology i. Therefore, if the first normalized survival curve is pointwise no larger than the second, its first hitting time of $\frac{1}{2}$ cannot exceed the second. When initial validities are equal, dividing the raw inequality by the common initial value yields the normalized inequality. □

5.2 Leakage monotonicity, finite half-life, and regeneration

The next result isolates semantic leakage.

Corollary 30 (Monotone discriminative decay under semantic leakage)**.** *Assume two ecologies have the same initial discriminative power and share a common declared regime schedule. Suppose that on every passive interval and for almost every* t*,*

$$S_t^{(A)} \ge S_t^{(B)}, \qquad N_t^{(A)} = N_t^{(B)}, \qquad F_t^{(A)} = F_t^{(B)}, \qquad U_t^{(A)} = U_t^{(B)}, \qquad J_t^{(A)} = J_t^{(B)}. \tag{43}$$

Assume further that any regime-time intervention is D*-order preserving in the sense that*

$$\mathcal{D}_A(\tau_k^-) \le \mathcal{D}_B(\tau_k^-) \quad \Longrightarrow \quad \mathcal{D}_A(\tau_k) \le \mathcal{D}_B(\tau_k). \tag{44}$$

Then under theorems 15 and 16,

$$\mathcal{D}_A(t) \le \mathcal{D}_B(t) \qquad \text{for all } t \ge 0. \tag{45}$$

Thus, ceteris paribus, more semantic leakage cannot improve benchmark discriminative power within the model.

Proof. Apply the comparison argument intervalwise using only (25), and propagate the ordering across regime times by the stated D-order-preserving jump condition. □

Theorem 31 (Finite passive half-life bound on a jump-free window)**.** *Assume theorem 10. Fix a passive interval* $[t_0, t_1]$ *on which no declared regime jump occurs and suppose there exists* $\underline{q} > 0$ *such that* $\underline{q}_B(t) \ge \underline{q}$ *for almost every* $t \in [t_0, t_1]$*. Then*

$$\mathcal{V}_B(t) \le \mathcal{V}_B(t_0) e^{-\underline{q}(t-t_0)} \qquad \text{for all } t \in [t_0, t_1]. \tag{46}$$

Consequently, if $t_0 + \log 2/\underline{q} \le t_1$*, then the process must cross half of its level at* t_0 *by time* $t_0 + \log 2/\underline{q}$*. In particular, if the same lower-rate condition holds from* 0 *onward and there are no upward validity jumps before time* $\log 2/\underline{q}$*, then*

$$\tau_{1/2}(B) \le \frac{\log 2}{\underline{q}}. \tag{47}$$

Proof. Integrate the right inequality in (19) from t_0 to t and use $\underline{q}_B(u) \ge \underline{q}$ almost everywhere on the passive interval. The hitting-time claim follows by solving $e^{-\underline{q}(t-t_0)} \le 1/2$. □

This theorem formalizes the basic point that a static public benchmark exposed to persistent positive decay pressure is a finite-lifetime instrument.

Corollary 32 (Certified no-redesign window)**.** *Under the setup of theorem 31, suppose governance declares a certified no-redesign window $[t_0, t_1]$ on which no upward validity jump is admissible, either because all interventions are prohibited or because every admissible intervention is guaranteed not to increase validity. Then the half-life conclusion of theorem 31 is ex ante certifiable on that window.*

This separates two use cases. Theorem 31 is an ex post passive-window theorem. Theorem 32 gives the ex ante design rule needed when one wants a governance guarantee rather than a retrospective decomposition.

Corollary 33 (Passive-regeneration threshold in the linear proxy)**.** *Assume the passive decay rate is bounded above by the linear proxy (29). If for almost every $t \in [0,T]$,*

$$a_N N_t \geq a_S S_t + a_J J_t + a_K K_t + a_F(1 - F_t) + a_U(1 - U_t), \tag{48}$$

then $q_B^{\mathrm{lin}}(t) = 0$ on $[0,T]$. Consequently the passive component of validity does not decay on any passive interval contained in $[0,T]$, so any change in $\mathcal{V}_B$ over that horizon must come from declared regime jumps such as regeneration, scope restriction, or benchmark redesign. If the inequality holds with additional slack $\delta > 0$, then the same no-passive-decay conclusion still holds, and the slack can be interpreted as regeneration headroom rather than as a negative passive hazard.

The interpretation is not that any churn prevents decay. The point is that *verified* regeneration must be strong enough to cancel passive decay pressure from leakage, coupling, drift, poor verification, and support erosion before any explicit redesign jump can improve the instrument.

Theorem 34 (Jump-aware average-budget longevity criterion)**.** *Assume theorem 13. Suppose the passive decay rate satisfies $q_B(t) \leq q_B^{\mathrm{lin}}(t)$ for almost every t on passive intervals contained in $[0,T]$. Let*

$$\underline{J}_B^{\mathrm{safe}}(t) \leq J_B(t) \tag{49}$$

be any declared lower envelope for the cumulative jump factor on $[0,T]$, and define the corresponding jump-loss budget

$$\Lambda_B^{\mathrm{safe}}(t) := -\log \underline{J}_B^{\mathrm{safe}}(t) \in [0,\infty] \tag{50}$$

with the convention $\Lambda_B^{\mathrm{safe}}(t) = +\infty$ if $\underline{J}_B^{\mathrm{safe}}(t) = 0$. Then for every $t \in [0,T]$,

$$\mathcal{V}_B(t) \geq \mathcal{V}_B(0)\, \underline{J}_B^{\mathrm{safe}}(t) \exp\Big(-\int_0^t q_B^{\mathrm{lin}}(u)\,\mathrm{d}u\Big). \tag{51}$$

If the lower envelope

$$\underline{E}_B(t) := \underline{J}_B^{\mathrm{safe}}(t) \exp\Big(-\int_0^t q_B^{\mathrm{lin}}(u)\,\mathrm{d}u\Big) \tag{52}$$

is nonincreasing on $[0,T]$ and satisfies

$$\int_0^T q_B^{\mathrm{lin}}(u)\,\mathrm{d}u + \Lambda_B^{\mathrm{safe}}(T) < \log 2, \tag{53}$$

then

$$\tau_{1/2}(B) > T \tag{54}$$

on the full càdlàg validity path.

Proof. Integrating (24) on each passive interval and using $q_B \le q_B^{\mathrm{lin}}$ yields the passive lower bound. Multiplying across passive intervals and then applying the declared lower jump envelope gives (51). If $\underline{E}_B$ is nonincreasing, its minimum on $[0,T]$ is attained at T, so

$$\mathcal{V}_B(t) \ge \mathcal{V}_B(0)\,\underline{E}_B(T) \qquad \text{for all } t \in [0,T].$$

The log-budget inequality is exactly the condition $\underline{E}_B(T) > 1/2$, which therefore prevents any half-life crossing by time T. □

The no-upward-jump case is the special choice $\underline{J}_B^{\mathrm{safe}}(t) = \prod_{k:\tau_k \le t} \xi_B(\tau_k) \le 1$. The advantage of the present form is that upward jumps are no longer excluded; they are incorporated through whatever conservative lower jump envelope governance can certify.

Proposition 35 (History-augmented robustness of the comparison argument)**.** *Let $\widetilde{X}_t = (X_t^{\mathrm{eco}}, M_t)$ be any augmented sufficient state, where M_t may summarize lagged leakage, evaluator-convention stock, or other memory variables taking values in a measurable space. Suppose*

$$\frac{\mathrm{d}}{\mathrm{d}t} \log \mathcal{V}_B(t) = -\widetilde{q}_V(\widetilde{X}_t) \tag{55}$$

for almost every t on the positive-validity region, and suppose $\widetilde{q}_V$ is coordinatewise monotone in the augmented coordinates in the same directional sense as theorem 14. Then the conclusions of theorems 24, 29 and 42 continue to hold after replacing X_t^{eco} by $\widetilde{X}_t$ and imposing the corresponding augmented state ordering.

Proof. The proofs of the comparison and interval results use only three ingredients: absolute continuity, pointwise rate ordering, and integration of $\log \mathcal{V}_B$. Once the augmented state is sufficient for the rate law and the same coordinatewise monotonicity argument yields pointwise rate ordering, the proofs go through unchanged. □

Proposition 36 (Targeted regeneration dominates untargeted regeneration in the linear proxy)**.** *Suppose the benchmark is partitioned into cells $i = 1, \dots, r$, and that on a horizon of interest the linear proxy can be written as*

$$q_B^{\mathrm{lin}}(t) = \Big[c_t + \sum_{i=1}^{r} (\lambda_i(t) - \rho_i n_i(t)) \Big]_+, \tag{56}$$

where c_t and $\lambda_i(t)$ are independent of the regeneration allocation, $\rho_i \ge 0$ is the marginal regeneration efficiency of cell i, and the allocation variables satisfy

$$0 \le n_i(t) \le \bar{n}_i(t), \qquad \sum_{i=1}^{r} n_i(t) = N_t^{\mathrm{tot}}. \tag{57}$$

Then any allocation maximizing $\sum_i \rho_i n_i(t)$ minimizes the instantaneous proxy hazard. In particular, an optimal policy is obtained by filling cells in weakly decreasing order of ρ_i up to the caps $\bar{n}_i(t)$. Consequently,

among all allocations with the same total budget process N_t^{tot}, a targeted regeneration policy weakly maximizes the lower bound on $\mathcal{V}_B(t)$ induced by theorem 34.

Proof. At each fixed t, minimizing (56) over the feasible set is equivalent to maximizing the linear objective $\sum_i \rho_i n_i(t)$ over a simplex with box constraints. Linear programs attain an optimum at an extreme point, and every extreme-point optimizer fills the highest-ρ_i cells first up to capacity before assigning mass to lower-ρ_i cells. The second claim follows because the lower bound in theorem 34 is monotone nonincreasing in the time integral of the proxy hazard, so pointwise minimization of the proxy hazard weakly maximizes the integrated lower bound. □

This proposition makes the design implication explicit: for a fixed novelty budget, benchmark maintenance should target the regions with the highest marginal validity return rather than spread updates uniformly by default.

5.3 Operational-class relativity and saturation

The same benchmark can have different lifetimes for different operational classes because the pair distribution Π_t changes.

Proposition 37 (Operational-class relative half-life)**.** *Consider the same benchmark score process $Y_B(m,t)$ and target functional $G_t(m)$ under two declared operational specifications that differ only through their pair distributions $\Pi_t^{(1)}$ and $\Pi_t^{(2)}$ (and hence potentially through the induced resolved-pair event and noise-floor interpretation). Assume $\mathcal{V}_B^{(1)}(0) > 0$ and $\mathcal{V}_B^{(2)}(0) > 0$. If for all t in a horizon of interest,*

$$\frac{\mathcal{D}_B^{(1)}(t)}{\mathcal{D}_B^{(1)}(0)} \le \frac{\mathcal{D}_B^{(2)}(t)}{\mathcal{D}_B^{(2)}(0)} \qquad \text{and} \qquad \frac{\mathcal{C}_B^{(1)}(t)}{\mathcal{C}_B^{(1)}(0)} \le \frac{\mathcal{C}_B^{(2)}(t)}{\mathcal{C}_B^{(2)}(0)}, \tag{58}$$

then

$$\widetilde{V}_B^{(1)}(t) \le \widetilde{V}_B^{(2)}(t) \qquad \text{and hence} \qquad \tau_{1/2}^{(1)}(B) \le \tau_{1/2}^{(2)}(B). \tag{59}$$

If, in addition, $\mathcal{D}_B^{(1)}(0) = \mathcal{D}_B^{(2)}(0)$ and $\mathcal{C}_B^{(1)}(0) = \mathcal{C}_B^{(2)}(0)$, then the same conclusion follows from the raw inequalities $\mathcal{D}_B^{(1)}(t) \le \mathcal{D}_B^{(2)}(t)$ and $\mathcal{C}_B^{(1)}(t) \le \mathcal{C}_B^{(2)}(t)$. In particular, a benchmark may saturate for a frontier-focused operational class while remaining instrumentally alive for a broader or weaker class.

Proof. Multiplying the normalized factor inequalities yields

$$\frac{\mathcal{V}_B^{(1)}(t)}{\mathcal{V}_B^{(1)}(0)} = \frac{\mathcal{D}_B^{(1)}(t)}{\mathcal{D}_B^{(1)}(0)} \cdot \frac{\mathcal{C}_B^{(1)}(t)}{\mathcal{C}_B^{(1)}(0)} \le \frac{\mathcal{D}_B^{(2)}(t)}{\mathcal{D}_B^{(2)}(0)} \cdot \frac{\mathcal{C}_B^{(2)}(t)}{\mathcal{C}_B^{(2)}(0)} = \frac{\mathcal{V}_B^{(2)}(t)}{\mathcal{V}_B^{(2)}(0)}. \tag{60}$$

The half-life ordering then follows from theorem 29. The equal-initial-value case reduces raw ordering to normalized ordering. □

This proposition formalizes why "benchmark death" is relative to a declared operational class rather than an absolute property of the artifact.

5.4 Exact overlap versus semantic leakage

Exact matching is an attractive operational shortcut, but it is systematically optimistic whenever semantic duplicates exist.

For each benchmark item b, let κ_b^{exact} and κ_b^{sem} be exact and semantic kernels satisfying

$$\kappa_b^{\text{exact}}(x) \le \kappa_b^{\text{sem}}(x) \qquad \text{for all } x \in \mathcal{X}. \tag{61}$$

Let the corresponding leakage masses be S_t^{exact} and S_t^{sem}.

Proposition 38 (Exact-match decontamination is optimistic)**.** *For all $t \ge 0$,*

$$S_t^{\text{exact}} \le S_t^{\text{sem}}. \tag{62}$$

Assume the two accounting schemes begin from the same initial benchmark state and differ only through the leakage coordinate, with all other state variables held fixed. Then under theorems 13 and 14,

$$\mathcal{V}_B^{\text{exact}}(t) \ge \mathcal{V}_B^{\text{sem}}(t) \qquad \textit{for all } t \ge 0, \tag{63}$$

and therefore

$$\tau_{1/2}^{\text{sem}}(B) \le \tau_{1/2}^{\text{exact}}(B). \tag{64}$$

Thus exact-match-only decontamination weakly overestimates benchmark survival whenever semantic duplicates or worked variants contribute additional leakage.

Proof. Kernel monotonicity implies the leakage inequality by integration against the nonnegative exposure measure μ_t. Because all other coordinates are fixed and q_V is nondecreasing in leakage, the semantic-accounting ecology has weakly larger decay rate at every time. The pointwise validity ordering then follows from theorem 24. Since the initial validities agree, theorem 29 yields the half-life ordering. □

This result gives a formal explanation for why exact-match filtering systematically overestimates benchmark longevity in the presence of semantic duplicates or worked variants.

6 Identification, Proxies, and Factor Attribution

6.1 Partial identification and the detection–estimation gap

In practice, S_t, J_t, K_t, and sometimes N_t or F_t are not observed exactly. Detectors disagree, provenance is incomplete, and benchmark-aware fine-tuning can obscure exposure [7, 8, 9, 19]. The correct object is therefore often an interval, not a point. More precisely, once a hazard map q_V has been calibrated, learned, or otherwise justified for the operational class of interest, bounded latent states propagate to bounded validity trajectories. Before stating such propagation results, it is useful to mark the identification boundary sharply.

Theorem 39 (Score-trace non-identifiability of state–hazard decomposition)**.** *Let $\mathcal{V}_B : [0,T] \to (0,\infty)$ be absolutely continuous and define the composite hazard $h(t) := -\frac{\mathrm{d}}{\mathrm{d}t} \log \mathcal{V}_B(t)$ for almost every t. Observing the path $\mathcal{V}_B(t)$ on $[0,T]$ identifies $h(t)$ almost everywhere, but in general does* not *point-identify a latent ecological state path X_t^{eco} and hazard map q_V satisfying (24). This remains true even if one restricts attention to interpretable six-dimensional states in a compact box, continuous hazard maps, and coordinatewise-monotone state paths.*

Proof. Choose any strictly increasing C^1 function $\phi : [0,T] \to [0,1]$ with $\phi(0) = 0$ and $\phi(T) < 1$, and define the interpretable ecological path

$$X_t^{(1)} := (\phi(t), 0, 1, 1, 0, 0) \in [0,1]^6.$$

Let $q_V^{(1)}$ be any continuous extension on $[0,1]^6$ satisfying

$$q_V^{(1)}(X_t^{(1)}) = h(t) \qquad \text{for almost every } t.$$

Now choose a different strictly increasing C^1 path

$$X_t^{(2)} := (0, \phi(t), 1, 1, 0, 0) \in [0,1]^6$$

and define a continuous extension $q_V^{(2)}$ satisfying

$$q_V^{(2)}(X_t^{(2)}) = h(t) \qquad \text{for almost every } t.$$

Both ecological paths remain in the same compact interpretable state box and are coordinatewise monotone, but they attribute the same observed hazard to different ecological coordinates. Since

$$q_V^{(1)}(X_t^{(1)}) = h(t) = q_V^{(2)}(X_t^{(2)})$$

almost everywhere, both pairs generate the same validity path. Varying the chosen coordinate path or its embedding yields infinitely many observationally equivalent decompositions. □

Thus score traces alone identify only the time-indexed composite hazard, not the latent benchmark-ecology state or the structural hazard map. Additional calibration structure is required.

Definition 40 (Proxy calibration correspondence)**.** Let $W_t \in \mathcal{W}$ be an observed proxy process. A proxy calibration correspondence is a measurable set-valued map

$$\mathcal{C}_t : \mathcal{W} \rightrightarrows \mathcal{S}$$

such that the latent state satisfies

$$X_t^{\text{eco}} \in \mathcal{C}_t(W_t) \qquad \text{for almost every } t. \tag{65}$$

The model is *point-calibrated* if $\mathcal{C}_t(W_t)$ is almost surely a singleton, and *set-calibrated* otherwise.

Proposition 41 (Point identification under injective calibrated proxies)**.** *Suppose theorem 13 holds, the hazard map q_V is known on $\mathcal{S}$, and the observed proxy process obeys*

$$W_t = g(X_t^{\text{eco}}) \tag{66}$$

for a known injective map $g : \mathcal{S} \to \mathcal{W}$. Suppose in addition that the declared regime-time jump multipliers $\xi_B(\tau_k)$ are point identified from benchmark revision logs or governance records. Then $X_t^{\text{eco}} = g^{-1}(W_t)$ is

point identified for almost every t, the effective hazard

$$q_B(t) = q_V(g^{-1}(W_t)) \tag{67}$$

is point identified, and

$$\mathcal{V}_B(t) = \mathcal{V}_B(0)\, J_B(t) \exp\Big(-\int_0^t q_V(g^{-1}(W_u))\,\mathrm{d}u\Big). \tag{68}$$

If g is not injective, then the fiber $g^{-1}(W_t)$ defines only a set-identified state unless additional restrictions are imposed. If the jump multipliers are not point identified, the passive hazard path remains point identified but the full validity path inherits jump-factor uncertainty.

Proof. Injectivity yields $X_t^{\mathrm{eco}} = g^{-1}(W_t)$ almost everywhere. Substituting into (24) identifies the passive hazard path. Integrating over each passive interval and multiplying by the identified jump multipliers at declared regime times gives the displayed representation with cumulative jump factor $J_B(t)$. Without injectivity, multiple latent ecological states may map to the same proxy observation. □

The next theorem gives the general set-identified counterpart.

Theorem 42 (Interval propagation under proxy calibration, declared jump factors, and a calibrated hazard map)**.** *Assume theorem 13. Let W_t be an observed proxy process with calibration correspondence $\mathcal{C}_t$. For each declared regime time $\tau_k \in (0, T]$, suppose the jump multiplier obeys a known envelope*

$$0 \le \underline{\xi}_B(\tau_k) \le \xi_B(\tau_k) \le \overline{\xi}_B(\tau_k) < \infty. \tag{69}$$

Define the passive hazard envelopes

$$\underline{q}_B(t) := \inf_{x \in \mathcal{C}_t(W_t)} q_V(x), \tag{70}$$

$$\overline{q}_B(t) := \sup_{x \in \mathcal{C}_t(W_t)} q_V(x). \tag{71}$$

Then for almost every t, $\underline{q}_B(t) \le q_B(t) \le \overline{q}_B(t)$, and for all $t \in [0, T]$,

$$\mathcal{V}_B(0)\, \underline{J}_B(t) \exp\Big(-\int_0^t \overline{q}_B(u)\,\mathrm{d}u\Big) \le \mathcal{V}_B(t) \le \mathcal{V}_B(0)\, \overline{J}_B(t) \exp\Big(-\int_0^t \underline{q}_B(u)\,\mathrm{d}u\Big). \tag{72}$$

Consequently, the full validity path is set identified by a jump-aware envelope pair. If the upper envelope is nonincreasing on a horizon of interest, for example because no upward jump multiplier above 1 is admissible on that horizon, then first-hitting-time objects such as half-life and threshold invalidity time are bracketed by the corresponding first hitting times of the lower and upper envelopes. The theorem should be read as a propagation result conditional on the supplied calibration correspondence, hazard map, and jump-factor envelopes, not as a claim that q_V itself is identifiable from arbitrary observations.

Proof. Because $X_t^{\mathrm{eco}} \in \mathcal{C}_t(W_t)$ almost everywhere, the definition of infimum and supremum gives

$$\underline{q}_B(t) \le q_V(X_t^{\mathrm{eco}}) = q_B(t) \le \overline{q}_B(t)$$

for almost every t. Integrating (24) on each passive interval yields passive exponential bounds. Multiplying those bounds across passive intervals and across declared jump multipliers, while using the jump envelopes

at each regime time, yields (72). □

Corollary 43 (Coordinatewise box calibration)**.** *Assume theorems 13 and 14. Suppose the calibration correspondence is the coordinatewise box determined by bounds*

$$\underline{S}_t \le S_t \le \overline{S}_t, \qquad \underline{N}_t \le N_t \le \overline{N}_t, \qquad \underline{F}_t \le F_t \le \overline{F}_t, \tag{73}$$

$$\underline{U}_t \le U_t \le \overline{U}_t, \qquad \underline{J}_t \le J_t \le \overline{J}_t, \qquad \underline{K}_t \le K_t \le \overline{K}_t. \tag{74}$$

Then the extremal hazards are attained at the coordinate corners,

$$\underline{q}_B(t) := q_V(\underline{S}_t, \overline{N}_t, \overline{F}_t, \overline{U}_t, \underline{J}_t, \underline{K}_t), \tag{75}$$

$$\overline{q}_B(t) := q_V(\overline{S}_t, \underline{N}_t, \underline{F}_t, \underline{U}_t, \overline{J}_t, \overline{K}_t), \tag{76}$$

and theorem 42 applies with these explicit formulas.

Proof. Coordinatewise monotonicity implies that q_V is minimized and maximized on the box at the displayed corners. □

Corollary 44 (Metrological value of lineage observability)**.** *Suppose the calibration-set diameter, or in the box-calibrated case the interval widths*

$$\Delta_S := \overline{S}_t - \underline{S}_t,\ \Delta_N := \overline{N}_t - \underline{N}_t,\ \Delta_F := \overline{F}_t - \underline{F}_t,\ \Delta_U := \overline{U}_t - \underline{U}_t,\ \Delta_J := \overline{J}_t - \underline{J}_t,\ \Delta_K := \overline{K}_t - \underline{K}_t,$$

is nonincreasing with lineage observability L_t, that the jump-multiplier interval widths $\log \overline{\xi}_B(\tau_k) - \log \underline{\xi}_B(\tau_k)$ are likewise nonincreasing with improved auditability, and that q_V is Lipschitz on the relevant state domain. Then the jump-aware log-envelope width

$$\widetilde{W}(t) := \log \frac{\overline{J}_B(t)}{\underline{J}_B(t)} + \int_0^t (\overline{q}_B(u) - \underline{q}_B(u))\, \mathrm{d}u \tag{77}$$

is nonincreasing with improved lineage observability. In particular, better provenance narrows the feasible validity and half-life intervals even when it does not directly change the latent benchmark quality.

Proof. Lipschitz continuity bounds $\overline{q}_B - \underline{q}_B$ by a constant times the diameter of the feasible calibration set; in the box-calibrated case this reduces to a nonnegative linear combination of coordinate widths. The jump contribution to $\widetilde{W}(t)$ is the sum of log-interval widths of the declared jump multipliers. If both the state-calibration widths and the jump-multiplier widths are nonincreasing in L_t, then so is the total jump-aware envelope width. □

6.2 Minimal calibration conditions for hazard-map identification

The interval results above are deliberately permissive: they allow broad calibration correspondences and therefore broad validity envelopes. One can ask when the stronger object, the structural hazard map q_V, becomes point identified rather than merely bounded.

Definition 45 (State-recovery regime)**.** A *state-recovery regime* is any declared identification design that point identifies the ecological path X_t^{eco} on a horizon of interest before any attempt is made to identify the hazard map itself. Examples include direct audit recovery, injective proxy inversion, calibrated latent-state

recovery from repeated audits, or experimentally designed support excitation together with an externally justified proxy-calibration map. Hazard-map identification in this paper is always conditional on belonging to such a regime; it is not claimed to arise from score traces alone.

Definition 46 (Support-dense calibrated design)**.** Fix a compact target set $K \subseteq \mathcal{S}$ and a horizon $[0, T]$. A calibration design is *support-dense on* K if:

1. the ecological state X_t^{eco} is point identified for almost every $t \in [0, T]$ from a declared state-recovery regime;
2. the composite hazard $h(t) := -\frac{\mathrm{d}}{\mathrm{d}t} \log \mathcal{V}_B(t)$ is identified for almost every $t \in [0, T]$; and
3. the identified state image

$$\mathcal{I}_T := \{X_t^{\text{eco}} : \ t \in [0, T] \text{ outside a null set}\}$$

is dense in K.

Theorem 47 (Hazard-map identification on calibrated support)**.** *Assume theorem 13. Let $K \subseteq \mathcal{S}$ be compact, and suppose q_V is continuous on K. If the calibration design is support-dense on K, then q_V is point identified on K.*

Proof. By theorem 13, the identified hazard and identified state satisfy

$$h(t) = q_V(X_t^{\text{eco}}) \qquad \text{for almost every } t \in [0, T].$$

Hence the values of q_V are identified on the dense set $\mathcal{I}_T \cap K$. Because q_V is continuous on the compact set K, there is at most one continuous extension from that dense set to all of K. Therefore q_V is point identified on K. □

Proposition 48 (Off-support non-identifiability of continuous hazard maps)**.** *Assume theorem 13. Let $K \subseteq \mathcal{S}$ be compact and suppose the identified state image $\mathcal{I}_T$ is not dense in K. Then, in general, q_V is not point identified on K even if X_t^{eco} and $h(t)$ are point identified for almost every $t \in [0, T]$. More precisely, there exist distinct continuous maps $q, q' : K \to [0, \infty)$ that agree on $\overline{\mathcal{I}_T} \cap K$ and therefore induce the same observations on $[0, T]$.*

Proof. Because $\mathcal{I}_T$ is not dense in K, there exists a nonempty open set $U \subseteq K$ disjoint from $\overline{\mathcal{I}_T}$. Choose any nonnegative continuous bump function ψ supported in U and not identically zero. For any continuous map q consistent with the observations on $\overline{\mathcal{I}_T} \cap K$, the perturbed map $q' := q + \psi$ is continuous, distinct from q, and agrees with q on $\overline{\mathcal{I}_T} \cap K$. Since the observed state path never enters U, both maps induce the same hazard values along the observed path and are observationally equivalent on $[0, T]$. □

Corollary 49 (Minimal calibration ingredients)**.** *Point identification of a hazard map on a target region requires three ingredients: point-calibrated state recovery, identification of the composite hazard along the observed path, and sufficiently rich support or experimental excitation to cover the target region. If any one of these ingredients fails, the structural hazard map is generally only set identified.*

The logical moral is slightly annoying but useful: one does not get a structural hazard map merely by drawing a smooth curve through validity points. One gets it only on the portion of state space that is first recovered by a declared state-recovery regime and then visited densely enough to pin the map down.

6.3 Projected-support extrapolation under minimal additional restrictions

Theorem 48 shows that off-support point identification is impossible without extra structure. The next results isolate weak additional restrictions that convert that impossibility into sharp *set* identification rather than unjustified point extrapolation.

Definition 50 (Identified support and Lipschitz hazard class)**.** Fix a compact target region $K \subseteq \mathcal{S}$ and let

$$E := \overline{\mathcal{I}_T} \cap K$$

denote the identified visited support inside K. For $L \geq 0$, let $\mathcal{Q}_L^+(K)$ be the class of nonnegative L-Lipschitz maps $q : K \to [0, \infty)$.

Theorem 51 (Sharp off-support interval under a known Lipschitz modulus)**.** *Assume theorem 13. Suppose q_V is point identified on E as q_E and that $q_V \in \mathcal{Q}_L^+(K)$ for a known constant L. Then for every $x \in K$,*

$$q_V(x) \in I_L(x) := \Big[\max\Big\{0, \sup_{y \in E}\big(q_E(y) - L\|x - y\|\big)\Big\},\ \inf_{y \in E}\big(q_E(y) + L\|x - y\|\big)\Big]. \tag{78}$$

Moreover the interval is sharp: for every $v \in I_L(x)$ there exists $\widetilde{q} \in \mathcal{Q}_L^+(K)$ that agrees with q_E on E and satisfies $\widetilde{q}(x) = v$.

Proof. If $q_V \in \mathcal{Q}_L^+(K)$, then for every $y \in E$,

$$q_V(x) \geq q_E(y) - L\|x - y\|, \qquad q_V(x) \leq q_E(y) + L\|x - y\|.$$

Taking the supremum and infimum over $y \in E$ gives the displayed interval, together with nonnegativity.

For sharpness, fix $v \in I_L(x)$ and define f_v on $E \cup \{x\}$ by $f_v|_E = q_E$ and $f_v(x) = v$. The membership condition $v \in I_L(x)$ is equivalent to

$$|v - q_E(y)| \leq L\|x - y\| \qquad \text{for all } y \in E,$$

so f_v is L-Lipschitz on $E \cup \{x\}$. By McShane's extension theorem [41], there exists an L-Lipschitz map $g_v : K \to \mathbb{R}$ extending f_v. Define

$$\widetilde{q}(z) := \max\{0, g_v(z)\}.$$

Because $z \mapsto \max\{0, z\}$ is 1-Lipschitz, $\widetilde{q}$ is again L-Lipschitz and nonnegative. It agrees with f_v on $E \cup \{x\}$ because all values of f_v are already nonnegative there. Hence $\widetilde{q} \in \mathcal{Q}_L^+(K)$ is observationally admissible and satisfies $\widetilde{q}(x) = v$. □

Definition 52 (Validity-improving partial order)**.** For $x, y \in \mathcal{S}$, write $x \preceq y$ if

$$x_S \leq y_S, \qquad x_N \geq y_N, \qquad x_F \geq y_F, \qquad x_U \geq y_U, \qquad x_J \leq y_J, \qquad x_K \leq y_K.$$

Thus $x \preceq y$ means that y is weakly better than x in every coordinate according to theorem 14.

Proposition 53 (Sharp monotone extrapolation bracket)**.** *Assume theorems 13 and 14. Let $x \in K$ and define*

$$L_\preceq(x) := \sup\{q_E(y) : y \in E,\ y \preceq x\}, \qquad U_\preceq(x) := \inf\{q_E(y) : y \in E,\ x \preceq y\}, \tag{79}$$

with the conventions $\sup \varnothing = 0$ *and* $\inf \varnothing = +\infty$. *Then every admissible coordinatewise-monotone extension of* q_E *to* K *satisfies*

$$L_{\preceq}(x) \le q_V(x) \le U_{\preceq}(x). \tag{80}$$

Moreover, if $L_{\preceq}(x) \le U_{\preceq}(x)$, *then every value* $v \in [L_{\preceq}(x), U_{\preceq}(x)]$ *is attained at* x *by some nonnegative coordinatewise-monotone extension on* K *consistent with* q_E *on* E.

Proof. If $y \preceq x$, coordinatewise monotonicity gives $q_E(y) \le q_V(x)$; taking the supremum yields the lower bound. The upper bound is analogous.

For sharpness, fix $v \in [L_{\preceq}(x), U_{\preceq}(x)]$ and define f_v on $E \cup \{x\}$ by $f_v|_E = q_E$ and $f_v(x) = v$. The bracket condition guarantees that whenever $y \preceq x$ one has $f_v(y) \le f_v(x)$, and whenever $x \preceq y$ one has $f_v(x) \le f_v(y)$. Since q_E itself is monotone on E, the map f_v is monotone on the partially ordered set $E \cup \{x\}$. Now define

$$\widetilde{q}(z) := \sup\{f_v(y) : y \in E \cup \{x\},\ y \preceq z\}. \tag{81}$$

This map is nonnegative and coordinatewise monotone by construction. If $z \in E \cup \{x\}$, monotonicity of f_v implies that every $y \preceq z$ in the support satisfies $f_v(y) \le f_v(z)$, while z itself belongs to the admissible set; hence $\widetilde{q}(z) = f_v(z)$. Therefore $\widetilde{q}$ is a monotone extension consistent with the observed support and attaining the prescribed value v at x. □

Corollary 54 (Minimal additional structure for off-support inference)**.** *Outside calibrated support, point identification is impossible without additional structure. A known Lipschitz modulus or a justified coordinatewise monotone order yields sharp set identification. If both are justified, the pointwise identified set contracts to*

$$I_L(x) \cap [L_{\preceq}(x), U_{\preceq}(x)]. \tag{82}$$

6.4 Factor-specific hazard identification

The benchmark-validity hazard is an aggregate object. If one wants to know whether decay is arriving primarily through loss of discriminative power or through loss of construct validity, the aggregate trace is not enough by itself.

Proposition 55 (Aggregate validity does not identify factor-specific hazards)**.** *Assume theorem 15. If only the aggregate validity path* $\mathcal{V}_B(t)$ *is observed on a time interval and neither* $\mathcal{D}_B(t)$ *nor* $\mathcal{C}_B(t)$ *is separately observed, then the factor-specific hazards* q_D *and* q_C *are generally not point identified even along the observed state path. More precisely, if* $\alpha(t)$ *is any measurable function satisfying*

$$-q_C\big(X_t^{\mathrm{eco}}\big) \le \alpha(t) \le q_D\big(X_t^{\mathrm{eco}}\big)$$

for almost every t, *then the perturbed hazards*

$$\widetilde{q}_D(t) := q_D\big(X_t^{\mathrm{eco}}\big) - \alpha(t), \tag{83}$$

$$\widetilde{q}_C(t) := q_C\big(X_t^{\mathrm{eco}}\big) + \alpha(t) \tag{84}$$

induce the same aggregate log-validity derivative

$$\frac{\mathrm{d}}{\mathrm{d}t} \log \mathcal{V}_B(t) = -(\widetilde{q}_D(t) + \widetilde{q}_C(t))$$

as the original decomposition.

Proof. Under theorem 15,

$$\frac{\mathrm{d}}{\mathrm{d}t}\log \mathcal{V}_B(t) = \frac{\mathrm{d}}{\mathrm{d}t}\log \mathcal{D}_B(t) + \frac{\mathrm{d}}{\mathrm{d}t}\log \mathcal{C}_B(t) = -(q_D + q_C)$$

along the observed path. The perturbation leaves the sum unchanged pointwise, so it leaves the aggregate derivative unchanged. Unless additional information pins down $\alpha(t)$, the separate factor hazards are therefore not point identified. □

Definition 56 (Projected calibrated support for factor hazards)**.** Because the baseline factor law allows spillovers across all ecological coordinates, the natural state objects for factor identification are the full ecological paths

$$X_t^D := X_t^{\mathrm{eco}}, \qquad X_t^C := X_t^{\mathrm{eco}}.$$

Fix compact target sets $K_D, K_C \subseteq \mathcal{S}$. A factor-calibration design is support-dense on (K_D, K_C) if:

1. X_t^D and X_t^C are point identified for almost every $t \in [0,T]$;
2. the factor hazards
$$h_D(t) := -\frac{\mathrm{d}}{\mathrm{d}t}\log \mathcal{D}_B(t), \qquad h_C(t) := -\frac{\mathrm{d}}{\mathrm{d}t}\log \mathcal{C}_B(t)$$
are identified for almost every $t \in [0,T]$; and
3. the identified images
$$\mathcal{I}_T^D := \{X_t^D : \ t \in [0,T] \text{ outside a null set}\}, \qquad \mathcal{I}_T^C := \{X_t^C : \ t \in [0,T] \text{ outside a null set}\}$$
are dense in K_D and K_C, respectively.

Theorem 57 (Factor-specific hazard identification on calibrated projected supports)**.** *Assume theorem 15. Let $K_D, K_C \subseteq \mathcal{S}$ be compact, and suppose q_D is continuous on K_D and q_C is continuous on K_C. If the factor-calibration design is support-dense on (K_D, K_C), then q_D is point identified on K_D and q_C is point identified on K_C.*

Proof. Under theorem 15, the identified projected states and identified factor hazards satisfy

$$h_D(t) = q_D(X_t^D), \qquad h_C(t) = q_C(X_t^C)$$

for almost every $t \in [0,T]$. Thus the values of q_D and q_C are identified on the dense sets $\mathcal{I}_T^D \cap K_D$ and $\mathcal{I}_T^C \cap K_C$. Continuity then yields unique continuous extensions to all of K_D and K_C. □

Corollary 58 (Minimal ingredients for factor-specific identification)**.** *Point identification of factor-specific hazard maps requires more information than point identification of the aggregate hazard. In addition to calibrated ecological-state recovery and support richness, one must separately observe or identify the discriminative-power and construct-validity traces. Without those factor traces, only the composite hazard is generally identified.*

6.5 Factor-specific intervention design

Passive observation is not the only route to factor attribution. Because some state coordinates enter only one factor hazard, matched interventions can sometimes recover factor-level effects even when only aggregate validity is observed.

Definition 59 (Exclusive and shared factor coordinates)**.** Under the baseline factor law, every ecological coordinate may affect both factor hazards. A coordinate block is called *D-exclusive* only under a declared design regime in which q_C is invariant to that block along the intervention family under study. A coordinate block is *C-exclusive* analogously. Coordinates not covered by such a restriction are *shared*. In many applications (S, U) are plausible candidates for approximate D-exclusive perturbations and K for approximate C-exclusive perturbations, but those claims are empirical design assumptions rather than part of the ambient model.

Proposition 60 (Exclusive-coordinate interventions identify factor-specific finite differences from aggregate validity)**.** *Assume theorem 15. Consider two matched benchmark ecologies A and B on a window $[t_0, t_1]$ with the same benchmark and operational specification.*

1. *If a declared design treats the perturbation block as D-exclusive, so that*

$$q_C(X^C_{A,u}) = q_C(X^C_{B,u}) \qquad \text{for almost every } u \in [t_0, t_1],$$

then

$$\log \frac{\mathcal{V}_A(t_1)}{\mathcal{V}_A(t_0)} - \log \frac{\mathcal{V}_B(t_1)}{\mathcal{V}_B(t_0)} = -\int_{t_0}^{t_1} \left[q_D(X^D_{A,u}) - q_D(X^D_{B,u}) \right] \mathrm{d}u, \tag{85}$$

where $X^D_{A,u} = X^{\mathrm{eco}}_{A,u}$ and similarly for B.

2. *If a declared design treats the perturbation block as C-exclusive, so that*

$$q_D(X^D_{A,u}) = q_D(X^D_{B,u}) \qquad \text{for almost every } u \in [t_0, t_1],$$

then

$$\log \frac{\mathcal{V}_A(t_1)}{\mathcal{V}_A(t_0)} - \log \frac{\mathcal{V}_B(t_1)}{\mathcal{V}_B(t_0)} = -\int_{t_0}^{t_1} \left[q_C(X^C_{A,u}) - q_C(X^C_{B,u}) \right] \mathrm{d}u, \tag{86}$$

where $X^C_{A,u} = X^{\mathrm{eco}}_{A,u}$ and similarly for B.

Hence matched interventions on exclusive coordinates identify factor-specific integrated finite differences even from aggregate validity alone.

Proof. Under theorem 15,

$$\frac{\mathrm{d}}{\mathrm{d}t} \log \mathcal{V}_B(t) = -q_D(X^{\mathrm{eco}}_t) - q_C(X^{\mathrm{eco}}_t)$$

on intervals where $\mathcal{V}_B(t) > 0$. In case (1), the q_C terms coincide pathwise and cancel after subtracting the two log-derivative equations and integrating over $[t_0, t_1]$. Case (2) is analogous with the q_D terms canceling instead. □

Proposition 61 (Shared-coordinate interventions do not isolate factor attribution from aggregate validity alone)**.** *Assume theorem 15. Consider matched interventions that vary only one shared coordinate among N, F, and J, while all other coordinates are held fixed. If only the aggregate validity path is observed,*

then the separate factor responses remain generally non-identified: the intervention identifies only the sum of the induced q_D and q_C shifts.

Proof. Under the stated matching, subtracting the two log-validity derivatives yields the integral of

$$[q_D(X_A^D) - q_D(X_B^D)] + [q_C(X_A^C) - q_C(X_B^C)].$$

Only the sum is observed. For any measurable perturbation $\alpha(t)$ small enough to preserve nonnegativity, one may redistribute part of the observed shift from the D factor to the C factor or conversely without changing the aggregate path, exactly as in theorem 55. Thus the separate factor responses are generally not point identified. □

Corollary 62 (Shared-coordinate interventions become factor-identifying once factor traces are available)**.** *Assume theorem 15. If a matched intervention varies only one coordinate while separate factor traces $\mathcal{D}_B(t)$ and $\mathcal{C}_B(t)$ are also observed, then the corresponding factor-specific integrated finite differences are directly identified from the separate log-derivative equations. In particular, shrinking-window matched interventions recover local directional derivatives whenever the relevant factor hazard is differentiable.*

Proof. With separate factor traces, one may apply the same subtraction-and-integration argument to the log-derivative equations for $\mathcal{D}_B$ and $\mathcal{C}_B$ individually. When only one coordinate varies and the hazard map is differentiable, dividing the identified finite difference by the coordinate step and letting the step size vanish yields the directional derivative. □

6.6 Shared-coordinate decomposition under exclusion and IV-style designs

Shared coordinates remain harder than exclusive ones because changing N, F, or J generally moves both factor hazards at once. Still, design can sometimes restore identification. The key is to distinguish passive shared-coordinate perturbations, which identify only sums, from *designed* perturbations that satisfy exclusion restrictions or generate linearly independent local transport across the two factors.

Definition 63 (Factor-excluding shared-coordinate design)**.** Fix a shared coordinate $Z \in \{N, F, J\}$ and a window $[t_0, t_1]$. A matched intervention family indexed by an arm $a \in \mathcal{A}$ is called *C-excluding for Z* if all non-Z coordinates coincide pathwise across arms, the induced Z paths may differ across arms, and

$$q_C(X_u^{C,a}) = q_C(X_u^{C,a'}) \qquad \text{for almost every } u \in [t_0, t_1] \text{ and all } a, a' \in \mathcal{A}. \tag{87}$$

Define *D-excluding* analogously with the roles of q_D and q_C reversed.

Proposition 64 (Exclusion on a shared coordinate identifies the nonexcluded factor from aggregate validity)**.** *Assume theorem 15. Under a C-excluding design for shared coordinate Z, any two arms $a, a' \in \mathcal{A}$ satisfy*

$$\log \frac{\mathcal{V}_a(t_1)}{\mathcal{V}_a(t_0)} - \log \frac{\mathcal{V}_{a'}(t_1)}{\mathcal{V}_{a'}(t_0)} = -\int_{t_0}^{t_1} \left[q_D(X_u^{D,a}) - q_D(X_u^{D,a'}) \right] \mathrm{d}u. \tag{88}$$

Analogously, a D-excluding design identifies the integrated C-factor response from aggregate validity.

Proof. Subtract the two log-validity derivative equations from theorem 15. Under a C-excluding design, the q_C terms coincide almost everywhere and cancel. Integrating over $[t_0, t_1]$ yields the displayed identity. The D-excluding case is symmetric. □

Corollary 65 (Local Wald identification under excluded shared-coordinate perturbations)**.** *Assume the hypotheses of theorem 64. Let $a = 0$ denote a baseline arm and suppose there is a one-parameter C-excluding perturbation family $\delta \mapsto a(\delta)$ such that only the shared coordinate Z changes and*

$$Z_u^{a(\delta)} = Z_u^0 + \delta\phi(u) + o(\delta) \tag{89}$$

in $L^1([t_0, t_1])$ for some integrable profile ϕ. If q_D is continuously differentiable in its Z coordinate on a neighborhood of the baseline path, then

$$-\frac{1}{\delta}\left[\log\frac{\mathcal{V}_{a(\delta)}(t_1)}{\mathcal{V}_{a(\delta)}(t_0)} - \log\frac{\mathcal{V}_0(t_1)}{\mathcal{V}_0(t_0)}\right] \to \int_{t_0}^{t_1} \partial_Z q_D\big(X_u^{D,0}\big)\,\phi(u)\,\mathrm{d}u \tag{90}$$

as $\delta \to 0$. Thus aggregate validity identifies a weighted average local D-factor sensitivity along the excluded design.

Proof. By theorem 64, the scaled log-validity contrast equals

$$\int_{t_0}^{t_1} \frac{q_D\big(X_u^{D,a(\delta)}\big) - q_D\big(X_u^{D,0}\big)}{\delta}\,\mathrm{d}u.$$

The stated differentiability and L^1 perturbation imply pointwise convergence of the difference quotient to $\partial_Z q_D(X_u^{D,0})\phi(u)$ along the path. On the compact image of the baseline path and sufficiently small perturbations, the derivative is bounded, so dominated convergence applies. □

Theorem 66 (Full-rank IV-style local decomposition for a shared coordinate)**.** *Assume theorem 15. Fix a shared coordinate $Z \in \{N, F, J\}$, a baseline path on $[t_0, t_1]$, and an integrable perturbation profile ϕ. Suppose two one-parameter matched intervention families $k = 1, 2$ generate local perturbations such that, as $\delta \to 0$,*

$$Z_u^{D,(k,\delta)} = Z_u + \delta a_k \phi(u) + o(\delta), \tag{91}$$

$$Z_u^{C,(k,\delta)} = Z_u + \delta b_k \phi(u) + o(\delta) \tag{92}$$

in $L^1([t_0, t_1])$, while all non-Z coordinates are held fixed up to $o(\delta)$. Assume further that q_D and q_C are continuously differentiable in Z along the baseline path. Define

$$\beta_D := \int_{t_0}^{t_1} \partial_Z q_D\big(X_u^D\big)\phi(u)\,\mathrm{d}u, \tag{93}$$

$$\beta_C := \int_{t_0}^{t_1} \partial_Z q_C\big(X_u^C\big)\phi(u)\,\mathrm{d}u, \tag{94}$$

and let $\Delta_k(\delta)$ denote the aggregate log-validity contrast between the perturbed and baseline arm in design k. Then

$$\frac{1}{\delta}\begin{pmatrix}\Delta_1(\delta)\\ \Delta_2(\delta)\end{pmatrix} \to -\begin{pmatrix}a_1 & b_1\\ a_2 & b_2\end{pmatrix}\begin{pmatrix}\beta_D\\ \beta_C\end{pmatrix}. \tag{95}$$

If the design matrix $A := \left(\begin{smallmatrix}a_1 & b_1\\ a_2 & b_2\end{smallmatrix}\right)$ has full rank, then (β_D, β_C) is locally identified from aggregate validity

by

$$\begin{pmatrix}\beta_D\\ \beta_C\end{pmatrix} = -A^{-1}\lim_{\delta\to 0}\frac{1}{\delta}\begin{pmatrix}\Delta_1(\delta)\\ \Delta_2(\delta)\end{pmatrix}. \tag{96}$$

The exclusion designs of theorem 64 are the special cases $(a_k, b_k) = (1, 0)$ *or* $(0, 1)$.

Proof. Apply a first-order expansion to the q_D and q_C terms in the aggregate log-validity derivative under each design k. The stated perturbation structure yields

$$\Delta_k(\delta) = -\delta(a_k\beta_D + b_k\beta_C) + o(\delta).$$

Collecting the two equations gives the matrix limit. If A is invertible, solve the resulting 2×2 linear system. □

Remark 67. The theorem is intentionally local. It does not pretend that arbitrary shared-coordinate interventions magically reveal factor decomposition. What it says is narrower and more useful: if the experimental protocol supplies either exclusion or two linearly independent local transport patterns across the factors, then aggregate validity can support factor attribution even without full factor traces. Without that design information, theorem 61 still rules.

6.7 Shared-coordinate sensitivity under imperfect exclusion and invalid IVs

Exact exclusion is a useful limiting case, not a default fact of life. A shared-coordinate intervention that is *intended* to move only one factor may still weakly perturb the other through hidden pathways, implementation spillovers, or judge adaptation. The right object is therefore often a sensitivity region indexed by declared violation budgets, in the same spirit as plausibly exogenous IV analysis [47].

Definition 68 (Exclusion-violation budget)**.** Fix a shared coordinate $Z \in \{N, F, J\}$, a window $[t_0, t_1]$, and two matched arms a, a'. Define the integrated factor contrasts

$$I_D(a, a') := \int_{t_0}^{t_1}\left[q_D(X_u^{D,a}) - q_D(X_u^{D,a'})\right]\mathrm{d}u, \tag{97}$$

$$I_C(a, a') := \int_{t_0}^{t_1}\left[q_C(X_u^{C,a}) - q_C(X_u^{C,a'})\right]\mathrm{d}u. \tag{98}$$

A declared C-exclusion budget $\bar{\Xi}_C \geq 0$ is *admissible* for the design if $|I_C(a, a')| \leq \bar{\Xi}_C$ for every matched arm pair used in the design. Define admissible D-exclusion budgets analogously.

Proposition 69 (Partial identification under imperfect exclusion)**.** *Assume theorem 15. For any two matched arms* a, a' *define the observed aggregate log-validity contrast*

$$\Delta_{aa'} := \log\frac{\mathcal{V}_a(t_1)}{\mathcal{V}_a(t_0)} - \log\frac{\mathcal{V}_{a'}(t_1)}{\mathcal{V}_{a'}(t_0)}. \tag{99}$$

Then

$$\Delta_{aa'} = -I_D(a, a') - I_C(a, a'). \tag{100}$$

If the design carries an admissible C*-exclusion budget* $\bar{\Xi}_C$*, then the integrated* D*-factor response is set identified by*

$$I_D(a, a') \in \left[-\Delta_{aa'} - \bar{\Xi}_C,\ -\Delta_{aa'} + \bar{\Xi}_C\right]. \tag{101}$$

Under an admissible D-exclusion budget $\bar{\Xi}_D$, the integrated C-factor response obeys the symmetric bracket

$$I_C(a,a') \in \Big[-\Delta_{aa'} - \bar{\Xi}_D,\ -\Delta_{aa'} + \bar{\Xi}_D\Big]. \tag{102}$$

Proof. Subtract the two log-validity identities from theorem 15 and integrate over $[t_0, t_1]$ to obtain $\Delta_{aa'} = -I_D(a,a') - I_C(a,a')$. If $|I_C(a,a')| \le \bar{\Xi}_C$, then $I_D(a,a') = -\Delta_{aa'} - I_C(a,a')$ must lie in the stated interval. The D-budget case is identical with the roles reversed. □

Corollary 70 (Exact exclusion is the zero-budget special case)**.** *If the admissible exclusion budget in theorem 69 is zero, then the interval collapses to the exact-identification statements of theorem 64.*

Proof. Set $\bar{\Xi}_C = 0$ or $\bar{\Xi}_D = 0$ in theorem 69. □

Definition 71 (Violation class for imperfect exclusion)**.** For a matched-arm design (a,a'), let $\mathcal{J}_C(a,a') \subseteq \mathbb{R}$ denote the admissible set of integrated spillover responses $I_C(a,a')$ from theorem 68. The associated D-factor identified set induced by an observed aggregate contrast $\Delta_{aa'}$ is

$$\mathcal{I}_D(\Delta_{aa'}; \mathcal{J}_C(a,a')) := \{-\Delta_{aa'} - \xi :\ \xi \in \mathcal{J}_C(a,a')\}. \tag{103}$$

Define $\mathcal{I}_C(\Delta_{aa'}; \mathcal{J}_D(a,a'))$ symmetrically.

Theorem 72 (Sharp identified set under imperfect exclusion)**.** *Assume theorem 15. Fix matched arms (a,a') and an observed aggregate contrast $\Delta_{aa'}$. Suppose the design knowledge restricts the nuisance spillover only through a nonempty closed set $\mathcal{J}_C(a,a') \subseteq \mathbb{R}$ and imposes no further restriction on $I_D(a,a')$ beyond the accounting identity*

$$\Delta_{aa'} = -I_D(a,a') - I_C(a,a'). \tag{104}$$

If every $\xi \in \mathcal{J}_C(a,a')$ is attainable by some admissible latent path consistent with the matched-arm design and the observed aggregate contrast, then the sharp identified set for the integrated D-factor response is exactly

$$I_D(a,a') \in \mathcal{I}_D(\Delta_{aa'}; \mathcal{J}_C(a,a')). \tag{105}$$

In particular, if $\mathcal{J}_C(a,a') = [\underline{\Xi}_C, \overline{\Xi}_C]$, then

$$I_D(a,a') \in [-\Delta_{aa'} - \overline{\Xi}_C,\ -\Delta_{aa'} - \underline{\Xi}_C], \tag{106}$$

and this interval is sharp. Under the symmetric budget class $\mathcal{J}_C(a,a') = [-\bar{\Xi}_C, \bar{\Xi}_C]$, the sharp identified set reduces to the bracket in theorem 69. The same statement holds for $I_C(a,a')$ with the roles of D and C reversed.

Proof. Necessity is immediate from the accounting identity: any admissible pair (I_D, I_C) with $I_C \in \mathcal{J}_C(a,a')$ must satisfy $I_D = -\Delta_{aa'} - I_C$, hence $I_D \in \mathcal{I}_D(\Delta_{aa'}; \mathcal{J}_C(a,a'))$. For sharpness, fix any $x \in \mathcal{I}_D(\Delta_{aa'}; \mathcal{J}_C(a,a'))$. By definition there exists $\xi \in \mathcal{J}_C(a,a')$ such that $x = -\Delta_{aa'} - \xi$. By attainability, some admissible latent path consistent with the design and observed aggregate contrast realizes $I_C(a,a') = \xi$; the accounting identity then forces $I_D(a,a') = x$. Thus every point in $\mathcal{I}_D(\Delta_{aa'}; \mathcal{J}_C(a,a'))$ is feasible, so the set is sharp. □

Remark 73. The point is not decorative set notation. Once exclusion is imperfect, the scientifically honest object is the image of an explicitly declared violation class. Symmetric budgets are only one special

case. Asymmetric or sign-restricted spillover classes can materially shrink the region, and the theory should expose that rather than bury it inside one generic interval.

Theorem 74 (Local shared-coordinate decomposition under bounded direct effects)**.** *Assume the perturbation structure of theorem 66. Let*

$$g := \lim_{\delta \to 0} \frac{1}{\delta} \begin{pmatrix} \Delta_1(\delta) \\ \Delta_2(\delta) \end{pmatrix} \tag{107}$$

exist, and suppose there is a known closed set $\mathcal{R} \subseteq \mathbb{R}^2$ such that every admissible direct-effect or exclusion-violation perturbation contributes a remainder vector $r \in \mathcal{R}$ satisfying

$$g = -A \begin{pmatrix} \beta_D \\ \beta_C \end{pmatrix} + r, \qquad A := \begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix}. \tag{108}$$

If A has full rank, then the identified set for the local factor sensitivities is

$$\mathcal{I}_\beta(g; A, \mathcal{R}) := \left\{ -A^{-1}(g - r) : \ r \in \mathcal{R} \right\}. \tag{109}$$

In particular, if $\mathcal{R} = [-\bar{r}_1, \bar{r}_1] \times [-\bar{r}_2, \bar{r}_2]$, then $\mathcal{I}_\beta(g; A, \mathcal{R})$ is a centrally symmetric parallelogram. Exact point identification is recovered only when $\mathcal{R} = \{0\}$.

Proof. The stated model is a linear system with unknown coefficient vector $(\beta_D, \beta_C)^\top$ and nuisance perturbation $r \in \mathcal{R}$. Since A is invertible, every admissible r implies

$$\begin{pmatrix} \beta_D \\ \beta_C \end{pmatrix} = -A^{-1}(g - r).$$

Taking the image of the admissible remainder set under the affine map $r \mapsto -A^{-1}(g - r)$ gives the displayed identified set. If $\mathcal{R} = \{0\}$ this image is a singleton, which is the exact-identification result of theorem 66. □

Remark 75. The lesson is slightly brutal but useful: once exclusion is imperfect, "IV-style" shared-coordinate attribution is not a theorem factory but a sensitivity-analysis problem. The point estimate survives only if the violation set collapses to a point. Otherwise the honest object is a region whose width is determined by explicit design budgets.

Corollary 76 (Sharp local sensitivity region under a rich violation class)**.** *Under the setup of theorem 74, suppose every $r \in \mathcal{R}$ is attainable by some admissible local perturbation design consistent with the observed limit vector g. Then the affine image $\mathcal{I}_\beta(g; A, \mathcal{R})$ is the sharp identified set for (β_D, β_C). In particular, a box remainder class yields a sharp parallelogram rather than merely an outer bound.*

Proof. Necessity was already shown in theorem 74. If every $r \in \mathcal{R}$ is attainable, then for each candidate coefficient vector of the form $-A^{-1}(g - r)$ there exists an admissible perturbation delivering exactly that vector. Hence every point in $\mathcal{I}_\beta(g; A, \mathcal{R})$ is feasible, so the set is sharp. □

6.8 Structural sharpness under imperfect exclusion

Violation classes need not be loose boxes. Many designs imply additional structure across matched contrasts: one-sided spillovers, monotone violation growth across exposure intensities, sparse contamination channels, or linear coupling induced by shared implementation pathways. Those restrictions should narrow the identified set transparently rather than being left as verbal folklore.

Definition 77 (Structured violation class across matched contrasts)**.** Fix $q \geq 1$ matched contrasts on the same shared coordinate, with observed aggregate log-validity contrast vector

$$\Delta := (\Delta_1, \ldots, \Delta_q)^\top \in \mathbb{R}^q. \tag{110}$$

Let

$$\theta_D := (I_D^{(1)}, \ldots, I_D^{(q)})^\top, \qquad \xi_C := (I_C^{(1)}, \ldots, I_C^{(q)})^\top \tag{111}$$

be the integrated target-factor and spillover-factor responses across these contrasts. A *structured C-violation class* is any nonempty closed set

$$\mathcal{J}_C^{\mathrm{str}} \subseteq \mathbb{R}^q \tag{112}$$

that encodes admissible spillover vectors through sign restrictions, monotonicity, sparsity, total-budget constraints, or more general linear relations such as $A\xi \leq b$. The induced identified set for the target-factor response vector is

$$\mathcal{I}_D^{\mathrm{str}}(\Delta; \mathcal{J}_C^{\mathrm{str}}) := \{-\Delta - \xi : \ \xi \in \mathcal{J}_C^{\mathrm{str}}\}. \tag{113}$$

Define $\mathcal{I}_C^{\mathrm{str}}$ symmetrically when the roles of D and C are reversed.

Theorem 78 (Sharp identified set under structural violation classes)**.** *Assume theorem 15. For the q matched contrasts from theorem 77, suppose the only restrictions on the spillover vector are that $\xi_C \in \mathcal{J}_C^{\mathrm{str}}$ and that the accounting identity*

$$\Delta = -\theta_D - \xi_C \tag{114}$$

holds componentwise. Then every admissible target-response vector must satisfy

$$\theta_D \in \mathcal{I}_D^{\mathrm{str}}(\Delta; \mathcal{J}_C^{\mathrm{str}}). \tag{115}$$

If, in addition, every $\xi \in \mathcal{J}_C^{\mathrm{str}}$ is attainable by some admissible latent path consistent with the matched designs and the observed contrast vector Δ, then $\mathcal{I}_D^{\mathrm{str}}(\Delta; \mathcal{J}_C^{\mathrm{str}})$ is the sharp identified set for θ_D. If $\mathcal{J}_C^{\mathrm{str}}$ is convex (respectively polyhedral), then so is the sharp identified set.

Proof. Necessity follows immediately from the accounting identity: every admissible pair (θ_D, ξ_C) satisfies $\theta_D = -\Delta - \xi_C$, hence lies in the displayed affine image. If every $\xi \in \mathcal{J}_C^{\mathrm{str}}$ is attainable, then each point $-\Delta - \xi$ in that image is realized by some admissible latent path, so the image is sharp. Convexity and polyhedrality are preserved by affine maps. □

Corollary 79 (Structural restrictions weakly shrink sharp regions)**.** *Let $\mathcal{J}_{C,1}^{\mathrm{str}} \subseteq \mathcal{J}_{C,2}^{\mathrm{str}}$ be two admissible structural violation classes for the same observed contrast vector Δ. Then*

$$\mathcal{I}_D^{\mathrm{str}}(\Delta; \mathcal{J}_{C,1}^{\mathrm{str}}) \subseteq \mathcal{I}_D^{\mathrm{str}}(\Delta; \mathcal{J}_{C,2}^{\mathrm{str}}). \tag{116}$$

If the inclusion of violation classes is strict and both classes are attainable, then the inclusion of sharp identified sets is also strict.

Proof. Affine images preserve set inclusion. Strictness follows because a strictly smaller attainable violation class omits at least one feasible nuisance vector whose image would otherwise appear in the larger identified set. □

Remark 80. This result is mathematically simple and operationally important. Once the violation class carries real design structure—for example, one-sided spillovers, nondecreasing direct effects across exposure levels, or a sparse shared-pathway model—the sharp region contracts by exactly the affine image of that structure. The honest object is therefore not "a sensitivity interval" in the abstract, but the sharp image of the declared structural violation class.

6.9 Construct-validity identification with noisy anchors

Construct validity is harder to estimate than discriminative power because the target-claim functional G_t is typically latent or only partially externalized. In the present paper the construct-validity component is $R_B(t)\mathcal{O}_B(t)$, so the identification problem has two parts: resolved-pair mass and order fidelity on those resolved pairs.

Let

$$T_t(m, m') := \operatorname{sgn}\big(G_t(m) - G_t(m')\big)$$

denote the latent target ordering on resolved pairs, and let $A_t(m, m') \in \{-1, +1\}$ be any claim-aligned anchor comparator defined on the same resolved-pair event $\mathcal{R}_t$. Define the anchor order error

$$e_A(t) := \mathbb{P}\big(A_t(m, m') \neq T_t(m, m') \mid (m, m') \in \mathcal{R}_t\big) \tag{117}$$

and the observed benchmark-anchor agreement

$$\Gamma_A(t) := \mathbb{P}\big(\operatorname{sgn}_{\delta_t}\big(Y_B(m, t) - Y_B(m', t)\big) = A_t(m, m') \mid (m, m') \in \mathcal{R}_t\big). \tag{118}$$

Definition 81 (Anchor-validation design)**.** An *anchor-validation design* is an auxiliary study, model, or calibration argument that targets the anchor error on the resolved-pair event by one of the following observable routes:

1. a gold-standard adjudication design, in which a sampled subset of resolved pairs is labeled by a higher-fidelity adjudication protocol;
2. a multi-anchor latent-label design, in which repeated imperfect anchors are fit under a latent-label model and the resulting identified set is reported;
3. a one-sided calibration design, in which domain knowledge certifies that some disagreement modes are impossible or bounded.

The design is *admissible* for theorem 83 on a declared analysis event if the resulting observable outputs imply an upper bound $\bar{e}_A(t)$ satisfying $e_A(t) \leq \bar{e}_A(t)$ on that event.

Theorem 82 (Observable sufficient conditions for admissible anchor-error budgets)**.** *Fix time t and the resolved-pair event $\mathcal{R}_t$. Each of the following is sufficient to produce an admissible upper bound on $e_A(t)$.*

1. ***Gold-standard adjudication subsample.*** *Suppose a sampled subset of resolved pairs is labeled by a reference adjudication T_t^{ref} such that $T_t^{\mathrm{ref}} = T_t$ on a declared event $\mathcal{E}_{\mathrm{ref}}$, and suppose the sampling or weighting scheme is declared so that the weighted reference disagreement targets $\mathbb{P}(A_t \neq T_t \mid \mathcal{R}_t)$ on that event. Then any valid upper confidence bound for the weighted disagreement probability is an admissible $\bar{e}_A(t)$ on $\mathcal{E}_{\mathrm{ref}}$.*

2. ***Multi-anchor latent-label design.*** *Suppose a declared latent-label model together with declared identifying restrictions yields an identified set $\mathcal{E}_A(t) \subseteq [0,1]$ satisfying $e_A(t) \in \mathcal{E}_A(t)$ on a declared event $\mathcal{E}_{\mathrm{lat}}$. Then $\bar{e}_A(t) := \sup \mathcal{E}_A(t)$ is admissible on $\mathcal{E}_{\mathrm{lat}}$.*

3. ***One-sided calibration design.*** *Suppose there is an observable partition $\mathcal{R}_t = Q_t \cup Q_t^c$ and declared bounds such that anchor error is impossible on Q_t and*

$$\mathbb{P}(Q_t^c \mid \mathcal{R}_t) \leq \bar{u}_t$$

on a declared event $\mathcal{E}_{\mathrm{cal}}$. Then $\bar{e}_A(t) := \bar{u}_t$ is admissible on $\mathcal{E}_{\mathrm{cal}}$. More generally, if on Q_t^c one has an admissible conditional error bound $\bar{e}_{A,Q^c}(t)$ and on Q_t an admissible conditional error bound $\bar{e}_{A,Q}(t)$, then

$$\bar{e}_A(t) := \mathbb{P}(Q_t \mid \mathcal{R}_t)\bar{e}_{A,Q}(t) + \mathbb{P}(Q_t^c \mid \mathcal{R}_t)\bar{e}_{A,Q^c}(t)$$

is admissible.

In each case the theory needs only the reported bound or identified set, not direct observation of the latent target order on the full resolved set.

Proof. For the adjudication design, on $\mathcal{E}_{\mathrm{ref}}$ the reference labels equal the latent target order, so the reported weighted disagreement probability is exactly $e_A(t)$; any valid upper confidence bound is therefore admissible. For the latent-label design, admissibility is immediate from the inclusion $e_A(t) \in \mathcal{E}_A(t)$ on $\mathcal{E}_{\mathrm{lat}}$. For the one-sided calibration design, decompose

$$e_A(t) = \mathbb{P}(Q_t \mid \mathcal{R}_t)\mathbb{P}(A_t \neq T_t \mid Q_t, \mathcal{R}_t) + \mathbb{P}(Q_t^c \mid \mathcal{R}_t)\mathbb{P}(A_t \neq T_t \mid Q_t^c, \mathcal{R}_t),$$

and then substitute the declared conditional or mass bounds. The final sentence is just the observation that downstream identification uses only the resulting error budget. □

Theorem 83 (Anchor-based partial identification of order fidelity and construct validity)**.** *For any anchor comparator A_t defined on resolved pairs,*

$$\left|\Gamma_A(t) - \mathcal{O}_B(t)\right| \leq e_A(t). \tag{119}$$

Hence if an admissible anchor-validation design yields an upper bound $e_A(t) \leq \bar{e}_A(t)$, then

$$\mathcal{O}_B(t) \in \Big[\max\{0, \Gamma_A(t) - \bar{e}_A(t)\},\ \min\{1, \Gamma_A(t) + \bar{e}_A(t)\}\Big]. \tag{120}$$

Consequently, if $R_B(t)$ is point identified, then

$$\mathcal{C}_B(t) \in R_B(t) \cdot \Big[\max\{0, \Gamma_A(t) - \bar{e}_A(t)\},\ \min\{1, \Gamma_A(t) + \bar{e}_A(t)\}\Big]. \tag{121}$$

More generally, if only $R_B(t) \in [\underline{R}_B(t), \overline{R}_B(t)]$ is available, then

$$\mathcal{C}_B(t) \in \Big[\underline{R}_B(t)\max\{0, \Gamma_A(t) - \bar{e}_A(t)\},\ \overline{R}_B(t)\min\{1, \Gamma_A(t) + \bar{e}_A(t)\}\Big]. \tag{122}$$

Proof. On the resolved-pair event, the indicators $\mathbf{1}\{\mathrm{sgn}_{\delta_t}(Y_B - Y'_B) = A_t\}$ and $\mathbf{1}\{\mathrm{sgn}_{\delta_t}(Y_B - Y'_B) = T_t\}$ can differ only when $A_t \neq T_t$. Therefore

$$|\Gamma_A(t) - \mathcal{O}_B(t)| \leq \mathbb{P}(A_t \neq T_t \mid \mathcal{R}_t) = e_A(t). \tag{123}$$

Multiplying the resulting interval for $\mathcal{O}_B(t)$ by any admissible interval for $R_B(t)$ yields the construct-validity interval because $\mathcal{C}_B(t) = R_B(t)\mathcal{O}_B(t)$ and all factors lie in $[0, 1]$. □

Definition 84 (Nondifferential directional anchor error on score-resolved resolved pairs)**.** Let

$$\mathcal{G}_t^\delta := \{(m, m') : |Y_B(m, t) - Y_B(m', t)| > \delta_t\} \tag{124}$$

be the benchmark score-resolved event. On $\mathcal{R}_t \cap \mathcal{G}_t^\delta$ define

$$B_t(m, m') := \mathrm{sgn}_{\delta_t}(Y_B(m, t) - Y_B(m', t)) \in \{-1, +1\}.$$

A directional anchor-error model is specified by the class-conditional error rates

$$e_A^+(t) := \mathbb{P}(A_t(m, m') = -1 \mid T_t(m, m') = +1,\ \mathcal{R}_t \cap \mathcal{G}_t^\delta), \tag{125}$$
$$e_A^-(t) := \mathbb{P}(A_t(m, m') = +1 \mid T_t(m, m') = -1,\ \mathcal{R}_t \cap \mathcal{G}_t^\delta). \tag{126}$$

It is called *nondifferential* if, on $\mathcal{R}_t \cap \mathcal{G}_t^\delta$,

$$\mathbb{P}(A_t = a \mid T_t, B_t, \mathcal{R}_t \cap \mathcal{G}_t^\delta) = \mathbb{P}(A_t = a \mid T_t, \mathcal{R}_t \cap \mathcal{G}_t^\delta) \qquad \text{for each } a \in \{-1, +1\}. \tag{127}$$

Define the score-resolved positive-order prevalence and score-resolved mass

$$p_{B,+}^\delta(t) := \mathbb{P}(B_t(m, m') = +1 \mid \mathcal{R}_t \cap \mathcal{G}_t^\delta), \qquad \rho_{B,\delta}(t) := \mathbb{P}(\mathcal{G}_t^\delta \mid \mathcal{R}_t), \tag{128}$$

and the score-resolved anchor agreement and score-resolved order fidelity

$$\Gamma_A^\delta(t) := \mathbb{P}(B_t(m, m') = A_t(m, m') \mid \mathcal{R}_t \cap \mathcal{G}_t^\delta), \qquad \mathcal{O}_B^\delta(t) := \mathbb{P}(B_t(m, m') = T_t(m, m') \mid \mathcal{R}_t \cap \mathcal{G}_t^\delta). \tag{129}$$

Whenever $\mathbb{P}(\mathcal{R}_t \cap \mathcal{G}_t^\delta) = 0$, the fail-closed output is the status label `insufficient-score-resolved` together with the trivial interval $\mathcal{O}_B(t) \in [0, 1]$.

Theorem 85 (Exact affine recovery on score-resolved resolved pairs under nondifferential directional anchor error)**.** *Assume the setup of theorem 84, suppose $\mathbb{P}(\mathcal{R}_t \cap \mathcal{G}_t^\delta) > 0$, and assume the anchor error is nondifferential with*

$$e_A^+(t) + e_A^-(t) < 1. \tag{130}$$

Then the score-resolved anchor agreement and the score-resolved order fidelity obey the exact affine relation

$$\Gamma_A^\delta(t) = (1 - e_A^+(t) - e_A^-(t))\mathcal{O}_B^\delta(t) + e_A^+(t)(1 - p_{B,+}^\delta(t)) + e_A^-(t)p_{B,+}^\delta(t), \tag{131}$$

and therefore

$$\mathcal{O}_B^\delta(t) = \frac{\Gamma_A^\delta(t) - e_A^+(t)(1 - p_{B,+}^\delta(t)) - e_A^-(t)p_{B,+}^\delta(t)}{1 - e_A^+(t) - e_A^-(t)}. \tag{132}$$

Moreover,

$$\mathcal{O}_B(t) = \rho_{B,\delta}(t)\,\mathcal{O}_B^\delta(t) \tag{133}$$

because on target-resolved pairs outside $\mathcal{G}_t^\delta$ the thresholded benchmark comparator equals 0 and cannot agree with $T_t \in \{-1, +1\}$. Hence if $R_B(t)$, $\rho_{B,\delta}(t)$, $\Gamma_A^\delta(t)$, and $p_{B,+}^\delta(t)$ are point identified or are supplied by a separately validated calibration study, then

$$\mathcal{C}_B(t) = R_B(t)\rho_{B,\delta}(t)\mathcal{O}_B^\delta(t) \tag{134}$$

is point identified as well. More generally, let Θ_t^δ be any admissible set for the tuple $(e_A^+(t), e_A^-(t), p_{B,+}^\delta(t), \rho_{B,\delta}(t), \Gamma_A^\delta(t))$. Pushing Θ_t^δ through (132) and (133) yields a valid identified set for $\mathcal{O}_B(t)$ and hence for $\mathcal{C}_B(t)$ after multiplication by an admissible interval for $R_B(t)$.

Proof. Write $x := \mathbb{P}(B_t = +1, T_t = +1 \mid \mathcal{R}_t \cap \mathcal{G}_t^\delta)$, $y := \mathbb{P}(B_t = -1, T_t = -1 \mid \mathcal{R}_t \cap \mathcal{G}_t^\delta)$, $u := \mathbb{P}(B_t = +1, T_t = -1 \mid \mathcal{R}_t \cap \mathcal{G}_t^\delta)$, and $v := \mathbb{P}(B_t = -1, T_t = +1 \mid \mathcal{R}_t \cap \mathcal{G}_t^\delta)$. Then

$$\mathcal{O}_B^\delta(t) = x + y, \qquad p_{B,+}^\delta(t) = x + u.$$

Under nondifferential anchor error on $\mathcal{R}_t \cap \mathcal{G}_t^\delta$,

$$\Gamma_A^\delta(t) = x(1 - e_A^+) + ue_A^- + y(1 - e_A^-) + ve_A^+.$$

Using $x + y = \mathcal{O}_B^\delta(t)$, $u + v = 1 - \mathcal{O}_B^\delta(t)$, and $x + u = p_{B,+}^\delta(t)$ gives (131). Since $e_A^+ + e_A^- < 1$, solving for $\mathcal{O}_B^\delta(t)$ yields (132). Finally,

$$\mathcal{O}_B(t) = \mathbb{P}(B_t = T_t \mid \mathcal{R}_t) = \mathbb{P}(\mathcal{G}_t^\delta \mid \mathcal{R}_t)\mathbb{P}(B_t = T_t \mid \mathcal{R}_t \cap \mathcal{G}_t^\delta)$$

because $B_t = 0$ on $\mathcal{R}_t \setminus \mathcal{G}_t^\delta$ while $T_t \in \{-1, +1\}$ on $\mathcal{R}_t$. This is (133). The set-valued statement follows by image propagation through the displayed formulas. □

Corollary 86 (Interval propagation for score-resolved directional anchor error)**.** *Assume the hypotheses of theorem 85. Let $I_\Gamma^\delta(t)$ be any valid interval for $\Gamma_A^\delta(t)$ and let $I_p^\delta(t)$ and $I_\rho^\delta(t)$ be valid intervals for $p_{B,+}^\delta(t)$ and $\rho_{B,\delta}(t)$. Then mapping the Cartesian product*

$$I_\Gamma^\delta(t) \times I_p^\delta(t) \times I_\rho^\delta(t) \times \mathcal{E}_t$$

through (132) and (133) yields a valid interval for $\mathcal{O}_B(t)$. If $R_B(t) \in [\underline{R}_B(t), \overline{R}_B(t)]$, multiplying that interval by $[\underline{R}_B(t), \overline{R}_B(t)]$ yields a valid interval for $\mathcal{C}_B(t)$.

Proof. The map from $(\Gamma_A^\delta, p_{B,+}^\delta, \rho_{B,\delta}, e_A^+, e_A^-)$ to $\mathcal{O}_B$ is explicit in (132) and (133). Validity of the pushed-forward interval follows by monotone image propagation over the admissible input sets. Multiplication by an admissible interval for $R_B(t)$ then yields a valid interval for $\mathcal{C}_B(t)$. □

Definition 87 (Benchmark-dependent anchor-error violation budget)**.** On $\mathcal{R}_t \cap \mathcal{G}_t^\delta$, define the nondifferential-violation budget

$$\Delta_{\mathrm{nd}}(t) := \sup_{a,b,\tau} \left| \mathbb{P}(A_t = a \mid T_t = \tau, B_t = b, \mathcal{R}_t \cap \mathcal{G}_t^\delta) - \mathbb{P}(A_t = a \mid T_t = \tau, \mathcal{R}_t \cap \mathcal{G}_t^\delta) \right|, \tag{135}$$

where the supremum ranges over $a, \tau, b \in \{-1, +1\}$. Thus $\Delta_{\mathrm{nd}}(t) = 0$ is the exact nondifferential case.

Theorem 88 (Sensitivity to benchmark-dependent anchor error)**.** *Assume the setup of theorem 84 and let $\Delta_{\mathrm{nd}}(t) \le \bar{\Delta}_{\mathrm{nd}}(t) < 1$ be an admissible violation budget. Then there exists a remainder term $\varepsilon_{\mathrm{nd}}(t)$ with*

$$|\varepsilon_{\mathrm{nd}}(t)| \le \bar{\Delta}_{\mathrm{nd}}(t) \tag{136}$$

such that the score-resolved agreement obeys

$$\Gamma_A^\delta(t) = (1 - e_A^+(t) - e_A^-(t))\mathcal{O}_B^\delta(t) + e_A^+(t)(1 - p_{B,+}^\delta(t)) + e_A^-(t)p_{B,+}^\delta(t) + \varepsilon_{\mathrm{nd}}(t). \tag{137}$$

Consequently, if $e_A^+(t) + e_A^-(t) < 1$, then

$$\mathcal{O}_B^\delta(t) \in \left[\frac{\Gamma_A^\delta(t) - \bar{\Delta}_{\mathrm{nd}}(t) - e_A^+(t)(1 - p_{B,+}^\delta(t)) - e_A^-(t)p_{B,+}^\delta(t)}{1 - e_A^+(t) - e_A^-(t)}, \frac{\Gamma_A^\delta(t) + \bar{\Delta}_{\mathrm{nd}}(t) - e_A^+(t)(1 - p_{B,+}^\delta(t)) - e_A^-(t)p_{B,+}^\delta(t)}{1 - e_A^+(t) - e_A^-(t)} \right], \tag{138}$$

and the interval propagates to $\mathcal{O}_B(t)$ and $\mathcal{C}_B(t)$ after multiplication by $\rho_{B,\delta}(t)$ and $R_B(t)$.

Proof. Let $E_t := \mathcal{R}_t \cap \mathcal{G}_t^\delta$. Expanding the score-resolved agreement by the joint law of (T_t, B_t) gives

$$\Gamma_A^\delta(t) = \sum_{\tau,b \in \{-1,+1\}} \mathbb{P}(A_t = b \mid T_t = \tau, B_t = b, E_t)\mathbb{P}(T_t = \tau, B_t = b \mid E_t).$$

Replace each conditional anchor law by the corresponding law given T_t alone, and denote the four replacement errors by

$$r_{\tau,b}(t) := \mathbb{P}(A_t = b \mid T_t = \tau, B_t = b, E_t) - \mathbb{P}(A_t = b \mid T_t = \tau, E_t).$$

By definition of $\Delta_{\mathrm{nd}}(t)$, each $|r_{\tau,b}(t)| \le \bar{\Delta}_{\mathrm{nd}}(t)$. Since the weights $\mathbb{P}(T_t = \tau, B_t = b \mid E_t)$ are nonnegative and sum to 1, the aggregate remainder

$$\varepsilon_{\mathrm{nd}}(t) := \sum_{\tau,b} r_{\tau,b}(t)\mathbb{P}(T_t = \tau, B_t = b \mid E_t)$$

satisfies $|\varepsilon_{\mathrm{nd}}(t)| \le \bar{\Delta}_{\mathrm{nd}}(t)$. After the replacement, the remaining expression is exactly the affine formula from theorem 85. Solving the perturbed affine relation yields the stated interval. □

Definition 89 (Qualified anchor with abstention)**.** Let $Q_t(m, m') \in \{0, 1\}$ be an observable qualification indicator on resolved pairs, with $Q_t = 1$ meaning that an anchor comparator is declared admissible for that pair and $Q_t = 0$ meaning abstention or out-of-scope status. Define the qualification mass

$$\kappa_A(t) := \mathbb{P}(Q_t(m, m') = 1 \mid (m, m') \in \mathcal{R}_t). \tag{139}$$

Whenever $Q_t = 1$, let $A_t^Q(m, m') \in \{-1, +1\}$ denote the qualified anchor comparator and define

$$\Gamma_{A,Q}(t) := \mathbb{P}(\mathrm{sgn}_{\delta_t}(Y_B(m,t) - Y_B(m',t)) = A_t^Q(m,m') \mid (m,m') \in \mathcal{R}_t,\ Q_t = 1). \tag{140}$$

Theorem 90 (Qualified-anchor partial identification with abstention)**.** *Assume A_t^Q is defined only on the qualified region $Q_t = 1$ from theorem 89. Let*

$$e_{A,Q}(t) := \mathbb{P}(A_t^Q(m,m') \neq T_t(m,m') \mid (m,m') \in \mathcal{R}_t,\ Q_t = 1). \tag{141}$$

If an anchor-validation study yields admissible bounds $e_{A,Q}(t) \le \bar{e}_{A,Q}(t)$ and $\kappa_A(t) \ge \underline{\kappa}_A(t)$, then

$$\mathcal{O}_B(t) \in \Big[\underline{\kappa}_A(t)\max\{0, \Gamma_{A,Q}(t) - \bar{e}_{A,Q}(t)\},\ 1 - \underline{\kappa}_A(t)(1 - \min\{1, \Gamma_{A,Q}(t) + \bar{e}_{A,Q}(t)\})\Big]. \tag{142}$$

Consequently, if $R_B(t) \in [\underline{R}_B(t), \overline{R}_B(t)]$, then

$$\mathcal{C}_B(t) \in \Big[\underline{R}_B(t)\underline{\kappa}_A(t)\max\{0, \Gamma_{A,Q}(t) - \bar{e}_{A,Q}(t)\},\ \overline{R}_B(t)\Big(1 - \underline{\kappa}_A(t)(1 - \min\{1, \Gamma_{A,Q}(t) + \bar{e}_{A,Q}(t)\})\Big)\Big]. \tag{143}$$

Proof. Condition on the resolved-pair event and decompose

$$\mathcal{O}_B(t) = \kappa_A(t)\,\mathcal{O}_{B,Q}(t) + (1 - \kappa_A(t))\,\mathcal{O}_{B,Q^c}(t), \tag{144}$$

where $\mathcal{O}_{B,Q}(t)$ is the order fidelity on the qualified region and $\mathcal{O}_{B,Q^c}(t) \in [0,1]$ is unrestricted on the abstention region. By theorem 83,

$$\mathcal{O}_{B,Q}(t) \in \Big[\max\{0, \Gamma_{A,Q}(t) - \bar{e}_{A,Q}(t)\},\ \min\{1, \Gamma_{A,Q}(t) + \bar{e}_{A,Q}(t)\}\Big]. \tag{145}$$

Using the lower bound $\kappa_A(t) \ge \underline{\kappa}_A(t)$ and the fact that the abstention-region fidelity is only known to lie in $[0,1]$ yields the displayed interval after minimizing and maximizing over the unknown abstention component. Multiplication by the admissible resolved-mass interval gives the construct-validity interval. □

Proposition 91 (Observed resolved-set mismatch adds an explicit error term)**.** *Let $\mathcal{R}_t^{(A)}$ be an anchor-resolved event constructed from observable anchor scores, and define the observable benchmark-anchor agreement*

$$\Gamma_A^{\mathrm{obs}}(t) := \mathbb{P}(\mathrm{sgn}_{\delta_t}(Y_B(m,t) - Y_B(m',t)) = A_t(m,m') \mid (m,m') \in \mathcal{R}_t^{(A)}). \tag{146}$$

Assume there exists $\pi_A(t) > 0$ such that

$$\mathbb{P}(\mathcal{R}_t) \ge \pi_A(t) \qquad \text{and} \qquad \mathbb{P}(\mathcal{R}_t^{(A)}) \ge \pi_A(t), \tag{147}$$

and let

$$\eta_A(t) := \mathbb{P}(\mathcal{R}_t \,\Delta\, \mathcal{R}_t^{(A)}) \tag{148}$$

be the resolved-set discrepancy. Then

$$|\Gamma_A^{\mathrm{obs}}(t) - \Gamma_A(t)| \le \frac{2\eta_A(t)}{\pi_A(t)}. \tag{149}$$

Consequently,

$$|\Gamma_A^{\mathrm{obs}}(t) - \mathcal{O}_B(t)| \le e_A(t) + \frac{2\eta_A(t)}{\pi_A(t)}. \tag{150}$$

If admissible upper bounds $e_A(t) \le \bar{e}_A(t)$ and $\eta_A(t) \le \bar{\eta}_A(t)$ are available, then

$$\mathcal{O}_B(t) \in \Big[\max\{0, \Gamma_A^{\mathrm{obs}}(t) - \bar{e}_A(t) - 2\bar{\eta}_A(t)/\pi_A(t)\},\ \min\{1, \Gamma_A^{\mathrm{obs}}(t) + \bar{e}_A(t) + 2\bar{\eta}_A(t)/\pi_A(t)\}\Big]. \tag{151}$$

If, in addition, $R_B(t) \in [\underline{R}_B(t), \overline{R}_B(t)]$, then multiplying the previous interval by that mass interval yields a valid interval for $\mathcal{C}_B(t)$.

Proof. Let E_t denote the event that benchmark and anchor comparators agree. Then

$$\Gamma_A^{\mathrm{obs}}(t) - \Gamma_A(t) = \frac{\mathbb{P}(E_t \cap \mathcal{R}_t^{(A)})}{\mathbb{P}(\mathcal{R}_t^{(A)})} - \frac{\mathbb{P}(E_t \cap \mathcal{R}_t)}{\mathbb{P}(\mathcal{R}_t)}. \tag{152}$$

Adding and subtracting $\mathbb{P}(E_t \cap \mathcal{R}_t)/\mathbb{P}(\mathcal{R}_t^{(A)})$ gives

$$|\Gamma_A^{\mathrm{obs}}(t) - \Gamma_A(t)| \le \frac{|\mathbb{P}(E_t \cap \mathcal{R}_t^{(A)}) - \mathbb{P}(E_t \cap \mathcal{R}_t)|}{\mathbb{P}(\mathcal{R}_t^{(A)})} + \mathbb{P}(E_t \cap \mathcal{R}_t)\Big|\frac{1}{\mathbb{P}(\mathcal{R}_t^{(A)})} - \frac{1}{\mathbb{P}(\mathcal{R}_t)}\Big| \tag{153}$$

$$\le \frac{\eta_A(t)}{\pi_A(t)} + \frac{|\mathbb{P}(\mathcal{R}_t^{(A)}) - \mathbb{P}(\mathcal{R}_t)|}{\pi_A(t)} \le \frac{2\eta_A(t)}{\pi_A(t)}. \tag{154}$$

Combining this with theorem 83 yields the order-fidelity interval. The construct-validity statement follows by multiplication with any admissible resolved-mass interval. □

Definition 92 (Directional resolved-set mismatch budgets)**.** Let $\mathcal{R}_t$ denote the latent target-resolved event and $\mathcal{R}_t^{(A)}$ the observable anchor-resolved event. Define the directional mismatch masses

$$\eta_A^+(t) := \mathbb{P}(\mathcal{R}_t^{(A)} \setminus \mathcal{R}_t), \qquad \eta_A^-(t) := \mathbb{P}(\mathcal{R}_t \setminus \mathcal{R}_t^{(A)}), \tag{155}$$

and let

$$\omega_A(t) := \mathbb{P}(\mathcal{R}_t \cap \mathcal{R}_t^{(A)}). \tag{156}$$

Thus $\mathbb{P}(\mathcal{R}_t^{(A)}) = \omega_A(t) + \eta_A^+(t)$ and $\mathbb{P}(\mathcal{R}_t) = \omega_A(t) + \eta_A^-(t)$.

Theorem 93 (Directional resolved-set mismatch decomposition)**.** *Let E_t denote the event that the thresholded benchmark comparator agrees with the anchor comparator. Assume*

$$\mathbb{P}(\mathcal{R}_t^{(A)}) > 0, \qquad \mathbb{P}(\mathcal{R}_t) > 0. \tag{157}$$

If in addition $\omega_A(t) = \mathbb{P}(\mathcal{R}_t \cap \mathcal{R}_t^{(A)}) > 0$, define

$$\gamma_{A,\cap}(t) := \mathbb{P}(E_t \mid \mathcal{R}_t \cap \mathcal{R}_t^{(A)}). \tag{158}$$

Then

$$|\Gamma_A^{\mathrm{obs}}(t) - \gamma_{A,\cap}(t)| \le \frac{\eta_A^+(t)}{\mathbb{P}(\mathcal{R}_t^{(A)})}, \qquad |\Gamma_A(t) - \gamma_{A,\cap}(t)| \le \frac{\eta_A^-(t)}{\mathbb{P}(\mathcal{R}_t)}. \tag{159}$$

Consequently,

$$|\Gamma_A^{\mathrm{obs}}(t) - \Gamma_A(t)| \le \frac{\eta_A^+(t)}{\mathbb{P}(\mathcal{R}_t^{(A)})} + \frac{\eta_A^-(t)}{\mathbb{P}(\mathcal{R}_t)}. \tag{160}$$

If admissible bounds

$$\eta_A^+(t) \le \bar{\eta}_A^+(t), \qquad \eta_A^-(t) \le \bar{\eta}_A^-(t), \qquad \mathbb{P}(\mathcal{R}_t) \ge \pi_A^{\mathrm{tar}}(t) > 0, \tag{161}$$

are available, then

$$|\Gamma_A^{\mathrm{obs}}(t) - \mathcal{O}_B(t)| \le e_A(t) + \frac{\bar{\eta}_A^+(t)}{\mathbb{P}(\mathcal{R}_t^{(A)})} + \frac{\bar{\eta}_A^-(t)}{\pi_A^{\mathrm{tar}}(t)}. \tag{162}$$

Hence

$$\mathcal{O}_B(t) \in \Big[\max\{0, \Gamma_A^{\mathrm{obs}}(t) - \bar{e}_A(t) - \bar{\eta}_A^+(t)/\mathbb{P}(\mathcal{R}_t^{(A)}) - \bar{\eta}_A^-(t)/\pi_A^{\mathrm{tar}}(t)\}, \tag{163}$$

$$\min\{1, \Gamma_A^{\mathrm{obs}}(t) + \bar{e}_A(t) + \bar{\eta}_A^+(t)/\mathbb{P}(\mathcal{R}_t^{(A)}) + \bar{\eta}_A^-(t)/\pi_A^{\mathrm{tar}}(t)\}\Big]. \tag{164}$$

If $R_B(t) \in [\underline{R}_B(t), \overline{R}_B(t)]$, multiplying the previous interval by that mass interval yields a valid interval for $\mathcal{C}_B(t)$. If $\omega_A(t) = 0$ or either denominator is zero, the fail-closed output is the status label `insufficient-overlap` *together with the trivial order-fidelity interval* $[0, 1]$.

Proof. Assume first that all denominators are positive. Write $A := \mathcal{R}_t^{(A)}$, $R := \mathcal{R}_t$, $O := A \cap R$, $M^+ := A \setminus R$, and $M^- := R \setminus A$. Then

$$\Gamma_A^{\mathrm{obs}}(t) = \frac{\mathbb{P}(E_t \cap O) + \mathbb{P}(E_t \cap M^+)}{\mathbb{P}(A)}, \qquad \gamma_{A,\cap}(t) = \frac{\mathbb{P}(E_t \cap O)}{\mathbb{P}(O)}. \tag{165}$$

Subtracting these two quantities and using $0 \le \mathbb{P}(E_t \cap M^+) \le \mathbb{P}(M^+) = \eta_A^+(t)$ gives

$$|\Gamma_A^{\mathrm{obs}}(t) - \gamma_{A,\cap}(t)| \le \frac{\eta_A^+(t)}{\mathbb{P}(A)}. \tag{166}$$

The bound for $\Gamma_A(t)$ is symmetric with M^- in place of M^+. The triangle inequality yields the displayed bound on $|\Gamma_A^{\mathrm{obs}}(t) - \Gamma_A(t)|$, and combining this with theorem 83 yields the order-fidelity interval. The construct-validity claim follows by interval multiplication. When $\omega_A(t) = 0$ or either denominator vanishes, the theorem deliberately returns only the fail-closed status and the trivial interval. □

Corollary 94 (Sharp nested-set identification for resolved-set mismatch)**.** *Assume the setup of theorem 92. If $\mathcal{R}_t^{(A)} \subseteq \mathcal{R}_t$ almost surely and the masses*

$$p_A(t) := \mathbb{P}(\mathcal{R}_t^{(A)}), \qquad p_R(t) := \mathbb{P}(\mathcal{R}_t) \tag{167}$$

are known, then the sharp identified set for $\Gamma_A(t)$ given $\Gamma_A^{\mathrm{obs}}(t)$ is

$$\Gamma_A(t) \in \Big[\frac{p_A(t)}{p_R(t)}\Gamma_A^{\mathrm{obs}}(t),\ 1 - \frac{p_A(t)}{p_R(t)}(1 - \Gamma_A^{\mathrm{obs}}(t))\Big]. \tag{168}$$

If instead $\mathcal{R}_t \subseteq \mathcal{R}_t^{(A)}$ almost surely and the same masses are known, then the sharp identified set is

$$\Gamma_A(t) \in \Big[\max\Big\{0, \frac{p_A(t)\Gamma_A^{\mathrm{obs}}(t) - (p_A(t) - p_R(t))}{p_R(t)}\Big\},\ \min\Big\{1, \frac{p_A(t)}{p_R(t)}\Gamma_A^{\mathrm{obs}}(t)\Big\}\Big]. \tag{169}$$

After adding an admissible anchor error budget $\bar{e}_A(t)$ through theorem 83, each interval propagates directly to a sharp or conservative interval for $\mathcal{O}_B(t)$, depending on whether the error budget is exact or only bounded.

Proof. If $\mathcal{R}_t^{(A)} \subseteq \mathcal{R}_t$, then the overlap event is exactly $\mathcal{R}_t^{(A)}$, so

$$\Gamma_A(t) = \frac{p_A(t)\Gamma_A^{\mathrm{obs}}(t) + u}{p_R(t)}, \qquad 0 \le u \le p_R(t) - p_A(t), \tag{170}$$

because the agreement rate on the unresolved remainder $\mathcal{R}_t \setminus \mathcal{R}_t^{(A)}$ is unrestricted on $[0,1]$. This yields the displayed interval, and every point in it is attainable by choosing an appropriate agreement rate on the remainder. The case $\mathcal{R}_t \subseteq \mathcal{R}_t^{(A)}$ is symmetric: now

$$p_A(t)\Gamma_A^{\mathrm{obs}}(t) = p_R(t)\Gamma_A(t) + u, \qquad 0 \le u \le p_A(t) - p_R(t), \tag{171}$$

which yields the second interval, again with attainability by appropriate choice of agreement rate on the excess anchor-resolved region. □

Remark 95. A symmetric-difference penalty is safe but blunt. Directional mismatch budgets expose whether the observable anchor threshold is too strict, too loose, or both. In the nested cases the identified set is not merely bounded more tightly; it is sharp.

Proposition 96 (Nonparametric multi-anchor intersection)**.** *Let $I_r(t)$ be any valid order-fidelity interval produced from anchor r, for example*

$$I_r(t) = \Big[\max\{0, \Gamma_r^{\mathrm{obs}}(t) - \bar{e}_r(t) - 2\bar{\eta}_r(t)/\pi_r(t)\},\ \min\{1, \Gamma_r^{\mathrm{obs}}(t) + \bar{e}_r(t) + 2\bar{\eta}_r(t)/\pi_r(t)\}\Big]. \tag{172}$$

Then

$$\mathcal{O}_B(t) \in \bigcap_{r=1}^{R} I_r(t) \tag{173}$$

whenever each $I_r(t)$ is valid.

Proof. Immediate, because $\mathcal{O}_B(t)$ belongs to every valid interval simultaneously. □

Corollary 97 (Multi-anchor amplification)**.** *Any valid shrinking of the order-fidelity interval induced by additional anchors immediately propagates to the construct-validity component after multiplication by an admissible interval for $R_B(t)$.*

Proof. The result is immediate from $\mathcal{C}_B(t) = R_B(t)\mathcal{O}_B(t)$ and monotonicity of interval multiplication on $[0,1]^2$. □

Theorem 98 (Shared-blind-spot dependence model for anchor panels)**.** *Suppose an odd-sized anchor panel aggregates to a majority-vote comparator A_t^{maj}, and suppose there exists a blind-spot event $B_t^\star$ on*

the resolved-pair event with

$$\mathbb{P}(B_t^\star \mid \mathcal{R}_t) \le \beta_t. \tag{174}$$

Assume that on the complement $(B_t^\star)^c$, the majority-vote comparator has conditional error at most $\bar{e}_{\mathrm{out}}(t)$. Then the overall majority-vote order error satisfies

$$e_{\mathrm{maj}}(t) := \mathbb{P}(A_t^{\mathrm{maj}} \neq T_t \mid \mathcal{R}_t) \le \beta_t + (1-\beta_t)\bar{e}_{\mathrm{out}}(t). \tag{175}$$

Consequently,

$$\mathcal{O}_B(t) \in \Big[\max\{0, \Gamma_{\mathrm{maj}}(t) - \beta_t - (1-\beta_t)\bar{e}_{\mathrm{out}}(t)\},\ \min\{1, \Gamma_{\mathrm{maj}}(t) + \beta_t + (1-\beta_t)\bar{e}_{\mathrm{out}}(t)\}\Big], \tag{176}$$

where $\Gamma_{\mathrm{maj}}(t) := \mathbb{P}(\mathrm{sgn}_{\delta_t}(Y_B - Y'_B) = A_t^{\mathrm{maj}} \mid \mathcal{R}_t)$.

Proof. By the law of total probability,

$$e_{\mathrm{maj}}(t) = \mathbb{P}(A_t^{\mathrm{maj}} \neq T_t \mid B_t^\star, \mathcal{R}_t)\mathbb{P}(B_t^\star \mid \mathcal{R}_t) + \mathbb{P}(A_t^{\mathrm{maj}} \neq T_t \mid (B_t^\star)^c, \mathcal{R}_t)\mathbb{P}((B_t^\star)^c \mid \mathcal{R}_t).$$

The first term is at most β_t and the second at most $(1-\beta_t)\bar{e}_{\mathrm{out}}(t)$. Apply theorem 83 with error budget $e_{\mathrm{maj}}(t)$. □

Proposition 99 (Pairwise-covariance bound for dependent anchor panels)**.** *Let $R \ge 3$ be odd and let $\Xi_r := \mathbf{1}\{A_t^{(r)} \neq T_t\}$ be the resolved-pair error indicator for anchor r. Suppose*

$$\mu_t := \sum_{r=1}^{R} \mathbb{E}[\Xi_r \mid \mathcal{R}_t] < m_R := \frac{R+1}{2} \tag{177}$$

and let

$$\sigma_t^2 := \sum_{r=1}^{R} \mathrm{Var}(\Xi_r \mid \mathcal{R}_t) + 2 \sum_{1 \le r < s \le R} \mathrm{Cov}(\Xi_r, \Xi_s \mid \mathcal{R}_t). \tag{178}$$

Then the majority-vote panel error is bounded by

$$e_{\mathrm{maj}}(t) := \mathbb{P}\Big(\sum_{r=1}^{R} \Xi_r \ge m_R \;\Big|\; \mathcal{R}_t\Big) \le \frac{\sigma_t^2}{\sigma_t^2 + (m_R - \mu_t)^2}. \tag{179}$$

Consequently,

$$\mathcal{O}_B(t) \in \Big[\max\{0, \Gamma_{\mathrm{maj}}(t) - \bar{e}_{\mathrm{maj}}(t)\},\ \min\{1, \Gamma_{\mathrm{maj}}(t) + \bar{e}_{\mathrm{maj}}(t)\}\Big] \tag{180}$$

for any declared upper bound $\bar{e}_{\mathrm{maj}}(t)$ on the right-hand side.

Proof. Conditional on $\mathcal{R}_t$, majority vote fails only if the centered sum $\sum_r (\Xi_r - \mathbb{E}\Xi_r)$ exceeds $m_R - \mu_t > 0$. Cantelli's inequality therefore gives

$$\mathbb{P}\Big(\sum_r \Xi_r \ge m_R \mid \mathcal{R}_t\Big) \le \frac{\mathrm{Var}(\sum_r \Xi_r \mid \mathcal{R}_t)}{\mathrm{Var}(\sum_r \Xi_r \mid \mathcal{R}_t) + (m_R - \mu_t)^2},$$

and the variance is exactly σ_t^2. No independence assumption is used here: Cantelli's one-sided Chebyshev bound depends only on the conditional first two moments, which is why the pairwise covariance terms enter explicitly. Apply theorem 83 to the resulting panel-error bound. □

Theorem 100 (Latent-factor dependence bound for anchor panels)**.** *Let Z_t be a latent difficulty factor on resolved pairs. Suppose that conditional on $(Z_t, \mathcal{R}_t)$ the anchor errors are independent with conditional error probabilities $e_r(z,t) := \mathbb{P}(A_t^{(r)} \neq T_t \mid Z_t = z, \mathcal{R}_t)$. For an odd panel of size R with majority threshold $m_R = (R+1)/2$, the majority-vote error rate is*

$$e_{\mathrm{maj}}(t) = \int \sum_{j=m_R}^{R} \sum_{\substack{S \subseteq \{1,\dots,R\} \\ |S|=j}} \prod_{r \in S} e_r(z,t) \prod_{r \notin S} (1 - e_r(z,t))\, \mu_t(\mathrm{d}z), \tag{181}$$

where μ_t is the conditional law of Z_t given $\mathcal{R}_t$. In the exchangeable case $e_r(z,t) \equiv e(z,t)$, this reduces to

$$e_{\mathrm{maj}}(t) = \int \sum_{j=m_R}^{R} \binom{R}{j} e(z,t)^j \big(1 - e(z,t)\big)^{R-j}\, \mu_t(\mathrm{d}z). \tag{182}$$

Consequently, any upper bound on the displayed integral can be inserted into theorem 83 to obtain a dependence-aware panel interval for $\mathcal{O}_B(t)$.

Proof. Conditional on $(Z_t = z, \mathcal{R}_t)$, the number of panel errors is a sum of independent Bernoulli variables with success probabilities $e_r(z,t)$. The conditional majority-error probability is therefore the stated Poisson-binomial upper tail. Averaging over the conditional law μ_t yields the unconditional panel error rate. The exchangeable formula is the binomial special case. □

7 Benchmark Portfolios and Coverage Uncertainty

7.1 Benchmark portfolio half-life

Single-benchmark half-life is often too brittle for real evaluation programs, which usually deploy families of benchmarks whose decay channels are only partly shared. This motivates a portfolio-level lifetime object.

Definition 101 (Weighted benchmark portfolio survival)**.** Let $\mathcal{P} = \{(B_i, w_i)\}_{i=1}^m$ be a benchmark portfolio with weights $w_i \geq 0$ and $\sum_{i=1}^m w_i = 1$, where each benchmark has positive initial validity. Define the normalized portfolio survival by

$$\widetilde{V}_{\mathcal{P}}(t) := \sum_{i=1}^{m} w_i\, \widetilde{V}_{B_i}(t), \tag{183}$$

and define the portfolio half-life by

$$\tau_{1/2}(\mathcal{P}) := \inf\{t \geq 0 : \widetilde{V}_{\mathcal{P}}(t) \leq \tfrac{1}{2}\}. \tag{184}$$

The normalization makes heterogeneous initial benchmark validities comparable before aggregation.

Proposition 102 (Portfolio half-life sandwich under monotone component survival)**.** *Assume each component survival curve $\widetilde{V}_{B_i}(t)$ is nonincreasing. Then*

$$\min_{1 \leq i \leq m} \tau_{1/2}(B_i) \leq \tau_{1/2}(\mathcal{P}) \leq \max_{1 \leq i \leq m} \tau_{1/2}(B_i). \tag{185}$$

Proof. If $t < \min_i \tau_{1/2}(B_i)$, then every component satisfies $\widetilde{V}_{B_i}(t) > \frac{1}{2}$, so their weighted average also exceeds $\frac{1}{2}$. Hence $\tau_{1/2}(\mathcal{P}) \geq \min_i \tau_{1/2}(B_i)$. If $t \geq \max_i \tau_{1/2}(B_i)$ and every component survival is

nonincreasing, then each component has already crossed below $\frac{1}{2}$ and cannot recover above it, so $\widetilde{V}_{B_i}(t) \le \frac{1}{2}$ for all i. Their weighted average is then at most $\frac{1}{2}$, proving $\tau_{1/2}(\mathcal{P}) \le \max_i \tau_{1/2}(B_i)$. □

Theorem 103 (Common-mode hazard ceiling for benchmark portfolios)**.** *Fix a horizon $[0,T]$. Suppose each component benchmark satisfies theorem 13, has no upward normalized-validity jumps on $[0,T]$, and has effective passive hazard decomposition*

$$q_i(t) = q_{\mathrm{com}}(t) + r_i(t) \tag{186}$$

for almost every passive time $t \in [0,T]$, where $q_{\mathrm{com}}(t) \ge 0$ is a common-mode decay channel and $r_i(t) \ge 0$ is an idiosyncratic remainder. Then

$$\widetilde{V}_{\mathcal{P}}(t) \le \exp\Big(-\int_0^t q_{\mathrm{com}}(u)\,\mathrm{d}u\Big) \qquad \text{for all } t \in [0,T]. \tag{187}$$

Consequently, if $q_{\mathrm{com}}(t) \ge \underline{q}_{\mathrm{com}} > 0$ almost everywhere on $[0,T]$, then

$$\tau_{1/2}(\mathcal{P}) \le \frac{\log 2}{\underline{q}_{\mathrm{com}}} \tag{188}$$

whenever the right-hand side does not exceed T. Thus diversification across benchmarks cannot outrun a sufficiently strong common-mode contamination channel.

Proof. For each component,

$$\widetilde{V}_{B_i}(t) = J_i(t)\exp\Big(-\int_0^t q_i(u)\,\mathrm{d}u\Big) = \Big[J_i(t)\exp\Big(-\int_0^t r_i(u)\,\mathrm{d}u\Big)\Big]\exp\Big(-\int_0^t q_{\mathrm{com}}(u)\,\mathrm{d}u\Big) \le \exp\Big(-\int_0^t q_{\mathrm{com}}(u)\,\mathrm{d}u\Big),$$

because $J_i(t) \le 1$ and $r_i(u) \ge 0$ on the certified horizon. Taking the weighted average over i yields the first claim. The half-life bound follows by comparing the common-mode envelope to $\frac{1}{2}$. □

Weighted portfolios are therefore useful, but not magical. They help mainly against idiosyncratic decay, not against shared leakage manifolds, shared judge families, or shared benchmark-generation pipelines.

7.2 Portfolio design under deterministic and scenario uncertainty

The portfolio definitions above describe a given benchmark family. A governance program must still decide how to allocate attention, refresh budget, and reporting weight across candidate benchmarks.

Proposition 104 (Deterministic horizon-survival design degenerates to a single benchmark)**.** *Fix a horizon $T > 0$ and suppose each candidate benchmark B_i has known normalized survival*

$$\sigma_i(T) := \widetilde{V}_{B_i}(T) \in [0,1].$$

Among all weights $w \in \Delta_m := \{w \in [0,1]^m : \ \sum_i w_i = 1\}$, the problem

$$\max_{w \in \Delta_m} \ \sum_{i=1}^m w_i \sigma_i(T) \tag{189}$$

is solved by any vertex weight vector concentrating all mass on an index in $\arg\max_i \sigma_i(T)$. *Hence diversification is not optimal under a single known deterministic survival profile unless additional constraints or uncertainty are imposed.*

Proof. The objective is linear in w over the simplex Δ_m. A linear objective on a simplex is maximized at an extreme point, and the extreme points are precisely the unit vectors. Any unit vector supported on an index with maximal $\sigma_i(T)$ is therefore optimal. □

Remark 105. This degeneracy is not a criticism of diversification; it clarifies why diversification needs a reason. In practice that reason is usually one of four things: scenario uncertainty about survival profiles, common-mode exposure caps, explicit coverage requirements, or risk aversion to tail invalidity rather than mean survival. The later portfolio results make those motives explicit rather than treating diversification as self-justifying.

Definition 106 (Scenario-constrained robust benchmark-portfolio design)**.** Fix a horizon $T > 0$, candidate benchmarks $B_1, \dots, B_m$, and scenario-specific normalized survival profiles

$$\sigma_i^{(s)} : [0, T] \to [0, \infty), \qquad s = 1, \dots, S,$$

where each $\sigma_i^{(s)}(\cdot)$ is declared and nonincreasing on $[0, T]$. Let $\sigma_i^{(s)}(T)$ denote its horizon value. With optional linear exposure caps $Aw \le c$ describing common-mode limits such as judge-family concentration, provenance overlap, or generation-pipeline overlap, the robust horizon-survival design problem is

$$\max_{w,\eta} \quad \eta \tag{190}$$

$$\text{subject to} \quad \sum_{i=1}^{m} w_i \sigma_i^{(s)}(T) \ge \eta, \qquad s = 1, \dots, S, \tag{191}$$

$$w \in \Delta_m, \tag{192}$$

$$Aw \le c. \tag{193}$$

The optimal value $\eta^*(T)$ is the maximal worst-case normalized portfolio survival at horizon T over the declared scenario set and exposure constraints.

Theorem 107 (Robust portfolio design is convex and certifies half-life targets)**.** *For any fixed horizon* T*, the robust design problem in theorem 106 is a feasible linear program whenever the constraint set is nonempty. An optimum exists whenever the feasible set is nonempty and compact. Moreover, if the optimal value satisfies* $\eta^*(T) > \frac{1}{2}$*, then the designed portfolio obeys*

$$\tau_{1/2}(\mathcal{P}_w) > T \tag{194}$$

under every scenario in the declared uncertainty set.

Proof. The objective is linear in (w, η) and every displayed constraint is linear. Hence the problem is a linear program. Compactness follows from the simplex constraint together with any finite exposure caps, so existence is standard whenever the feasible set is nonempty. If $\eta^*(T) > 1/2$, then by construction every feasible optimum satisfies

$$\widetilde{V}_{\mathcal{P}_w}^{(s)}(T) = \sum_i w_i \sigma_i^{(s)}(T) \ge \eta^*(T) > \tfrac{1}{2}$$

for each scenario s. Because each declared scenario survival profile is nonincreasing on $[0,T]$, the corresponding portfolio survival profile is also nonincreasing on $[0,T]$. Therefore remaining above one half at time T implies remaining above one half at every earlier time on that horizon, which yields $\tau_{1/2}(\mathcal{P}_w) > T$ under every scenario. □

Remark 108. Theorems 104 and 107 explain a small but important conceptual point. Benchmark diversification is not automatically a theorem of good governance. It becomes rational only after one acknowledges scenario uncertainty, common-mode concentration limits, or explicit coverage constraints. Otherwise the optimal horizon-survival rule is embarrassingly greedy: use the single benchmark with the best known survival profile at the target horizon.

7.3 Portfolio design with explicit coverage constraints

Robust survival is not the only design objective. A benchmark family may also be required to cover multiple task types, modalities, demographic slices, attack surfaces, or capability regimes. Those requirements can be formalized without breaking convexity.

Definition 109 (Coverage-constrained robust portfolio design)**.** Fix coverage cells $g = 1, \dots, G$ and declared benchmark-coverage coefficients

$$\kappa_{ig} \in [0,1], \qquad i = 1, \dots, m.$$

For a portfolio $w \in \Delta_m$, define its cellwise coverage by

$$\mathrm{Cov}_g(w) := \sum_{i=1}^{m} w_i \kappa_{ig}. \tag{195}$$

Given target coverages $\rho \in [0,1]^G$, the coverage-constrained robust design problem is

$$\begin{aligned}
v(\rho;T) := \max_{w,\eta} \quad & \eta && && (196)\\
\text{subject to} \quad & \sum_{i=1}^{m} w_i \sigma_i^{(s)}(T) \geq \eta, && s = 1, \dots, S, && (197)\\
& w \in \Delta_m, && && (198)\\
& Aw \leq c, && && (199)\\
& \mathrm{Cov}_g(w) \geq \rho_g, && g = 1, \dots, G. && (200)
\end{aligned}$$

Theorem 110 (Coverage-constrained design remains convex and its value frontier is concave)**.** *For any fixed horizon T, the problem in theorem 109 is a linear program whenever the feasible set is nonempty. The value function $\rho \mapsto v(\rho;T)$ on the feasible coverage region is componentwise nonincreasing and concave.*

Proof. Every displayed constraint is linear in (w,η), so the problem is a linear program. If $\rho' \geq \rho$ componentwise, then every portfolio feasible for ρ' is also feasible for ρ, so $v(\rho';T) \leq v(\rho;T)$.

To prove concavity, let $\rho^{(1)}$ and $\rho^{(2)}$ be feasible and choose optimal portfolios $w^{(1)}$ and $w^{(2)}$. For any $\lambda \in [0,1]$, the convex combination $w^{(\lambda)} := \lambda w^{(1)} + (1-\lambda) w^{(2)}$ satisfies

$$\mathrm{Cov}_g(w^{(\lambda)}) = \lambda \mathrm{Cov}_g(w^{(1)}) + (1-\lambda)\mathrm{Cov}_g(w^{(2)}) \geq \lambda \rho_g^{(1)} + (1-\lambda)\rho_g^{(2)}$$

for every g, and likewise preserves the linear exposure caps. The objective

$$f_T(w) := \min_{1\le s\le S} \sum_{i=1}^{m} w_i \sigma_i^{(s)}(T)$$

is the minimum of finitely many linear functions and is therefore concave in w. Hence

$$v(\lambda\rho^{(1)} + (1-\lambda)\rho^{(2)}; T) \ge f_T(w^{(\lambda)}) \ge \lambda f_T(w^{(1)}) + (1-\lambda) f_T(w^{(2)}) = \lambda v(\rho^{(1)}; T) + (1-\lambda) v(\rho^{(2)}; T).$$

Thus $v(\rho; T)$ is concave. □

Proposition 111 (Coverage constraints can force diversification)**.** *Assume the feasible set in theorem 109 is nonempty. If no single benchmark i satisfies both the exposure caps and the coverage requirements $\mathrm{Cov}_g(e_i) \ge \rho_g$ for all g, then every feasible portfolio has support size at least two. In particular, explicit coverage constraints can break the single-benchmark degeneracy from theorem 104 even under a deterministic survival model.*

Proof. A unit vector e_i corresponds to a single-benchmark portfolio. By assumption, no such e_i is feasible. Therefore any feasible portfolio must assign positive weight to at least two benchmarks. □

Remark 112. The coverage frontier $v(\rho; T)$ is a governance object, not just an optimization artifact. Its shadow prices quantify how much worst-case horizon survival must be sacrificed to maintain representational breadth across declared evaluation cells. That tradeoff should be reported whenever benchmark-family design claims to balance longevity against scope.

Remark 113 (Computational scale)**.** For fixed horizon T, the deterministic and scenario-robust portfolio programs remain linear in the number of benchmarks, scenarios, and coverage cells. A direct LP formulation has $O(m + S + G)$ primal variables before robustification and adds only one block of dual variables per uncertain coverage cell in the box or budgeted models. When (m, G, S) are large, one should work with frontier slices, weighted-sum support-hyperplane methods, or column-generation style pruning rather than enumerate the entire high-dimensional coverage frontier.

Definition 114 (Frontier-efficient portfolio action)**.** At a decision epoch, let a redesign or regeneration action choose a benchmark-family portfolio w together with any non-portfolio controls. Suppose the reporting layer provides a certified lower coverage vector $\underline{\mathrm{Cov}}(w)$ and a certified lower survival summary $\underline{s}_T(w)$ at the planning horizon T. A portfolio action w is called *frontier dominated* if there exists another feasible portfolio w' such that

$$\underline{\mathrm{Cov}}(w') \ge \underline{\mathrm{Cov}}(w) \quad \text{componentwise}, \qquad \underline{s}_T(w') \ge \underline{s}_T(w), \tag{201}$$

with at least one inequality strict.

Theorem 115 (Coverage-frontier action reduction for safe control)**.** *Let $\Psi(w) := (\underline{\mathrm{Cov}}(w), \underline{s}_T(w))$ denote the certified portfolio summary. Assume portfolio choice enters the certified safe-control problem only through $\Psi(w)$ in the following control-sufficient sense:*

1. *for every non-portfolio control component, the immediate reward can be written as a measurable function of the current control state, the non-portfolio action, and $\Psi(w)$, and is nondecreasing in $\Psi(w)$;*

2. *the feasible post-action control states, transition kernels, and downstream action constraints depend on w only through $\Psi(w)$;*

3. *if $\Psi(w') \geq \Psi(w)$ componentwise, then the certified continuation value under any admissible downstream policy is weakly larger under w' than under w.*

Then removing frontier-dominated portfolio actions in the sense of theorem 114 does not reduce the optimal safe-control value. If, in addition, the certified lower survival summary is given by the coverage-frontier value $v(\rho; T)$ from theorem 110 and every feasible target ρ admits a measurable frontier-attaining selector $w^\star(\rho)$, then the controller can replace the full portfolio vector by the frontier target ρ together with its certified value $v(\rho; T)$ without loss relative to the certified action class.

Proof. If w is frontier dominated by w', then $\Psi(w') \geq \Psi(w)$ componentwise with at least one strict inequality. By assumptions (1)–(3), every immediate reward, feasible next-state law, and downstream certified continuation value available after choosing w is weakly dominated by the corresponding object after choosing w'. Hence w cannot improve the safe-control value and may be removed without loss. For the second claim, theorem 110 states that $v(\rho; T)$ is the maximal certified lower survival attainable among portfolios satisfying coverage target ρ. The measurable selector assumption lets the planner implement a frontier-attaining portfolio for each feasible ρ, while control sufficiency says no other portfolio feature matters beyond $\Psi(w)$. Therefore optimization over portfolios can be replaced by optimization over feasible coverage targets and their frontier values without loss relative to the certified action class. □

Remark 116. This reduction is operationally useful because portfolio weights are often high dimensional while the coverage target is low dimensional and policy relevant. The reduction is only as honest as the certified lower frontier itself; if the frontier is learned with uncertainty, that uncertainty must be carried into control rather than buried in a point frontier.

7.4 Coverage-matrix uncertainty

Coverage coefficients are rarely literal facts. They are usually outputs of audits, taxonomies, or evaluator judgments, and those objects are themselves uncertain. Treating the coverage matrix as exact can therefore produce fake diversification.

Definition 117 (Robust coverage under matrix uncertainty)**.** Let $\mathcal{U}_\kappa$ be a declared uncertainty set of admissible coverage matrices $\kappa = (\kappa_{ig})_{i,g}$. The robust coverage of a portfolio $w \in \Delta_m$ on cell g is

$$\underline{\mathrm{Cov}}_g(w) := \inf_{\kappa \in \mathcal{U}_\kappa} \sum_{i=1}^{m} w_i \kappa_{ig}. \tag{202}$$

The robust coverage-constrained design problem replaces the nominal constraints $\mathrm{Cov}_g(w) \geq \rho_g$ in theorem 109 by $\underline{\mathrm{Cov}}_g(w) \geq \rho_g$ for all g.

Proposition 118 (Box-uncertain coverage reduces to lower-envelope coverage)**.** *Suppose*

$$\mathcal{U}_\kappa^{\mathrm{box}} := \{\kappa : \underline{\kappa}_{ig} \leq \kappa_{ig} \leq \overline{\kappa}_{ig} \text{ for all } i, g\}. \tag{203}$$

Then the robust coverage constraints from Definition 117 are equivalent to

$$\sum_{i=1}^{m} w_i \underline{\kappa}_{ig} \geq \rho_g, \qquad g = 1, \ldots, G. \tag{204}$$

Hence box-robust coverage design remains a linear program.

Proof. For fixed w and g, the map $\kappa \mapsto \sum_i w_i \kappa_{ig}$ is linear and nondecreasing in each coordinate because $w_i \geq 0$. The infimum over the box is therefore attained at the lower corner $\underline{\kappa}_{ig}$. □

Theorem 119 (Budgeted coverage uncertainty admits a linear robust counterpart)**.** *Fix nominal coverages $\bar{\kappa}_{ig} \in [0,1]$, nonnegative deviations d_{ig}, and cellwise uncertainty budgets $\Gamma_g \in [0, m]$. Consider the Bertsimas–Sim style uncertainty set*

$$\mathcal{U}^{\Gamma}_{\kappa,g} := \left\{ \kappa_{\cdot g} : \ \kappa_{ig} = \bar{\kappa}_{ig} - d_{ig} z_{ig}, \ 0 \leq z_{ig} \leq 1, \ \sum_{i=1}^{m} z_{ig} \leq \Gamma_g \right\} \tag{205}$$

for each cell g. Then the robust coverage constraint

$$\inf_{\kappa_{\cdot g} \in \mathcal{U}^{\Gamma}_{\kappa,g}} \sum_{i=1}^{m} w_i \kappa_{ig} \geq \rho_g \tag{206}$$

is equivalent to the existence of dual variables $u_g \geq 0$ and $v_{ig} \geq 0$ such that

$$\sum_{i=1}^{m} w_i \bar{\kappa}_{ig} - \Gamma_g u_g - \sum_{i=1}^{m} v_{ig} \geq \rho_g, \tag{207}$$

$$u_g + v_{ig} \geq d_{ig} w_i, \qquad i = 1, \ldots, m. \tag{208}$$

Consequently, scenario-constrained portfolio design with budgeted coverage uncertainty remains a linear program.

Proof. For fixed w and g, the worst-case coverage problem is

$$\inf_{0 \leq z \leq 1, \ \sum z_i \leq \Gamma_g} \sum_i w_i \bar{\kappa}_{ig} - \sum_i d_{ig} w_i z_i.$$

This is a bounded linear program in z: the box constraints give compact feasibility, and the objective is finite even when some $d_{ig} = 0$ or $\Gamma_g = 0$. The equivalent maximization of $\sum_i d_{ig} w_i z_i$ over the same feasible set is therefore a feasible bounded primal LP, so strong duality applies. Dualizing that maximization under the budget and box constraints yields the stated counterpart. When $d_{ig} = 0$, the corresponding dual constraint reduces to $u_g + v_{ig} \geq 0$ and is harmless. When $\Gamma_g = 0$, the budget variable can be set to zero and the robust constraint collapses to nominal coverage. □

Proposition 120 (Coverage uncertainty can force additional diversification)**.** *Assume the robust feasible set from Definition 117 is nonempty. If for every benchmark i there exists some admissible coverage matrix $\kappa \in \mathcal{U}_\kappa$ and some cell g such that $\sum_j (e_i)_j \kappa_{jg} = \kappa_{ig} < \rho_g$, then no single-benchmark portfolio is robustly feasible. Hence every robustly feasible portfolio has support size at least two.*

Proof. A single-benchmark portfolio is a unit vector e_i. By assumption, each e_i violates at least one robust coverage constraint under some admissible matrix, so none is robustly feasible. Any robustly feasible portfolio must therefore place positive weight on at least two benchmarks. □

7.5 Learned uncertainty sets for coverage matrices

Fixed box or budget sets are useful when governance wants explicit worst-case knobs. When coverage claims are themselves estimated from audits, classifier outputs, or rater studies, however, the uncertainty set should ideally be learned from data rather than chosen by hand. This is the benchmark-governance analog of data-driven robust optimization [48].

Definition 121 (Learned coverage uncertainty set)**.** Let $\mathcal{D}_n$ denote audit data used to estimate the unknown true coverage matrix $\kappa^\star \in [0,1]^{m\times G}$. A random set-valued map $\widehat{\mathcal{U}}_{\kappa,1-\delta}(\mathcal{D}_n)$ is called a $(1-\delta)$ *learned coverage uncertainty set* if

$$\mathbb{P}\big(\kappa^\star \in \widehat{\mathcal{U}}_{\kappa,1-\delta}(\mathcal{D}_n)\big) \geq 1-\delta. \tag{209}$$

The learned robust coverage design problem replaces the fixed set $\mathcal{U}_\kappa$ in theorem 117 by $\widehat{\mathcal{U}}_{\kappa,1-\delta}(\mathcal{D}_n)$.

Theorem 122 (Finite-sample transfer guarantee for learned coverage sets)**.** *Fix targets $\rho_1,\dots,\rho_G$. Let $w(\mathcal{D}_n)$ be any measurable portfolio rule such that, for the realized audit data,*

$$\inf_{\kappa\in\widehat{\mathcal{U}}_{\kappa,1-\delta}(\mathcal{D}_n)} \sum_{i=1}^{m} w_i(\mathcal{D}_n)\kappa_{ig} \geq \rho_g, \qquad g=1,\dots,G. \tag{210}$$

Then, with probability at least $1-\delta$ over the audit data,

$$\sum_{i=1}^{m} w_i(\mathcal{D}_n)\kappa^\star_{ig} \geq \rho_g, \qquad g=1,\dots,G. \tag{211}$$

Thus coverage feasibility against the learned uncertainty set transfers directly into finite-sample coverage feasibility against the true matrix on the confidence event.

Proof. On the event $\{\kappa^\star \in \widehat{\mathcal{U}}_{\kappa,1-\delta}(\mathcal{D}_n)\}$, the true matrix is one admissible candidate inside the inner infimum, so the robust constraint implies the true coverage inequality for every cell g. The event occurs with probability at least $1-\delta$ by definition. □

Proposition 123 (Polyhedral learned sets preserve linear tractability)**.** *Suppose the realized learned uncertainty set can be written as a nonempty polytope*

$$\widehat{\mathcal{U}}_{\kappa,1-\delta}(\mathcal{D}_n) = \{\kappa \in [0,1]^{m\times G} : A\,\mathrm{vec}(\kappa) \leq b\} \tag{212}$$

for some data-dependent matrix A and vector b. Then each robust coverage constraint

$$\inf_{\kappa\in\widehat{\mathcal{U}}_{\kappa,1-\delta}(\mathcal{D}_n)} \sum_{i=1}^{m} w_i\kappa_{ig} \geq \rho_g \tag{213}$$

admits a linear-program dual representation. Consequently, scenario-constrained portfolio design with polyhedral learned coverage sets remains a tractable linear program.

Proof. For fixed g and w, the inner optimization is a linear program in κ. Strong duality for nonempty bounded polytopes converts the robust constraint into linear inequalities in the primal weights and dual variables. □

Remark 124. The theorem gives a transfer guarantee, not a fairy tale. If the learned set has poor finite-sample coverage, or if the audit data are not representative of the taxonomy actually used in deployment, the resulting robust portfolio can still be epistemically crooked. The logic is conditional on honest uncertainty-set construction.

Definition 125 (Dependence-aware sequential learned coverage sets)**.** Let $(\mathcal{G}_n)_{n\geq 1}$ be the filtration generated by a possibly dependent sequence of audit batches. A sequence of random set-valued maps

$$\widehat{\mathcal{U}}_{\kappa,n,1-\delta} = \widehat{\mathcal{U}}_{\kappa,n,1-\delta}(\mathcal{D}_n) \tag{214}$$

is called a *dependence-aware* $(1-\delta)$ *sequential learned coverage set* if each set is $\mathcal{G}_n$-measurable and

$$\mathbb{P}\Big(\forall n \geq 1 :\ \kappa^\star \in \widehat{\mathcal{U}}_{\kappa,n,1-\delta}\Big) \geq 1-\delta. \tag{215}$$

The definition is agnostic about how the guarantee is obtained. It may come from blockwise arguments, cluster-robust calibration, martingale methods, or other dependence-aware constructions. What matters for the downstream design theorem is the uniform containment event itself.

Theorem 126 (Stopping-time transfer under dependent audits)**.** *Fix targets $\rho_1, \dots, \rho_G$. Let τ be any almost surely finite stopping time with respect to $(\mathcal{G}_n)$, and let w_τ be any $\mathcal{G}_\tau$-measurable portfolio rule such that, on the realized audit history,*

$$\inf_{\kappa \in \widehat{\mathcal{U}}_{\kappa,\tau,1-\delta}} \sum_{i=1}^{m} w_{\tau,i}\kappa_{ig} \geq \rho_g, \qquad g = 1, \dots, G. \tag{216}$$

If theorem 125 holds, then

$$\mathbb{P}\Big(\sum_{i=1}^{m} w_{\tau,i}\kappa^\star_{ig} \geq \rho_g \text{ for all } g = 1, \dots, G\Big) \geq 1-\delta. \tag{217}$$

Thus dependence-aware learned uncertainty sets preserve valid robust coverage guarantees even when audit sizes and design choices are selected adaptively.

Proof. On the uniform containment event from theorem 125, the true matrix $\kappa^\star$ belongs to $\widehat{\mathcal{U}}_{\kappa,\tau,1-\delta}$ at the random time τ. Therefore the robust feasibility inequality at time τ also holds with $\kappa = \kappa^\star$, which yields the true coverage constraints. The containment event has probability at least $1-\delta$. □

Proposition 127 (Polyhedral sequential learned sets preserve tractability)**.** *Suppose that for every n the realized set $\widehat{\mathcal{U}}_{\kappa,n,1-\delta}$ is almost surely a nonempty polytope of the form*

$$\widehat{\mathcal{U}}_{\kappa,n,1-\delta} = \{\kappa \in [0,1]^{m\times G} : A_n \operatorname{vec}(\kappa) \leq b_n\}, \tag{218}$$

with A_n and b_n measurable with respect to $\mathcal{G}_n$. Then at each monitoring epoch n, and in particular at any stopping time τ, the robust coverage design problem against $\widehat{\mathcal{U}}_{\kappa,n,1-\delta}$ remains a linear program after dualization.

Proof. Condition on the realized audit history. The inner minimization over κ is then a deterministic linear program over a nonempty bounded polytope, so strong duality converts each robust constraint into linear inequalities in the portfolio weights and dual variables. This argument applies pathwise at every deterministic epoch and therefore also at any stopping time. □

Remark 128. The point of the dependence-aware extension is plain: audit logs are often clustered by benchmark family, evaluator, time window, or provenance chain. Treating them as i.i.d. fairy dust is unnecessary and often wrong. What the downstream theory needs is not independence but a set sequence whose coverage claim already respects the dependence actually present.

7.6 Dependent audits under mixing and martingale regimes

The previous subsection treated dependence-aware learned sets abstractly through their uniform containment property. We now give two broad sufficient audit regimes under which that property can be instantiated without pretending the audit rows are independent.

Definition 129 (Mixing and martingale audit regimes)**.** For each coverage coordinate $c = (i, g)$, let $Y_{n,c} \in [0, 1]$ denote the audit contribution at audit epoch n, with target mean $\kappa_c^\star$. Consider either of the following regimes.

1. **Geometric β-mixing regime.** The process $(Y_{n,c})_{n\geq 1}$ is strictly stationary with $\mathbb{E}[Y_{n,c}] = \kappa_c^\star$ and β-mixing coefficients satisfying
$$\beta_c(k) \leq \bar{\beta}_c \rho_c^k, \qquad k \geq 1, \tag{219}$$
for some constants $\bar{\beta}_c < \infty$ and $\rho_c \in (0, 1)$.

2. **Bounded martingale-noise regime.** There exists an adapted process $(\xi_{n,c})_{n\geq 1}$ such that
$$Y_{n,c} = \kappa_c^\star + \xi_{n,c}, \qquad \mathbb{E}[\xi_{n,c} \mid \mathcal{G}_{n-1}] = 0, \qquad |\xi_{n,c}| \leq b_c \text{ a.s.}, \tag{220}$$
and the predictable quadratic variation
$$V_{n,c} := \sum_{t=1}^{n} \mathbb{E}[\xi_{t,c}^2 \mid \mathcal{G}_{t-1}] \tag{221}$$
is almost surely finite for every n.

Theorem 130 (Mixing- and martingale-based construction of sequential learned coverage sets)**.** *Assume theorem 129. For each coordinate $c = (i, g)$ choose an error budget $\delta_c > 0$ with $\sum_c \delta_c \leq \delta$. Suppose that under the actually declared regime for coordinate c one constructs an adapted confidence sequence*
$$[L_{n,c}, U_{n,c}]_{n\geq 1} \tag{222}$$
for $\kappa_c^\star$ satisfying
$$\mathbb{P}\Big(\forall n \geq 1 : \ \kappa_c^\star \in [L_{n,c}, U_{n,c}]\Big) \geq 1 - \delta_c. \tag{223}$$
Define the random set
$$\widehat{\mathcal{U}}^{\mathrm{dep}}_{\kappa,n,1-\delta} := \Big\{\kappa \in [0, 1]^{m\times G} : \ L_{n,c} \leq \kappa_c \leq U_{n,c} \text{ for every coordinate } c = (i, g)\Big\}. \tag{224}$$

Then

$$\mathbb{P}\Big(\forall n \geq 1 :\ \kappa^\star \in \widehat{\mathcal{U}}^{\mathrm{dep}}_{\kappa,n,1-\delta}\Big) \geq 1-\delta, \tag{225}$$

so $\widehat{\mathcal{U}}^{\mathrm{dep}}_{\kappa,n,1-\delta}$ *is a dependence-aware* $(1-\delta)$ *sequential learned coverage set in the sense of theorem 125.*

Proof. For each coordinate c, let E_c denote the event that its confidence sequence covers $\kappa^\star_c$ for all times. By assumption, $\mathbb{P}(E_c) \geq 1-\delta_c$. On the intersection $\cap_c E_c$, every coordinate of $\kappa^\star$ lies in its corresponding interval for all times, so $\kappa^\star \in \widehat{\mathcal{U}}^{\mathrm{dep}}_{\kappa,n,1-\delta}$ for all n. A union bound yields

$$\mathbb{P}\Big(\bigcap_c E_c\Big) \geq 1 - \sum_c \delta_c \geq 1-\delta.$$

□

Remark 131. The theorem is deliberately modular. Under geometric mixing one may use blockwise confidence sequences whose validity depends on the declared decay rate of the mixing coefficients, while under the martingale regime one may use Freedman-style or nonnegative-supermartingale confidence sequences driven by the predictable variance process [51, 52, 49]. The theory here does not privilege one construction; it only insists that the downstream portfolio guarantee inherit whatever dependence-aware validity the chosen construction actually has.

Corollary 132 (Adaptive portfolio updates remain valid under declared mixing or martingale regimes)**.** *Under the conditions of theorem 130, every stopping-time portfolio rule that is robust feasible against* $\widehat{\mathcal{U}}^{\mathrm{dep}}_{\kappa,n,1-\delta}$ *inherits the stopping-time transfer guarantee of theorem 126.*

Proof. The claim is immediate because theorem 130 verifies the containment property required by theorem 126. □

7.7 Nonstationary and regime-switching audits

The previous subsection covered stationary mixing and martingale regimes. Real audit programs can also be nonstationary because benchmark taxonomies, evaluator pools, provenance channels, or benchmark families move between distinct regimes. The safe object is then a time-indexed or regime-indexed uncertainty set rather than a single stationary one.

Definition 133 (Observed-regime and posterior-mixture audit models)**.** For each coverage coordinate $c = (i, g)$ and regime $r \in \{1, \ldots, R\}$, let $\kappa^\star_{c,r} \in [0,1]$ denote the regime-r target mean.

1. **Observed-regime model.** An adapted process $(R_n)_{n\geq 1}$ taking values in $\{1, \ldots, R\}$ is observed at each audit epoch, and the current target is

$$\kappa^\star_{n,c} = \kappa^\star_{c,R_n}. \tag{226}$$

2. **Posterior-mixture model.** The regime is latent, but posterior weights

$$p_{n,r} := \mathbb{P}(R_n = r \mid \mathcal{G}_n), \qquad r = 1, \ldots, R, \tag{227}$$

are available from a declared regime filter. The current target is the conditional mean

$$\kappa^\star_{n,c} = \sum_{r=1}^{R} p_{n,r}\kappa^\star_{c,r}. \tag{228}$$

For every coordinate c and regime r, let $[L^{(r)}_{n,c}, U^{(r)}_{n,c}]_{n\geq 1}$ be an adapted confidence sequence for $\kappa^\star_{c,r}$ satisfying

$$\mathbb{P}\Big(\forall n \geq 1 :\ \kappa^\star_{c,r} \in [L^{(r)}_{n,c}, U^{(r)}_{n,c}]\Big) \geq 1 - \delta_{c,r}, \tag{229}$$

with $\sum_{c,r} \delta_{c,r} \leq \delta$.

Theorem 134 (Sequential learned sets for regime-switching audits)**.** *Assume theorem 133. In the observed-regime model define*

$$\widehat{\mathcal{U}}^{\mathrm{reg}}_{\kappa,n,1-\delta} := \Big\{\kappa \in [0,1]^{m\times G} :\ L^{(R_n)}_{n,c} \leq \kappa_c \leq U^{(R_n)}_{n,c} \text{ for every coordinate } c\Big\}. \tag{230}$$

In the posterior-mixture model define

$$\widehat{\mathcal{U}}^{\mathrm{mix}}_{\kappa,n,1-\delta} := \Big\{\kappa \in [0,1]^{m\times G} :\ \kappa = \sum_{r=1}^{R} p_{n,r}\kappa^{(r)},\ \kappa^{(r)}_c \in [L^{(r)}_{n,c}, U^{(r)}_{n,c}] \text{ for all } c, r\Big\}. \tag{231}$$

Then

$$\mathbb{P}\Big(\forall n \geq 1 :\ \kappa^\star_n \in \widehat{\mathcal{U}}^{\mathrm{reg}}_{\kappa,n,1-\delta}\Big) \geq 1 - \delta \tag{232}$$

under the observed-regime model, and

$$\mathbb{P}\Big(\forall n \geq 1 :\ \kappa^\star_n \in \widehat{\mathcal{U}}^{\mathrm{mix}}_{\kappa,n,1-\delta}\Big) \geq 1 - \delta \tag{233}$$

under the posterior-mixture model.

Proof. Let $E_{c,r}$ denote the event that the confidence sequence for coordinate c and regime r covers $\kappa^\star_{c,r}$ for all times. By a union bound, the event $E := \cap_{c,r} E_{c,r}$ has probability at least $1-\delta$. On E, the observed-regime claim is immediate because whenever $R_n = r$, one has $\kappa^\star_{n,c} = \kappa^\star_{c,r} \in [L^{(r)}_{n,c}, U^{(r)}_{n,c}]$ for every coordinate. For the posterior-mixture model, on the same event each regime vector $\kappa^{\star,(r)}$ lies in its regime-specific box for all times, so the convex combination $\kappa^\star_n = \sum_r p_{n,r}\kappa^{\star,(r)}$ belongs to $\widehat{\mathcal{U}}^{\mathrm{mix}}_{\kappa,n,1-\delta}$ for every n. □

Corollary 135 (Safe stopping-time portfolio updates under regime-switching audits)**.** *Any stopping-time portfolio rule that is robust feasible against $\widehat{\mathcal{U}}^{\mathrm{reg}}_{\kappa,n,1-\delta}$ or $\widehat{\mathcal{U}}^{\mathrm{mix}}_{\kappa,n,1-\delta}$ inherits the same stopping-time transfer logic as in theorem 126.*

Proof. On the corresponding $1-\delta$ containment event from theorem 134, the true current coverage vector lies in the learned set at every time. Pathwise robust feasibility then implies pathwise satisfaction of the portfolio constraints, and the probability statement follows directly. □

Remark 136. This result is intentionally fail-closed. It does not claim that arbitrary adversarial drift is harmless. It says only that if one declares either observable regimes or a posterior-weighted regime model and constructs regime-specific confidence sequences honestly, then the downstream coverage guarantee can track that nonstationarity without pretending the audit world is stationary.

8 Sequential Monitoring and Safe Control

8.1 Time-to-invalidity beyond half-life

Half-life is useful, but it hard-codes the threshold $\alpha = 1/2$ and ignores the distributional shape of failure times under scenario uncertainty. Operational governance often cares instead about the first time a benchmark or portfolio becomes invalid relative to a task-specific threshold.

Definition 137 (Threshold invalidity time and risk measures)**.** Fix a threshold $\alpha \in (0,1)$ and a portfolio $w \in \Delta_m$. Under scenario s, define the threshold invalidity time

$$T_\alpha^{(s)}(w) := \inf\{t \geq 0 : \ \widetilde{V}_{\mathcal{P}_w}^{(s)}(t) \leq \alpha\}, \tag{234}$$

with the convention $\inf \emptyset = \infty$. If scenarios carry probabilities $p_s > 0$ summing to one, define the lower $(1-\beta)$ safe invalidity quantile by

$$Q_{1-\beta,\alpha}^{\text{safe}}(w) := \sup\{t \geq 0 : \ \sum_{s=1}^{S} p_s \, \mathbf{1}\{T_\alpha^{(s)}(w) \geq t\} \geq 1-\beta\}, \tag{235}$$

and, for a target horizon $H > 0$, define the invalidity shortfall loss

$$L_{H,\alpha}^{(s)}(w) := \left(H - T_\alpha^{(s)}(w)\right)_+. \tag{236}$$

The corresponding tail-risk summary is $\mathrm{CVaR}_\beta(L_{H,\alpha}(w))$ under the scenario law.

Proposition 138 (Scenariowise survival dominance improves invalidity-time risk)**.** *Fix $\alpha \in (0,1)$. If two portfolios w and w' satisfy*

$$\widetilde{V}_{\mathcal{P}_w}^{(s)}(t) \geq \widetilde{V}_{\mathcal{P}_{w'}}^{(s)}(t) \qquad \text{for all } t \geq 0 \text{ and every scenario } s, \tag{237}$$

then $T_\alpha^{(s)}(w) \geq T_\alpha^{(s)}(w')$ for every s. Consequently,

$$Q_{1-\beta,\alpha}^{\text{safe}}(w) \geq Q_{1-\beta,\alpha}^{\text{safe}}(w') \tag{238}$$

and

$$\mathrm{CVaR}_\beta(L_{H,\alpha}(w)) \leq \mathrm{CVaR}_\beta(L_{H,\alpha}(w')) \tag{239}$$

for every confidence level $\beta \in (0,1)$ and horizon $H > 0$.

Proof. Pointwise survival dominance implies that the crossing time below level α cannot occur earlier for w than for w' in any scenario, yielding the first claim. The safe quantile is monotone under first-order stochastic dominance of T_α, and the shortfall loss $L_{H,\alpha}^{(s)}(w)$ is pointwise smaller than $L_{H,\alpha}^{(s)}(w')$ in every scenario. Monotonicity of CVaR with respect to almost-sure dominance then gives the tail-risk inequality. □

Corollary 139 (Threshold-generalized robust survival certification)**.** *Fix a horizon $T > 0$ and threshold $\alpha \in (0,1)$. Suppose the declared scenario survival profiles are nonincreasing on $[0,T]$. If a portfolio w satisfies*

$$\widetilde{V}_{\mathcal{P}_w}^{(s)}(T) > \alpha \qquad \text{for every declared scenario } s, \tag{240}$$

then $T_\alpha^{(s)}(w) > T$ *for every scenario. In particular, the certificate in theorem 107 is the special case* $\alpha = \frac{1}{2}$.

Proof. If the survival curve is nonincreasing and remains above α at time T in every scenario, then it remained above α at every earlier time in that horizon. Therefore the first crossing time below α must be later than T in every scenario. □

Remark 140. The point of these risk measures is not to make finance cosplay out of benchmark governance. It is to stop pretending that one threshold at one confidence level tells the whole story. A benchmark family can have the same half-life as another while exhibiting far worse lower-tail invalidity risk at stricter thresholds or longer horizons.

8.2 Sequential monitoring and stopping rules

Static lifetime summaries are not enough once benchmark governance becomes a live process. Audits, anchors, and contamination evidence arrive over time, and people will peek whenever they feel like it. The correct safety object is therefore an anytime-valid envelope rather than a one-shot interval [49, 50].

Definition 141 (Anytime-valid validity envelope)**.** Let Z denote either a single benchmark B or a benchmark portfolio $\mathcal{P}_w$, and fix a thresholded validity object $\widetilde{V}_Z(t)$ of interest. An adapted pair of observable processes $(\underline{V}_t, \overline{V}_t)_{t\geq 0}$ is called a $(1-\delta)$ *anytime-valid validity envelope* if

$$\mathbb{P}\Big(\forall t \geq 0 :\ \underline{V}_t \leq \widetilde{V}_Z(t) \leq \overline{V}_t\Big) \geq 1 - \delta. \tag{241}$$

Proposition 142 (Epoch-sampled envelope transfer)**.** *Let* $(t_n)_{n=0}^H$ *be any increasing sequence of monitoring epochs. If* $(\underline{V}_t, \overline{V}_t)_{t\geq 0}$ *is an anytime-valid envelope in the sense of theorem 141, then the sampled process* $(\underline{V}_{t_n}, \overline{V}_{t_n})_{n=0}^H$ *is a discrete-time* $(1-\delta)$ *envelope for* $(\widetilde{V}_Z(t_n))_{n=0}^H$*. If deployments occur only at the epochs* t_n *and the benchmark is held fixed between epochs except for declared regeneration or retirement actions, then no additional discretization error enters the deployment-safety statements.*

Proof. The sampling claim is immediate because the event $\{\forall t \geq 0 : \underline{V}_t \leq \widetilde{V}_Z(t) \leq \overline{V}_t\}$ implies the same inequalities at every sampled epoch. If deployment decisions are taken only at those epochs, the control layer uses only the sampled envelope and therefore inherits the same coverage event without an extra approximation term. □

Theorem 143 (Safe retirement by upper-envelope crossing)**.** *Fix a governance threshold* $\alpha \in (0,1)$ *and define the stopping time*

$$\tau_{\alpha,\delta}^{\mathrm{ret}} := \inf\{t \geq 0 :\ \overline{V}_t \leq \alpha\}. \tag{242}$$

If $(\underline{V}_t, \overline{V}_t)$ *is a* $(1-\delta)$ *anytime-valid validity envelope for* $\widetilde{V}_Z(t)$*, then*

$$\mathbb{P}\big(\tau_{\alpha,\delta}^{\mathrm{ret}} < \infty,\ \widetilde{V}_Z(\tau_{\alpha,\delta}^{\mathrm{ret}}) > \alpha\big) \leq \delta. \tag{243}$$

If, in addition, $\widetilde{V}_Z(t)$ *is nonincreasing, then on the envelope-coverage event the threshold invalidity time satisfies*

$$T_\alpha(Z) \leq \tau_{\alpha,\delta}^{\mathrm{ret}}. \tag{244}$$

Thus the sequential retirement time is an upper confidence bound on the true invalidity time.

Proof. On the event from theorem 141, we have $\widetilde{V}_Z(t) \le \overline{V}_t$ for all t. Therefore if $\tau^{\mathrm{ret}}_{\alpha,\delta} < \infty$ then

$$\widetilde{V}_Z(\tau^{\mathrm{ret}}_{\alpha,\delta}) \le \overline{V}_{\tau^{\mathrm{ret}}_{\alpha,\delta}} \le \alpha,$$

so false retirement can occur only on the complement of the envelope-coverage event, whose probability is at most δ. If $\widetilde{V}_Z$ is nonincreasing, then once $\widetilde{V}_Z(\tau^{\mathrm{ret}}_{\alpha,\delta}) \le \alpha$, the first crossing time below α must have occurred no later than $\tau^{\mathrm{ret}}_{\alpha,\delta}$. □

Proposition 144 (Safe continuation by lower-envelope certification)**.** *Under the same setup, let σ be any stopping time and additionally assume that $\widetilde{V}_Z(t)$ is nonincreasing. Then*

$$\mathbb{P}(\underline{V}_\sigma > \alpha \text{ and } T_\alpha(Z) \le \sigma) \le \delta. \tag{245}$$

Equivalently, whenever a reported lower envelope satisfies $\underline{V}_\sigma > \alpha$, one may continue using the benchmark or portfolio through time σ with error probability at most δ.

Proof. On the coverage event from theorem 141, the implication $\underline{V}_\sigma > \alpha \Rightarrow \widetilde{V}_Z(\sigma) > \alpha$ holds for every stopping time σ. If additionally $T_\alpha(Z) \le \sigma$ and $\widetilde{V}_Z$ is nonincreasing, then $\widetilde{V}_Z(\sigma) \le \alpha$, a contradiction. Hence the bad event implies coverage failure, whose probability is at most δ. □

Definition 145 (Operational life-cycle status)**.** Fix a governance threshold $\alpha \in (0,1)$ and an anytime-valid envelope $(\underline{V}_n, \overline{V}_n)$. At monitoring epoch n, classify the current benchmark or portfolio as

- *active* if $\underline{V}_n > \alpha$,
- *watchlist* if $\underline{V}_n \le \alpha < \overline{V}_n$,
- *regenerate-or-replace* if $\overline{V}_n \le \alpha$ and at least one admissible repair action exists,
- *retire-or-scope-restrict* if $\overline{V}_n \le \alpha$ and no admissible repair action preserves the current operational class.

The last category includes frontier-only failure cases in which the benchmark should remain alive only after an explicit restriction of the operational class.

Remark 146. These statements are intentionally generic. The paper does not derive a universal confidence sequence for benchmark validity from raw samples because that depends on the estimator class and dependence structure. What it does show is how any valid online envelope can be converted into governance actions without being invalidated by continuous monitoring or optional stopping.

8.3 Sequential retirement, regeneration, and safe dynamic control

Sequential monitoring becomes operational only when it is coupled to actions. A live benchmark program may continue deploying the current instrument, regenerate or replace it, or retire it outright. The decision problem is sequential and partially observed, but the safety constraint is simple: deployment decisions must respect the anytime-valid envelope rather than a convenient point estimate.

Definition 147 (Envelope-admissible policy)**.** Fix a governance threshold $\alpha \in (0,1)$ and monitoring epochs $n = 0, 1, \ldots, H-1$. Let I_n be any $\mathcal{G}_n$-measurable information state that includes the current anytime-valid envelope bounds $(\underline{V}_n, \overline{V}_n)$ for the deployed benchmark or benchmark portfolio, and let

$\mathcal{R}(I_n)$ denote the feasible regeneration or replacement actions at epoch n. A policy π is called *envelope admissible* if it satisfies two rules at every epoch:

1. it deploys the current benchmark for another epoch only when $\underline{V}_n > \alpha$;
2. whenever $\overline{V}_n \le \alpha$, it does not carry the current benchmark forward without either retiring it or applying a regeneration/replacement action before any further deployment.

A regeneration or replacement action is not counted as continued operational use of the current benchmark; it changes the instrument before the next deployment epoch.

Theorem 148 (Envelope-admissible policies are δ-safe)**.** *Suppose $(\underline{V}_n, \overline{V}_n)_{n=0}^{H-1}$ is a discrete-time $(1-\delta)$ anytime-valid validity envelope for the deployed benchmark or portfolio at the monitoring epochs. Let π be any envelope-admissible policy in the sense of theorem 147. Then*

$$\mathbb{P}\Big(\exists n \in \{0,\dots,H-1\} :\ \pi \text{ deploys the current benchmark at epoch } n \text{ and } \widetilde{V}_n \le \alpha\Big) \le \delta. \tag{246}$$

If the deployed benchmark's validity is nonincreasing between regeneration or replacement actions, then on the envelope-coverage event no envelope-admissible policy retains the current benchmark past its true invalidity time without either retiring it or replacing it.

Proof. On the envelope-coverage event, $\underline{V}_n \le \widetilde{V}_n \le \overline{V}_n$ holds for every epoch n. If the policy deploys the current benchmark at epoch n, envelope admissibility requires $\underline{V}_n > \alpha$, which on the coverage event implies $\widetilde{V}_n > \alpha$. Hence unsafe deployment can occur only if the coverage event fails, which has probability at most δ. For the second claim, if validity is nonincreasing between regeneration or replacement actions and $\overline{V}_n \le \alpha$, then on the coverage event the current benchmark is already invalid at epoch n, so envelope admissibility forbids carrying that instrument forward without retirement or replacement. □

Definition 149 (Safe control objective)**.** Fix a finite horizon H. Let $r_n(I,a)$ be a bounded immediate reward for action a taken in information state I at epoch n, where regeneration or replacement costs are encoded as negative reward. Let $P_n(\cdot \mid I, a)$ be the controlled transition kernel on next-period information states. Among envelope-admissible policies, define the value of a policy by

$$J^\pi(I_0) := \mathbb{E}^\pi\Big[\sum_{n=0}^{H-1} r_n(I_n, a_n) + g(I_H)\,\Big|\, I_0\Big], \tag{247}$$

where g is a bounded terminal value function.

Theorem 150 (Bellman recursion for optimal safe retirement–regeneration control)**.** *Under theorem 149, the optimal value among envelope-admissible policies satisfies the dynamic program*

$$W_H(I) = g(I), \tag{248}$$

$$W_n(I) = \max_{a \in \mathcal{A}_\alpha^{\mathrm{safe}}(I)} \big\{ r_n(I,a) + \mathbb{E}\big[W_{n+1}(I_{n+1}) \mid I_n = I, a_n = a\big] \big\}, \qquad n = H-1,\dots,0, \tag{249}$$

where the safe action set is

$$\mathcal{A}_\alpha^{\mathrm{safe}}(I) := \begin{cases} \{\text{continue}, \text{retire}\} \cup \mathcal{R}(I), & \underline{V}(I) > \alpha, \\ \{\text{retire}\} \cup \mathcal{R}(I), & \underline{V}(I) \le \alpha. \end{cases} \tag{250}$$

Consequently, optimal retirement and regeneration planning can be carried out by standard dynamic programming after restricting the action set to the envelope-admissible region.

Proof. The admissible action set depends only on the current information state, so the problem is a finite-horizon Markov control problem on the information state space after unsafe actions are removed. Backward induction over the bounded reward-to-go yields the displayed Bellman recursion, and the standard principle of optimality applies because the future problem after any admissible action is again a safe control problem of the same form. □

Definition 151 (Ambiguity-set safe control)**.** Under the safe-control setup of theorem 149, suppose that for each epoch n, information state I, and admissible action a, the next-state transition law is only known to lie in a nonempty ambiguity set $\mathfrak{P}_n(I,a)$ of probability kernels on information states. The robust safe value of a policy is

$$J^{\pi}_{\mathrm{rob}}(I_0) := \inf_{P\in\mathfrak{P}(\pi)} \mathbb{E}^{\pi}_{P}\Big[\sum_{n=0}^{H-1} r_n(I_n,a_n) + g(I_H)\,\Big|\, I_0\Big], \tag{251}$$

where $\mathfrak{P}(\pi)$ is the set of rectangular kernel selections consistent with the per-stage ambiguity sets and policy π.

Theorem 152 (Robust Bellman recursion for ambiguity-set safe control)**.** *Under theorem 151, assume the ambiguity sets are rectangular across stages and states. Then the optimal robust value among envelope-admissible policies satisfies*

$$W^{\mathrm{rob}}_H(I) = g(I), \tag{252}$$

$$W^{\mathrm{rob}}_n(I) = \max_{a\in\mathcal{A}^{\mathrm{safe}}_\alpha(I)} \left\{ r_n(I,a) + \inf_{P\in\mathfrak{P}_n(I,a)} \mathbb{E}_P\big[W^{\mathrm{rob}}_{n+1}(I_{n+1}) \mid I_n = I\big] \right\}, \qquad n = H-1,\dots,0. \tag{253}$$

Thus transition uncertainty can be handled by max–min dynamic programming without relaxing the explicit envelope-based safety filter.

Proof. After trimming the unsafe actions, the remaining problem is a finite-horizon rectangular robust Markov decision problem on the information state. Rectangularity lets the adversary choose the worst admissible kernel stagewise and statewise, so the standard robust principle of optimality applies. Backward induction yields the displayed max–min Bellman recursion. □

Proposition 153 (Plug-in control is weakly optimistic relative to robust safe control)**.** *Let $\widehat{P}_n(\cdot \mid I,a) \in \mathfrak{P}_n(I,a)$ be any nominated plug-in kernel for every safe state-action pair. Then*

$$W^{\mathrm{rob}}_n(I) \le W^{\mathrm{plug}}_n(I) \tag{254}$$

for every epoch n and reachable safe information state I, where W^{plug} denotes the value obtained by solving the nominal safe Bellman recursion with $\widehat{P}_n$ in place of the ambiguity set. Hence the nominal planner is a weakly optimistic special case of the robust safe planner.

Proof. For every fixed action, the robust continuation value takes an infimum over a set containing the nominated plug-in kernel. Therefore the robust continuation value is no larger than the plug-in continuation value. Maximizing over the same safe action set preserves the inequality. □

Proposition 154 (Finite-sample transition ambiguity from empirical confidence sets)**.** *Suppose the safe-control state and action spaces have been reduced to a finite abstraction and that for each safe state-action pair (I, a) one observes $n_{I,a}$ transitions with empirical next-state distribution $\widehat{P}_n(\cdot \mid I, a)$. Let $\beta_{I,a}(\delta) \geq 0$ be any radius such that*

$$\mathbb{P}\Big(\|P^\star(\cdot \mid I, a) - \widehat{P}_n(\cdot \mid I, a)\|_1 \leq \beta_{I,a}(\delta) \text{ for all safe } (I, a)\Big) \geq 1 - \delta. \tag{255}$$

Then the ambiguity sets

$$\mathfrak{P}_n(I, a) := \Big\{P : \ \|P(\cdot \mid I, a) - \widehat{P}_n(\cdot \mid I, a)\|_1 \leq \beta_{I,a}(\delta)\Big\} \tag{256}$$

constitute a finite-sample $(1-\delta)$ transition-coverage certificate for the robust Bellman recursion. The same logic applies to rectangular confidence sets built from multinomial exact intervals, empirical-Bernstein bounds, or Bayesian credible sets with validated frequentist coverage.

Proof. On the stated high-probability event, the true transition law belongs to every reported ambiguity set simultaneously. The robust Bellman recursion of theorem 152 is then evaluated over a class that contains the true kernel, which is exactly the coverage condition needed downstream. □

Proposition 155 (One-step dominance of regeneration actions)**.** *Assume theorem 14. Let $G : \mathcal{S} \to \mathbb{R}$ be any post-regeneration utility that is nonincreasing in $S, J, K, 1-F, 1-U$ and nondecreasing in N. Consider two feasible regeneration actions r and r' from the same benchmark-ecology state X, with costs $c(r)$ and $c(r')$, producing post-action states X^r and $X^{r'}$. If, almost surely,*

$$S^r \leq S^{r'}, \quad N^r \geq N^{r'}, \quad F^r \geq F^{r'}, \quad U^r \geq U^{r'}, \quad J^r \leq J^{r'}, \quad K^r \leq K^{r'}, \tag{257}$$

and $c(r) \leq c(r')$, then

$$\mathbb{E}[G(X^r)] - c(r) \geq \mathbb{E}[G(X^{r'})] - c(r'). \tag{258}$$

Hence r' is weakly dominated in any myopic regeneration step whose objective respects the benchmark-quality ordering induced by theorem 14.

Proof. Coordinatewise dominance of X^r over $X^{r'}$ in the monotone order implied by theorem 14 yields $G(X^r) \geq G(X^{r'})$ almost surely. Taking expectations and subtracting the weakly smaller cost of r gives the claim. □

8.4 Safe control under partial observability

The previous control results treated the information state as if it were already the correct control state. In practice the underlying benchmark ecology is only partially observed through audits, anchor outcomes, contamination diagnostics, and regeneration logs. The natural control object is therefore a belief state.

Definition 156 (Safe partially observed benchmark control)**.** Let $X_n \in \mathcal{X}$ denote the latent benchmark-ecology state at epoch n, let O_{n+1} denote the next observation, and let H_n denote the observed history. Two control-state conventions are admissible.

1. *Exact-belief convention.* Under a declared generative model, define the Bayesian belief state

$$b_n(\cdot) := \mathbb{P}(X_n \in \cdot \mid H_n), \tag{259}$$

with exact update

$$b_{n+1} = \Phi_n(b_n, a_n, O_{n+1}). \tag{260}$$

2. *Declared information-state convention.* Let $I_n^{\mathrm{P}} := \Psi_n(H_n)$ be any measurable summary updated by

$$I_{n+1}^{\mathrm{P}} = \widetilde{\Phi}_n(I_n^{\mathrm{P}}, a_n, O_{n+1}). \tag{261}$$

The declared state is called *control sufficient* if the conditional law of future observations, rewards, and the reported lower-validity functional given history depends on the past only through (I_n^{P}, a_n).

Let Z_n denote whichever control state is being used, either b_n or I_n^{P}. Let $\underline{\mathcal{V}}(Z_n)$ be any measurable lower-validity functional such that on the declared coverage event,

$$\underline{\mathcal{V}}(Z_n) \le \widetilde{V}_n \qquad \text{for every epoch } n, \tag{262}$$

where $\widetilde{V}_n$ is the true currently deployed validity object. Define the safe action set

$$\mathcal{A}_\alpha^{\mathrm{safe,P}}(Z) := \begin{cases} \{\text{continue}, \text{retire}\} \cup \mathcal{R}(Z), & \underline{\mathcal{V}}(Z) > \alpha, \\ \{\text{retire}\} \cup \mathcal{R}(Z), & \underline{\mathcal{V}}(Z) \le \alpha. \end{cases} \tag{263}$$

If $Z = b$ is the exact belief state and $r_n(x, a)$ is the bounded latent-state reward, define its belief average by

$$\bar{r}_n(b, a) := \int r_n(x, a)\, b(\mathrm{d}x). \tag{264}$$

More generally, for a control-sufficient declared information state let $\bar{r}_n(Z, a)$ denote the corresponding reported one-step reward functional.

Theorem 157 (Belief-state or control-sufficient-state Bellman recursion for safe POMDP control)**.** *Under theorem 156, suppose either that $Z_n = b_n$ is the exact Bayesian belief state under the declared generative model, or that $Z_n = I_n^{\mathrm{P}}$ is a declared information state that is control sufficient. Then the finite-horizon optimal value among control-state-admissible policies satisfies*

$$W_H^{\mathrm{P}}(Z) = g(Z), \tag{265}$$

$$W_n^{\mathrm{P}}(Z) = \max_{a \in \mathcal{A}_\alpha^{\mathrm{safe,P}}(Z)} \left\{ \bar{r}_n(Z, a) + \mathbb{E}\big[W_{n+1}^{\mathrm{P}}(Z_{n+1}) \mid Z_n = Z, a_n = a\big] \right\}, \qquad n = H-1, \dots, 0. \tag{266}$$

Hence safe benchmark control under partial observability is an exact dynamic program on the true belief state, and more generally on any declared information state only when that state is genuinely control sufficient.

Proof. In the exact-belief case, the usual POMDP sufficiency property implies that conditional on the current belief and action, the law of future observations and rewards does not depend on the earlier history. In the declared-state case, this same property is assumed as control sufficiency. After restricting the feasible action set to $\mathcal{A}_\alpha^{\mathrm{safe,P}}(Z)$, the problem is therefore a finite-horizon Markov control problem on the state Z_n, and backward induction yields the displayed Bellman recursion. □

Corollary 158 (Safety of control-state-admissible deployment)**.** *If* $(\underline{\mathcal{V}}(Z_n))_{n=0}^{H-1}$ *is a* $(1-\delta)$ *anytime-valid lower bound sequence for the deployed validity object, then any control-state-admissible policy satisfies*

$$\mathbb{P}\Big(\exists n \in \{0,\dots,H-1\} : \text{ the policy deploys at epoch } n \text{ and } \widetilde{V}_n \le \alpha\Big) \le \delta. \tag{267}$$

Proof. On the validity-envelope event, admissibility implies $\underline{\mathcal{V}}(Z_n) > \alpha$ whenever the policy continues deployment. By assumption this yields $\widetilde{V}_n > \alpha$ for every deployed epoch. Unsafe deployment is therefore possible only if the anytime-valid lower-bound event fails, which has probability at most δ. □

Definition 159 (Robust safe partially observed benchmark control)**.** Under the setup of theorem 156, let Z denote the exact belief state or a declared control-sufficient information state. Suppose that for each epoch n, control state Z, and safe action a, the relevant joint transition–observation law and induced state-update rule are only known to lie in a nonempty ambiguity set $\mathfrak{Q}_n(Z,a)$. Let $\bar{r}_n(Z,a)$ be the reported one-step reward functional for the chosen control-state convention. The robust safe value of a control-state-admissible policy π is

$$J^{\pi}_{\mathrm{RP}}(Z_0) := \inf_{Q\in\mathfrak{Q}(\pi)} \mathbb{E}^{\pi}_{Q}\Big[\sum_{n=0}^{H-1} \bar{r}_n(Z_n,a_n) + g(Z_H)\,\Big|\, Z_0\Big], \tag{268}$$

where $\mathfrak{Q}(\pi)$ is the set of stagewise kernel selections consistent with the reported ambiguity sets and policy π.

Theorem 160 (Rectangular robust Bellman recursion for safe POMDP control)**.** *Under theorem 159, assume the ambiguity sets are stagewise rectangular and that each admissible model* $Q \in \mathfrak{Q}_n(Z,a)$ *induces a measurable next-state law on the chosen control-state convention. Suppose also that the convention is exact-belief or control sufficient in the sense of theorem 156. Then the optimal robust value among control-state-admissible policies satisfies*

$$W^{\mathrm{RP}}_H(Z) = g(Z), \tag{269}$$

$$W^{\mathrm{RP}}_n(Z) = \max_{a\in\mathcal{A}^{\mathrm{safe,P}}_{\alpha}(Z)} \left\{ \inf_{Q\in\mathfrak{Q}_n(Z,a)} \Big[\bar{r}_n(Z,a) + \mathbb{E}_Q\big[W^{\mathrm{RP}}_{n+1}(Z_{n+1}) \mid Z_n = Z, a_n = a\big]\Big] \right\}, \tag{270}$$

$$n = H-1,\dots,0. \tag{271}$$

Thus partial observability and model ambiguity can be handled jointly through a max–min dynamic program on the true belief state, or more generally on any declared state representation only when that representation is control sufficient.

Proof. After restricting actions to the safe set, the controller faces a finite-horizon partially observed control problem. Exact-belief sufficiency or declared control sufficiency reduces the problem to a Markov control problem on Z_n. Stagewise rectangularity then lets the adversary choose the worst admissible model separately at each reachable control-state/action pair, so the robust principle of optimality applies and backward induction yields the displayed recursion. □

Proposition 161 (Safety and reward optimality separate under partial observability)**.** *In the partially observed setting, the deployment-safety guarantee from theorem 158 requires only that the lower-validity sequence remain anytime valid. Robust value optimality claims, however, additionally require that the true*

joint transition–observation law belong to the reported ambiguity sets used by the planner. Hence model misspecification can invalidate reward guarantees without necessarily invalidating the safety certificate.

Proof. Theorem 158 depends only on the implication that deployment occurs only when the reported lower-validity quantity exceeds the governance threshold on the coverage event. The robust value statements from theorem 160, by contrast, are defined relative to the ambiguity set itself; if the true law falls outside that set, the pessimistic value comparison need not hold. □

Remark 162. Rectangularity is not bookkeeping trivia. Without a stagewise rectangular ambiguity structure, the exact stagewise Bellman separation can fail, and the honest object is a nested minimax or scenario-tree control problem rather than a one-line recursion.

Definition 163 (Rectangular outer hull of a nonrectangular ambiguity family)**.** Let $\widetilde{\Omega}$ be any declared family of admissible joint transition–observation laws for the safe POMDP problem, not assumed rectangular. Fix a declared planning domain $\mathcal{D}_n$ of control states at each epoch n such that every control state reachable under some law in $\widetilde{\Omega}$ belongs to $\mathcal{D}_n$. A *rectangular outer hull* of $\widetilde{\Omega}$ on $(\mathcal{D}_n)$ is any stagewise rectangular family $\mathrm{Rect}(\widetilde{\Omega})$ such that for every (n, Z, a) with $Z \in \mathcal{D}_n$, every one-step conditional kernel generated by some law in $\widetilde{\Omega}$ from (n, Z, a) belongs to the corresponding stagewise set in $\mathrm{Rect}(\widetilde{\Omega})$.

Theorem 164 (Conservative rectangular-hull planning under nonrectangular ambiguity)**.** *Assume the setup of theorem 156. Let $\widetilde{W}_n^{\mathrm{NR}}$ denote the finite-horizon robust value when the adversary ranges over a declared nonrectangular ambiguity family $\widetilde{\Omega}$ on a planning domain $(\mathcal{D}_n)$, and let $W_n^{\square}$ denote the value produced by the rectangular Bellman recursion of theorem 160 using any rectangular outer hull $\mathrm{Rect}(\widetilde{\Omega})$ from theorem 163. Then*

$$W_n^{\square}(Z) \le \widetilde{W}_n^{\mathrm{NR}}(Z) \qquad \text{for every control state } Z \text{ that is reachable under some law in } \widetilde{\Omega} \text{ at epoch } n. \tag{272}$$

Hence rectangular-hull planning is a conservative lower-value surrogate even when exact rectangularity is not justified. The explicit validity-based safety filter remains unchanged because it depends on the lower-validity envelope rather than on dynamic rectangularity.

Proof. Every admissible law in $\widetilde{\Omega}$ uses one-step conditional kernels that, by construction, belong to the stagewise sets in $\mathrm{Rect}(\widetilde{\Omega})$ on the declared domain $(\mathcal{D}_n)$. The rectangular outer hull therefore permits every adversarial law in $\widetilde{\Omega}$ and possibly additional history-inconsistent combinations of one-step kernels. Taking an infimum over the larger rectangular class can only decrease the value. Applying theorem 160 to the outer hull yields the displayed lower bound on the original-family reachable states. Safety follows from the same envelope-admissibility logic as before and does not require equality of the two values. □

Remark 165. The point of the rectangular hull is not metaphysical purity. It is to separate two claims that are often sloppily fused: exact dynamic programming requires rectangularity, whereas conservative planning only needs a reported outer hull. The latter is weaker, less elegant, and often more honest.

Proposition 166 (Rectangular-hull pessimism gap under one-step mismatch budgets)**.** *Under the setup of theorem 164, suppose that for each epoch n there exists $\epsilon_n \ge 0$ such that for every reachable control state Z and safe action a,*

$$\inf_{Q \in \widetilde{\Omega}_n(Z,a)} \mathbb{E}_Q[f(Z_{n+1})] \ge \inf_{Q \in \mathrm{Rect}(\widetilde{\Omega})_n(Z,a)} \mathbb{E}_Q[f(Z_{n+1})] - \epsilon_n \tag{273}$$

for every continuation function f bounded by 1 in sup norm. Then

$$0 \le \widetilde{W}_n^{\mathrm{NR}}(Z) - W_n^{\square}(Z) \le \sum_{t=n}^{H-1} \epsilon_t. \tag{274}$$

Thus the conservatism of rectangular-hull planning is controlled by the cumulative one-step hull mismatch rather than left implicit.

Proof. Apply backward induction. At the terminal stage the values agree. If the one-step pessimism gap is at most ϵ_n for every bounded continuation function and the next-stage gap is bounded by $\sum_{t=n+1}^{H-1} \epsilon_t$, then the stage-$n$ Bellman infimum under the rectangular hull can undershoot the nonrectangular value by at most ϵ_n plus the next-stage gap. Summing the recursion gives the bound. □

8.5 Safe exploration–exploitation for regeneration policy

Learning which regeneration actions work best is itself a sequential problem. Exploration is useful because hazard responses and regeneration effects are rarely known exactly, but exploration without safety control is just a polite word for breaking the instrument on purpose.

Definition 167 (Optimistic safe regeneration on a finite abstraction)**.** Fix an episodic horizon H and a finite abstraction $\mathcal{Z}$ of reachable safe belief-action cells $z = (b, a)$. For episode k and stage h, let $\widehat{Q}_{k,h}(z)$ be a safe estimated action-value and let $\beta_{k,h}(z) \ge 0$ be a confidence bonus such that on a declared confidence event,

$$Q_h^\star(z) \in \big[\widehat{Q}_{k,h}(z) - \beta_{k,h}(z),\ \widehat{Q}_{k,h}(z) + \beta_{k,h}(z)\big] \qquad \text{for every reachable safe cell } z. \tag{275}$$

The optimistic safe policy chooses, at each visited safe belief state, an action in

$$\arg \max_{a \in \mathcal{A}_\alpha^{\mathrm{safe,P}}(b)} \{\widehat{Q}_{k,h}(b, a) + \beta_{k,h}(b, a)\}. \tag{276}$$

Let $\varepsilon_{\mathrm{abs}} \ge 0$ denote any additional per-episode abstraction or planning error relative to the exact safe belief-state problem. Let $\mathfrak{D}_K \ge 0$ denote any declared cumulative nonstationarity budget measuring the total deviation of the true safe action-values from the reference values used to build the bonuses across the first K episodes.

Theorem 168 (Confidence-width regret bound for safe optimistic exploration)**.** *Under theorem 167, let* Reg_K *denote cumulative regret over K episodes relative to the optimal safe policy for the true model. On the confidence event,*

$$\mathrm{Reg}_K \le 2 \sum_{k=1}^{K} \sum_{h=0}^{H-1} \beta_{k,h}(z_{k,h}) + K\varepsilon_{\mathrm{abs}} + \mathfrak{D}_K, \tag{277}$$

where $z_{k,h}$ is the safe belief-action cell visited by the optimistic policy at episode k, stage h.

Proof. Fix a visited safe belief state. By optimism, the optimistic action has estimated upper value at least as large as the upper confidence value of the true optimal safe action. On the confidence event, both the optimistic action and the oracle safe action differ from the reference safe action values by at most the corresponding bonus widths. The additional discrepancy between those reference values and the true possibly drifting values is, by declaration, absorbed by the cumulative nonstationarity budget $\mathfrak{D}_K$. Therefore the instantaneous regret at each visited safe cell is bounded by twice the visited bonus width

plus any declared abstraction or planning error, with the total drift contribution added across episodes. Summing this bound over stages and episodes yields the claim. □

Corollary 169 (Square-root regret under inverse-count safe bonuses)**.** *Assume the finite abstraction $\mathcal{Z}$ and suppose*

$$\beta_{k,h}(z) \le c\sqrt{\frac{\log(1/\delta)}{1 \vee N_{k,h}(z)}} \qquad \text{for every } z \in \mathcal{Z}, \tag{278}$$

where $N_{k,h}(z)$ is the visit count to cell z before episode k, stage h. Then on the confidence event,

$$\mathrm{Reg}_K \le 4c\sqrt{|\mathcal{Z}|HK\log(1/\delta)} + K\varepsilon_{\mathrm{abs}} + \mathfrak{D}_K. \tag{279}$$

Proof. The counting bound $\sum_{t=1}^{N} t^{-1/2} \le 2\sqrt{N}$ applied cellwise implies

$$\sum_{k,h} \beta_{k,h}(z_{k,h}) \le 2c\sqrt{|\mathcal{Z}|HK\log(1/\delta)}. \tag{280}$$

Substituting this into theorem 168 yields the result. □

Remark 170. This theorem is not pretending to solve all of safe reinforcement learning. It isolates the operational moral needed here: once regeneration is learned online, one should report not just that the policy is "adaptive" but the confidence system that keeps exploration inside the safe action set, together with either an episodically frozen-dynamics justification or an explicit drift budget $\mathfrak{D}_K$ for nonstationarity.

8.6 Approximate safe dynamic programming and regret bounds

Exact safe dynamic programming can still be expensive once the information state includes uncertainty envelopes, audit summaries, and portfolio structure. The next results show that approximate solvers can be analyzed without sacrificing the explicit safety layer, provided one reports approximation error or Bellman residuals honestly.

Definition 171 (Approximate safe value recursion)**.** Fix the safe control problem from theorem 149. Let $(\widehat{W}_n)_{n=0}^{H}$ be bounded surrogate value functions on the information state space with $\widehat{W}_H = g$. Define the surrogate-greedy safe policy $\widehat{\pi}$ recursively by choosing, at each epoch n,

$$\widehat{\pi}_n(I) \in \arg\max_{a \in \mathcal{A}_\alpha^{\mathrm{safe}}(I)} \left\{ r_n(I,a) + \mathbb{E}\big[\widehat{W}_{n+1}(I_{n+1}) \mid I_n = I, a_n = a\big] \right\}. \tag{281}$$

For each epoch n, let

$$\varepsilon_n := \sup_{I \in \mathcal{I}_n^{\mathrm{reach}}} \big|\widehat{W}_n(I) - W_n(I)\big| \tag{282}$$

denote the worst-case value approximation error on the set $\mathcal{I}_n^{\mathrm{reach}}$ of reachable safe information states.

Theorem 172 (Finite-horizon regret bound for surrogate-greedy safe control)**.** *Under theorem 171, let $J_n^{\widehat{\pi}}(I)$ denote the true value-to-go from epoch n onward under the surrogate-greedy safe policy $\widehat{\pi}$, and let W_n be the optimal safe value function from theorem 150. Then for every reachable safe state I at epoch n,*

$$W_n(I) - J_n^{\widehat{\pi}}(I) \le 2\sum_{t=n+1}^{H} \varepsilon_t. \tag{283}$$

In particular, if $\widehat{W}_H = g$ exactly, then the initial regret obeys

$$W_0(I_0) - J_0^{\widehat{\pi}}(I_0) \le 2 \sum_{t=1}^{H-1} \varepsilon_t. \tag{284}$$

Proof. Fix epoch $n < H$ and a reachable safe state I. Write

$$\begin{aligned} W_n(I) - J_n^{\widehat{\pi}}(I) = &\max_{a \in \mathcal{A}_\alpha^{\mathrm{safe}}(I)} \Big\{ r_n(I,a) + \mathbb{E}[W_{n+1}(I_{n+1}) \mid I, a] \Big\} \\ &- \Big\{ r_n(I, \widehat{\pi}_n(I)) + \mathbb{E}[J_{n+1}^{\widehat{\pi}}(I_{n+1}) \mid I, \widehat{\pi}_n(I)] \Big\}. \end{aligned}$$

Add and subtract $\widehat{W}_{n+1}$ inside the conditional expectations. Because $\widehat{\pi}_n(I)$ is greedy with respect to $\widehat{W}_{n+1}$ over the safe action set, the resulting surrogate Bellman gap is nonpositive. The remaining two approximation terms are each bounded by ε_{n+1}, yielding

$$W_n(I) - J_n^{\widehat{\pi}}(I) \le 2\varepsilon_{n+1} + \mathbb{E}[W_{n+1}(I_{n+1}) - J_{n+1}^{\widehat{\pi}}(I_{n+1}) \mid I, \widehat{\pi}_n(I)]. \tag{285}$$

Backward induction from $W_H = J_H^{\widehat{\pi}} = g$ proves the displayed bound. □

Proposition 173 (Bellman-residual certificate implies regret bound)**.** *Under theorem 171, define the safe Bellman residuals*

$$\begin{aligned} \eta_n := \sup_{I \in \mathcal{I}_n^{\mathrm{reach}}} \Big| \widehat{W}_n(I) - \max_{a \in \mathcal{A}_\alpha^{\mathrm{safe}}(I)} \Big\{ r_n(I,a) \\ + \mathbb{E}[\widehat{W}_{n+1}(I_{n+1}) \mid I_n = I, a_n = a] \Big\} \Big|, \qquad n = 0, \dots, H-1. \end{aligned} \tag{286}$$

with $\widehat{W}_H = g$. Then

$$\varepsilon_n \le \sum_{t=n}^{H-1} \eta_t, \qquad n = 0, \dots, H-1, \tag{287}$$

and therefore the surrogate-greedy safe policy satisfies

$$W_0(I_0) - J_0^{\widehat{\pi}}(I_0) \le 2 \sum_{t=1}^{H-1} t\, \eta_t. \tag{288}$$

Proof. Let T_n^{safe} denote the safe Bellman operator at epoch n. Then

$$\widehat{W}_n - W_n = (\widehat{W}_n - T_n^{\mathrm{safe}} \widehat{W}_{n+1}) + T_n^{\mathrm{safe}}(\widehat{W}_{n+1} - W_{n+1}). \tag{289}$$

The safe Bellman operator is nonexpansive in the sup norm because it is a maximum of conditional expectation operators. Hence

$$\varepsilon_n \le \eta_n + \varepsilon_{n+1}. \tag{290}$$

Backward induction with $\varepsilon_H = 0$ yields $\varepsilon_n \le \sum_{t=n}^{H-1} \eta_t$. Substituting this into theorem 172 gives

$$W_0(I_0) - J_0^{\widehat{\pi}}(I_0) \le 2 \sum_{s=1}^{H-1} \varepsilon_s \le 2 \sum_{s=1}^{H-1} \sum_{t=s}^{H-1} \eta_t = 2 \sum_{t=1}^{H-1} t\, \eta_t. \tag{291}$$

□

Remark 174. The Bellman residual η_n is a sup-norm object over reachable safe states. In continuous spaces one rarely computes it exactly. The intended operational use is through auditable upper bounds obtained from holdout grids, Lipschitz envelopes, certified function-approximation bounds, or reachability-restricted sampling certificates. The theory does not require exact computation; it requires an honest upper certificate.

Theorem 175 (Grid-based safe control approximation under projected Lipschitz dynamics)**.** *Let $(\mathcal{I}, d)$ be a metric information-state space. Suppose there exists a nonexpansive projection $P_h : \mathcal{I} \to \mathcal{I}_h$ onto a finite grid such that*

$$\sup_{I \in \mathcal{I}_n^{\mathrm{reach}}} d(I, P_h(I)) \le h \tag{292}$$

for every epoch n. Assume the immediate rewards satisfy

$$|r_n(I, a) - r_n(I', a)| \le L_r \, d(I, I') \tag{293}$$

for all safe actions, and that the controlled transition kernels satisfy the Wasserstein-Lipschitz bound

$$\big|\mathbb{E}[f(I_{n+1}) \mid I, a] - \mathbb{E}[f(I'_{n+1}) \mid I', a]\big| \le L_P \, \mathrm{Lip}(f) \, d(I, I') \tag{294}$$

for every bounded Lipschitz test function f. Assume also that the lower-validity functional used to define the safe action set is Lipschitz,

$$|\underline{V}(I) - \underline{V}(I')| \le L_V d(I, I'), \tag{295}$$

and that there exists a safety margin $\zeta_n > 0$ on each reachable layer such that

$$|\underline{V}(I) - \alpha| \ge \zeta_n \qquad \text{for all } I \in \mathcal{I}_n^{\mathrm{reach}}, \tag{296}$$

with $L_V h < \zeta_n$. Suppose further that the true optimal safe value functions W_n are Lipschitz on reachable safe states with known constants C_n^W:

$$|W_n(I) - W_n(I')| \le C_n^W d(I, I'), \qquad I, I' \in \mathcal{I}_n^{\mathrm{reach}}. \tag{297}$$

Define the projected-grid surrogate by $\widehat{W}_H := g$ and, for each grid state $z \in \mathcal{I}_h$ and epoch $n < H$,

$$\widehat{W}_n(z) := \max_{a \in \mathcal{A}_\alpha^{\mathrm{safe}}(z)} \Big\{ r_n(z, a) + \mathbb{E}\big[\widehat{W}_{n+1}(P_h(I_{n+1})) \mid I_n = z, a_n = a\big] \Big\}, \tag{298}$$

with extension $\widehat{W}_n(I) := \widehat{W}_n(P_h(I))$ to reachable states. Define bias constants by

$$B_H := 0, \qquad B_n := B_{n+1} + L_r + (1 + L_P) C_{n+1}^W, \qquad n = H-1, \dots, 0. \tag{299}$$

Then

$$\varepsilon_n \le B_n h, \qquad n = 0, \dots, H. \tag{300}$$

Consequently,

$$W_0(I_0) - J_0^{\widehat{\pi}}(I_0) \le 2h \sum_{t=1}^{H-1} B_t. \tag{301}$$

Proof. The safety-margin condition and Lipschitz continuity of $\underline{V}$ imply

$$|\underline{V}(I) - \underline{V}(P_h(I))| \le L_V h < \zeta_n,$$

so $\underline{V}(I)$ and $\underline{V}(P_h(I))$ lie on the same side of the threshold α for every reachable state at epoch n. Hence the safe action sets coincide.

Because $\widehat{W}_H = g = W_H$, we have $\varepsilon_H = 0$. Suppose inductively that $\varepsilon_{n+1} \le B_{n+1}h$. For any reachable safe state I and safe action a,

$$\begin{aligned}
&\Big|r_n(P_h(I),a) + \mathbb{E}[\widehat{W}_{n+1}(P_h(I_{n+1})) \mid P_h(I),a] - (r_n(I,a) + \mathbb{E}[W_{n+1}(I_{n+1}) \mid I,a])\Big| \\
&\le |r_n(P_h(I),a) - r_n(I,a)| \\
&\quad + \mathbb{E}\Big[|\widehat{W}_{n+1}(P_h(I_{n+1})) - W_{n+1}(P_h(I_{n+1}))| \;\Big|\; P_h(I),a\Big] \\
&\quad + \Big|\mathbb{E}[W_{n+1}(P_h(I_{n+1})) \mid P_h(I),a] - \mathbb{E}[W_{n+1}(P_h(I_{n+1})) \mid I,a]\Big| \\
&\quad + \mathbb{E}\Big[|W_{n+1}(P_h(I_{n+1})) - W_{n+1}(I_{n+1})| \;\Big|\; I,a\Big] \\
&\le L_r h + \varepsilon_{n+1} + L_P C^W_{n+1} h + C^W_{n+1} h.
\end{aligned}$$

Taking maxima over the identical safe action sets yields

$$\varepsilon_n \le \varepsilon_{n+1} + (L_r + (1+L_P)C^W_{n+1})h. \tag{302}$$

Backward induction gives $\varepsilon_n \le B_n h$. The regret claim then follows from theorem 172. □

Remark 176. The theorem is intentionally local to reachable states with an interior safety margin. It does not claim that the global safe value function must be Lipschitz across safety-boundary kinks where the admissible action set changes discontinuously. The role of the margin condition $L_V h < \zeta_n$ is precisely to keep the projected recursion inside a layer on which the safe action set is locally stable.

8.7 From estimation to control: a joint certificate

The practical handoff from measurement to action is a common place for hidden wishful thinking. A benchmark program estimates validity, estimates transitions, solves a planner, and then quietly acts as if the whole chain inherited one undifferentiated truth value. It does not. The handoff itself needs a certificate.

Definition 177 (Estimation-to-control certificate)**.** Fix either the information-state control setup from theorem 151 or the belief-state control setup from theorem 159. An *estimation-to-control certificate* at horizon H consists of

1. an anytime-valid validity envelope with failure budget δ_V,
2. reported transition or joint transition–observation ambiguity sets with coverage budget δ_P,
3. if portfolio redesign is part of the action space, a certified lower coverage frontier or certified lower coverage/survival summaries with failure budget δ_C,
4. a computed safe policy $\widehat{\pi}$ together with an optimization or approximation certificate $\varepsilon_{\text{plan}} \ge 0$ for the reported robust control problem.

The certificate is called *jointly valid* if the true validity process lies in the reported envelope on an event $\mathcal{E}_V$ with probability at least $1-\delta_V$, the true controlled kernel law lies in the reported ambiguity sets on an event $\mathcal{E}_P$ with probability at least $1-\delta_P$, and any reported coverage frontier or certified portfolio summaries are valid on an event $\mathcal{E}_C$ with probability at least $1-\delta_C$.

Theorem 178 (Joint estimation-to-control guarantee)**.** *Consider a policy $\widehat{\pi}$ produced from a jointly valid certificate in the sense of theorem 177. Suppose $\widehat{\pi}$ is envelope admissible and is $\varepsilon_{\text{plan}}$-optimal for the reported robust control problem over the certified action class. Then:*

1. *with probability at least $1-\delta_V$, the policy never deploys the current benchmark at an epoch where the true deployed validity is at or below the governance threshold α;*
2. *on the event $\mathcal{E}_V \cap \mathcal{E}_P$, the* expected value *of $\widehat{\pi}$ under the true kernel law is at least the reported robust value minus $\varepsilon_{\text{plan}}$;*
3. *on the event $\mathcal{E}_C$, any action pruning justified by theorem 115 is lossless relative to the certified action class.*

Consequently, the safety claim, the pessimistic expected-value claim, and the frontier-pruning claim hold jointly with probability at least $1-(\delta_V+\delta_P+\delta_C)$.

Proof. The first claim is immediate from envelope admissibility and the anytime-validity event $\mathcal{E}_V$, exactly as in theorems 148 and 158. For the second claim, on $\mathcal{E}_P$ the true kernel law is one admissible element of the ambiguity set used to compute the reported robust value. Therefore, for any policy π, its robust value is no larger than its expected value under the true law on that event. Since $\widehat{\pi}$ is $\varepsilon_{\text{plan}}$-optimal for the reported robust problem, its expected true value on $\mathcal{E}_P$ is at least the reported optimal robust value minus $\varepsilon_{\text{plan}}$. For the third claim, theorem 115 is deterministic conditional on the correctness of the reported certified lower frontier or certified lower portfolio summaries, so it holds on $\mathcal{E}_C$. Intersecting the three events and applying the union bound gives the stated joint probability. □

Corollary 179 (Modular union-bound certification)**.** *If the validity envelope, ambiguity set, coverage/frontier certificate, and planner approximation certificate are produced by separate procedures with failure budgets δ_V, δ_P, δ_C, and δ_{plan}, then all reported claims hold simultaneously with probability at least*

$$1-(\delta_V+\delta_P+\delta_C+\delta_{\text{plan}}). \tag{303}$$

Proof. Apply the union bound to the four failure events. □

9 Estimation Protocol

The theory becomes operational only if both the validity components and the state variables can be estimated or bounded. This section sketches a practical protocol rather than a single mandated estimator.

9.1 Identification regimes before estimation

Before computing any estimate, one should classify the benchmark into an identification regime. If one observes only the validity trace or score trace, then theorem 39 implies that only the composite hazard path is identified; latent ecological coordinates are not. If proxies are available through a known injective

map, then theorem 41 yields point identification of the ecological state and hazard along the observed path. If, in addition, the protocol belongs to a declared state-recovery regime and attains a support-dense calibrated design in the sense of theorems 47 and 49, then the structural aggregate hazard map is identified on the visited target region; if not, off-support values remain non-identified by theorem 48. If one is willing to impose only weak shape restrictions outside support, then theorems 51, 53 and 54 replace point extrapolation by sharp set-valued extrapolation. If one separately observes the discriminative-power and construct-validity traces, then theorems 55, 57 and 58 determine whether factor-specific hazards are non-identified, point identified on calibrated support, or merely bounded. If only a calibration correspondence is available, then the benchmark is set identified and the correct outputs are intervals as in theorems 42 and 43. If matched interventions on benchmark construction or exposure are feasible, then theorems 27, 60 to 62 and 64 to 66 determine whether one can recover only a sign, an exclusive-regime factor effect, an exclusion-based shared-coordinate effect, or a full local shared-coordinate decomposition. At the portfolio level, one should also classify whether coverage coefficients are treated as exact, box uncertain, or budget uncertain in the sense of theorems 117 to 119, and whether the reporting target is a half-life, a threshold invalidity time, or a tail-risk metric from Definition 137. This is a partial-identification mindset in the sense of Manski-style analysis: the inferential target should match the information actually available [31, 32, 37].

9.2 Calibrating the hazard map

A hazard map should not be treated as a freely chosen smooth summary. To justify it as a structural object rather than a curve-fitting convenience, the empirical protocol should separately document three ingredients from theorem 49: point-calibrated state recovery, identification or sufficiently tight smoothing of the composite hazard path, and support-rich coverage of the target state region. In practice this means combining proxy inversion or calibration studies for $(S_t, N_t, F_t, U_t, J_t, K_t)$ with smoothed estimates of $-\frac{\mathrm{d}}{\mathrm{d}t}\log \mathcal{V}_B(t)$ and, when possible, deliberately varied benchmark-construction or exposure conditions that excite the relevant coordinates.

The key caution is off-support extrapolation. Even a beautifully estimated hazard surface along observed states does not identify what the surface would have been in unvisited regions of state space; theorem 48 shows that continuous alternatives can agree perfectly on the observed path and still diverge elsewhere. Weak shape restrictions can tighten the admissible set without creating magic point identification: theorems 51 and 53 yield sharp brackets, not warrant for unsupported certainty. Hence any reported hazard map should explicitly distinguish *identified-on-support* regions from *structurally extrapolated* regions and should name the restriction class used to produce the latter.

9.3 Designing factor-targeted interventions

If the empirical goal is factor attribution rather than passive monitoring, intervention design should distinguish baseline factor laws from optional exclusivity regimes. Perturbations of S or U may isolate discriminative-power effects in aggregate validity only when the design justifies treating the induced path as approximately D-exclusive in the sense of theorem 60; perturbations of K may isolate construct-validity effects only under an analogous C-exclusive justification. Interventions on shared coordinates (N, F, J) are still useful, but aggregate validity then identifies only a sum of factor responses by theorem 61. To recover a factor decomposition for shared coordinates, the protocol must either measure factor traces

separately, impose a factor-excluding design as in theorem 64, or build at least two linearly independent local perturbation families as in theorem 66.

A practical design rule is therefore simple: use exclusive-regime interventions only when their zero- or bounded-spillover justification is written down, use exclusion-style shared-coordinate interventions when one can justify them substantively, and use general shared-coordinate interventions for stress-testing. The first class can pin down which factor moves, the second can recover local shared-coordinate sensitivities, and the third reveals where the factor split remains underdetermined and therefore where extra instrumentation is needed. Any claimed exclusion should be written down explicitly rather than implied by vibes.

9.4 Sensitivity analysis for imperfect exclusion

When shared-coordinate attribution relies on exclusion or IV-style logic, the protocol should never jump directly from a design story to a point estimate. First declare the intended exclusion, then declare an admissible violation budget or direct-effect set, and only then compute the induced identification region using theorems 69, 72, 74 and 76. In practice this means writing down which implementation channels could leak into the nominally excluded factor, assigning either deterministic bounds or statistically learned bounds to those channels, and reporting how the factor estimate widens as those budgets increase. If the conclusion changes under small plausible budgets, the design is fragile and should be reported as such rather than laundered into a single coefficient.

9.5 Constructing sharp violation classes

A violation budget should be treated as a model of the admissible spillover class, not as a cosmetic sensitivity knob. Whenever domain knowledge supports sign restrictions, asymmetric bounds, or a finite-dimensional remainder polytope, the appropriate output is the sharp image set from theorems 72 and 76 rather than the looser symmetric interval from theorem 69. Operationally, this means documenting which spillover paths are genuinely believed possible and which are ruled out by the intervention mechanics.

9.6 Estimating discriminative power

Given a declared operational model class $\mathcal{M}_t^{\mathrm{obs}}$ and an empirical pair distribution $\widehat{\Pi}_t$, estimate discriminative power by

$$\widehat{\mathcal{D}}_B(t;\delta) := \frac{1}{|\mathcal{P}_t|} \sum_{(m_i,m_j)\in\mathcal{P}_t} \mathbf{1}\{|Y_B(m_i,t) - Y_B(m_j,t)| > \delta\}, \tag{304}$$

where $\mathcal{P}_t$ is a sample of ordered model pairs drawn according to $\widehat{\Pi}_t$. Bootstrap or Bayesian hierarchical intervals should be reported because leaderboard sampling is often sparse and nonstationary. Benchmark^2-style discriminability diagnostics are useful sanity checks here, but the key governance point is that $\widehat{\mathcal{D}}_B$ is class-relative: changing the operational class changes the estimate [24].

9.7 Estimating construct validity

Construct validity cannot be read off the benchmark alone. One needs one or more claim-aligned reference instruments or anchors. Let $Z_t^{(r)}(m)$ denote anchor r, where an anchor may be an external task outcome,

an adjudicated expert score, a deployment-side KPI, or a separately validated benchmark known to target the same claim more directly. Each anchor induces an order comparator

$$A_t^{(r)}(m_i, m_j) := \operatorname{sgn}(Z_t^{(r)}(m_i) - Z_t^{(r)}(m_j)). \tag{305}$$

The observable anchor-resolved event attached to anchor r is

$$\mathcal{R}_t^{(r)}(\varepsilon) := \{(m_i, m_j) : |Z_t^{(r)}(m_i) - Z_t^{(r)}(m_j)| > \varepsilon\}, \tag{306}$$

with empirical resolved mass

$$\widehat{R}_r^{\mathrm{obs}}(t;\varepsilon) := \frac{1}{|\mathcal{P}_t|} \sum_{(m_i,m_j)\in\mathcal{P}_t} \mathbf{1}\{|Z_t^{(r)}(m_i) - Z_t^{(r)}(m_j)| > \varepsilon\}. \tag{307}$$

When $\widehat{R}_r^{\mathrm{obs}}(t;\varepsilon) = 0$, the protocol should return the status label `under-resolved` and should not fabricate a numeric order-fidelity estimate. Otherwise define the empirical benchmark-anchor agreement

$$\widehat{\Gamma}_r^{\mathrm{obs}}(t;\varepsilon) := \frac{\displaystyle\sum_{(m_i,m_j)\in\mathcal{P}_t} \mathbf{1}\{|Z_t^{(r)}(m_i) - Z_t^{(r)}(m_j)| > \varepsilon\} \cdot \mathbf{1}\{\operatorname{sgn}_{\delta_t}(Y_B(m_i,t) - Y_B(m_j,t)) = A_t^{(r)}(m_i,m_j)\}}{\sum_{(m_i,m_j)\in\mathcal{P}_t} \mathbf{1}\{|Z_t^{(r)}(m_i) - Z_t^{(r)}(m_j)| > \varepsilon\}}. \tag{308}$$

This estimator targets the observable quantity $\Gamma_r^{\mathrm{obs}}(t;\varepsilon)$ from theorems 91 and 93, not the latent order-fidelity component directly. The next step is therefore not "trust the anchor," but "declare the anchor-validation design." In the sense of theorems 81 and 82, that design may be a gold-standard adjudication subsample, a multi-anchor latent-label fit, or a one-sided calibration study. If such a design or domain knowledge provides an admissible order-error budget $\bar{e}_r(t)$, directional resolved-set discrepancy budgets $\bar{\eta}_r^+(t), \bar{\eta}_r^-(t)$, and an admissible latent resolved-mass interval $R_B(t) \in [\underline{R}_r(t), \overline{R}_r(t)]$, then theorem 93 yields the sharper order-fidelity interval

$$\begin{aligned}\mathcal{O}_B(t) \in \Big[&\max\{0, \widehat{\Gamma}_r^{\mathrm{obs}}(t) - \bar{e}_r(t) - \bar{\eta}_r^+(t)/\widehat{p}_r^{\mathrm{obs}}(t) - \bar{\eta}_r^-(t)/\underline{p}_r^{\mathrm{tar}}(t)\},\\ &\min\{1, \widehat{\Gamma}_r^{\mathrm{obs}}(t) + \bar{e}_r(t) + \bar{\eta}_r^+(t)/\widehat{p}_r^{\mathrm{obs}}(t) + \bar{\eta}_r^-(t)/\underline{p}_r^{\mathrm{tar}}(t)\}\Big].\end{aligned} \tag{309}$$

where $\widehat{p}_r^{\mathrm{obs}}(t)$ is the observed anchor-resolved mass and $\underline{p}_r^{\mathrm{tar}}(t)$ is a declared lower bound for the latent target-resolved mass. If the design implies a nesting relation between resolved sets, theorem 94 should be used instead of a generic additive bound because the resulting interval can be sharp rather than merely conservative. Up to sampling uncertainty, this yields the construct-validity interval

$$\mathcal{C}_B(t) \in \Big[\underline{R}_r(t) \cdot \underline{O}_r(t),\ \overline{R}_r(t) \cdot \overline{O}_r(t)\Big], \tag{310}$$

where $[\underline{O}_r(t), \overline{O}_r(t)]$ denotes the previous order-fidelity interval.

When directional class-conditional anchor errors are defensible, the protocol can be sharper still, but only after being honest about the conditioning event. Specifically, estimate or externally calibrate the score-resolved target-resolved mass $\rho_{B,\delta}(t)$, the score-resolved positive-order prevalence $p_{B,+}^{\delta}(t)$, and the score-resolved anchor agreement $\Gamma_A^{\delta}(t)$ on a labeled or otherwise separately identified resolved-pair sample;

these are not obtained for free from raw anchor agreement alone. Under the nondifferential-error condition in theorem 84, theorem 85 then converts those inputs into either a point estimate or a valid identified set for $\mathcal{O}_B(t)$. If benchmark-dependent anchor error is plausible, the protocol should additionally declare a violation budget of the form in theorem 87 and widen the estimate using theorem 88. Whenever only intervals for the score-resolved inputs are available, theorem 86 should be used to push those intervals through the affine recovery map rather than pretending that the latent prevalences are observed exactly.

When several imperfect anchors are available, the default nonparametric procedure is to construct one order-fidelity interval from each anchor and then intersect them as in theorem 96. This already tightens identification without assuming independence or any specific latent-noise model. An odd-sized majority-vote panel provides a stronger but more model-dependent route: estimate the panel agreement $\widehat{\Gamma}_{\mathrm{maj}}(t)$ and then combine it with an explicit panel-error bound, for example the shared-blind-spot bound from theorem 98, the covariance-based bound from theorem 99, or the latent-factor mixture formula from theorem 100. In practice, both an optimistic conditional-independence benchmark and a dependence-robust benchmark should be reported. More structured alternatives include weighted panels, latent-label models, or expert-adjudicated anchor hierarchies in the spirit of Dawid–Skene style observer-error modeling [38]; the theory here only requires that the resulting anchor system come with a defensible order-error, resolved-mass, and dependence budget.

Cross-benchmark corroboration metrics such as Benchmark^2's ranking consistency can be useful auxiliary diagnostics [24], but they are not by themselves construct validity, because peer benchmarks can share the same blind spots. If no claim-aligned anchor exists, broad construct-validity claims should be weakened or withheld. This is precisely where many benchmark papers overclaim relative to their evidence [14, 16].

9.8 Qualified anchors and abstention budgets

In many real audits an anchor is only trusted on a visibly qualified subset of resolved pairs. The protocol should therefore record the qualification rule Q_t, the lower qualification mass $\underline{\kappa}_A(t)$, and the qualified-region error budget $\bar{e}_{A,Q}(t)$, then apply theorem 90 rather than imputing accuracy on abstained pairs. If no auditable lower bound on qualification mass is available, the abstention region must be treated as unconstrained and the resulting order-fidelity interval will be correspondingly wide. Zero qualified resolved pairs should be reported as `insufficient-resolved-pairs`, not converted into a point estimate by denominator patching.

9.9 Estimating semantic leakage

For each benchmark item b, estimate $\kappa_b(x)$ using a stack that combines exact matching, lexical overlap, embedding-based semantic retrieval, template similarity, and solution-structure similarity. The goal is not to decide whether x is "the same item" in some binary metaphysical sense, but to assign leakage mass. A sampled estimator takes the form

$$\widehat{S}_t = \frac{1}{|B|} \sum_{b \in B} \sum_{x \in \widehat{C}_t} w_t(x)\, \widehat{\kappa}_b(x), \tag{311}$$

where $\widehat{C}_t$ is an indexed proxy for the recursive corpus and $w_t(x)$ are exposure weights. Because soft and variant contamination are easy to miss, $\widehat{S}_t$ should normally be treated as a proxy observation feeding a

calibration set for S_t, not as an automatically identified latent state. At minimum it should be reported with an uncertainty envelope rather than as a lone point estimate [18, 19].

9.10 Estimating verified novelty inflow

Novelty inflow is not mere timestamp churn. A new item contributes to N_t only if it is temporally or procedurally new relative to plausible exposure, materially distinct from prior benchmark manifolds, and verifiable enough to support evaluation. A useful decomposition is

$$N_t = \Lambda_t \cdot E_t \cdot Q_t, \tag{312}$$

where Λ_t is a raw arrival rate of candidate items, E_t is an effective novelty factor, and Q_t is a verifiability gate. Dynamic benchmark systems can be read as attempts to raise Λ_t and E_t while keeping Q_t acceptably high [10, 11, 13].

9.11 Estimating verification fidelity, support diversity, and coupling

Verification fidelity F_t can be estimated from objective-answer availability, validator agreement, adversarial stress tests, judge substitution sensitivity, and label-audit quality. Reliability-focused work on platinum benchmarks is particularly relevant here because hidden label errors directly lower F_t even when the benchmark item format itself looks sound [23]. Support diversity U_t can be estimated from coverage metrics over benchmark subdomains, tails, difficulty strata, or perturbation families. Evaluator coupling J_t can be proxied by shared judge families, overlap between benchmark-generation and evaluation pipelines, or benchmark-specific prompt conventions. None of these proxies is perfect, which is precisely why interval methods are more faithful than point claims.

9.12 Estimating construct drift

Construct drift K_t measures misalignment between benchmark behavior and the target claim functional. Indicators include weak transfer from benchmark gains to deployment outcomes, instability under small task reframings, persistent "accuracy-on-the-line" style misspecification when the benchmark is meant to test robustness, and evidence that scores are being driven by benchmark-specific exploitability rather than the intended capability [14, 16, 15, 26]. Adversarial or expert-generated live benchmarks that correlate with external capability measures can serve as drift audits rather than as replacements for the original benchmark [25]. The formal recommendation is to build a construct-drift audit map

$$\Psi_K(\text{transfer audits}, \text{reframing audits}, \text{external validity checks}) \rightsquigarrow [\underline{K}_t, \overline{K}_t], \tag{313}$$

and to carry K_t as an interval-valued ecological coordinate unless a stronger calibration study is available. This keeps drift on the same inferential footing as leakage and coupling rather than leaving it as a verbal warning.

9.13 Lineage observability and interval tightening

Lineage observability L_t is estimated from timestamp coverage, source-chain completeness, revision logging, benchmark-construction disclosure, and, when available, cryptographically anchored provenance. Within

the present theory, stronger lineage observability does not automatically make a benchmark better; it makes the benchmark's life-cycle state more identifiable and can enlarge the set of actions that can be justified safely. This is why L_t enters the observability state and narrows the feasible set in theorems 9, 42 and 44 rather than appearing directly in the base decay rate.

9.14 Portfolio monitoring

Real evaluation programs often track a family of benchmarks rather than one instrument. For such programs, the natural object to report is the weighted normalized portfolio survival $\widetilde{V}_{\mathcal{P}}(t)$ from theorem 101 together with its current half-life estimate or bound. The single most important diagnostic is whether the apparent portfolio resilience comes from genuine diversification or from averaging over benchmarks that all share the same dominant decay channel. Theorem theorem 103 implies that common-mode leakage, shared judge families, or shared generation pipelines can cap portfolio lifetime even when individual benchmarks look superficially diverse.

The design question should also be reported, not merely the monitoring statistic. Theorem 104 shows that with a single known deterministic hazard profile, horizon-survival optimization collapses to the single best benchmark. Diversification becomes rational only once uncertainty sets, common-mode exposure caps, or coverage requirements are declared. Theorems 107, 110, 111 and 118 to 120 then supply tractable design problems together with explicit survival–coverage frontiers and robust coverage audits that should be monitored rather than hidden inside ad hoc weights.

Operationally, this means benchmark families should be designed to reduce common-mode exposure as well as to increase item count. Different provenance chains, different judge families, different generation pipelines, and different task manifolds matter because they weaken common hazards, not because diversity is aesthetically pleasing. It also means the report should include threshold invalidity times and, when a scenario law is declared, safe quantiles or shortfall-tail summaries from theorems 137 to 139 rather than a lone half-life. If monitoring is ongoing rather than one-shot, those summaries should be paired with anytime-valid envelopes and stopping logic as in theorems 141, 143 and 144.

9.15 Coverage uncertainty and invalidity-risk reporting

Coverage should be audited as an uncertain matrix, not as a decorative table of point claims. In the simplest case, one can elicit lower and upper coverage coefficients and report the box-robust design from theorem 118. When coverage auditing is noisier or partly adversarial, a budgeted uncertainty model of the form in theorem 119 is often more faithful: only some cells or benchmark–cell entries are allowed to be maximally wrong at once, and the uncertainty budget Γ_g is itself a governance parameter that should be disclosed.

Reporting should also move beyond a single half-life scalar whenever the benchmark family supports more than one operational threshold. For each declared threshold α, one should report the current lower or interval estimate of T_α, together with at least one scenario-based safe quantile or shortfall-tail summary from Definition 137. The point is not ornamentation. It is that a benchmark may look acceptable at $\alpha = 1/2$ while already being operationally unsafe at a stricter threshold or under the lower tail of the scenario law.

9.16 Dependence-aware learned coverage sets

When coverage audits are clustered by time, evaluator, benchmark family, or provenance chain, the learned uncertainty set should itself be calibrated under that dependence structure. In practice this means first declaring which regime from theorem 129 is supposed to justify the sequential set construction: a blockwise or geometrically mixing regime, or a bounded-increment martingale regime. The resulting coordinatewise confidence sequences should then be combined into the product set from theorem 130 rather than treating the dependence correction as an afterthought. Once that is done, theorems 126 and 132 allow adaptive monitoring or stopping-time portfolio updates without losing the coverage guarantee.

9.17 Nonstationary and regime-switching audit calibration

If audits are not merely dependent but also nonstationary, the protocol should stop pretending that one immutable coverage vector governs all times. Instead it should declare either observable audit regimes or a posterior-weighted regime filter and then maintain regime-specific confidence sequences as in theorems 133 and 134. Operationally, that means the audit log should record the regime label or filtering model, the regime-specific sample allocation, and the rule used to map those regime-level intervals into the current uncertainty set. Portfolio updates performed at stopping times should then be checked against the appropriate observed-regime or posterior-mixture set, not against a stationary box lifted from a different epoch.

9.18 Sequential monitoring and stopping

A live benchmark program should maintain an online upper and lower envelope for the validity quantity that matters operationally. The exact construction may come from confidence sequences for pairwise separation rates, anchor-agreement proportions, or portfolio survival summaries; the theory here only requires the result to be anytime valid. Once such an envelope is in place, retirement can be triggered by the upper-envelope crossing rule in theorem 143, continuation can be justified only while the lower envelope remains above the governance threshold as in theorem 144, and the monitoring dashboard can expose the life-cycle labels from theorem 145 rather than a raw score stream. This avoids the usual p-hacking swamp where a benchmark is repeatedly audited until someone decides the answer looked convenient enough to act on.

9.19 Dynamic retirement and regeneration planning

Monitoring alone does not choose actions. After constructing an anytime-valid envelope, the program should declare the reward model, feasible regeneration actions, and safe action set used in the Bellman recursion of theorem 150. In practical terms, this means recording the immediate operational benefit of continued use, the direct cost of regeneration, and the expected state shift induced by candidate regeneration policies. If the transition model is learned rather than known, the program should also publish the ambiguity set used by the robust recursion in theorems 152 and 160; otherwise the planner is merely a plug-in controller in the sense of theorem 153. Dominated regeneration actions should be pruned before optimization using order arguments such as theorem 155. If exact safe dynamic programming is computationally expensive, the program should report either direct value-approximation errors or Bellman-residual certificates so that theorems 172, 173 and 175 can be invoked to translate solver error into an explicit regret budget.

9.20 Transition ambiguity sets for safe control

When the transition kernel in the safe-control layer is estimated from finite audit or intervention data, the protocol should output an ambiguity set $\mathfrak{P}_n(I, a)$ or, under partial observability, a joint transition–observation ambiguity set $\mathfrak{Q}_n(b, a)$ rather than a single plug-in kernel. In finite abstractions this can be built directly from empirical confidence sets as in theorem 154; in richer spaces it can be built from confidence rectangles, posterior credible regions with validated coverage, regime-specific uncertainty sets, or declared distributional ambiguity families. What matters is that the construction be declared and that the resulting family be rectangular enough for theorems 152 and 160 to apply. If only a nonrectangular family is defensible, the protocol should publish a rectangular outer hull together with a pessimism-gap budget in the sense of theorems 164 and 166. Reporting only the nominal planner hides model uncertainty exactly where the action logic is most consequential. If only a nominal kernel is reported, it should be labeled explicitly as a plug-in planner and compared against the robust lower value from theorem 153.

9.21 Belief-state control under partial observability

In many benchmark programs the true ecology state is not directly observed. The control object is then a belief state built from audits, anchors, contamination signals, and regeneration logs. The protocol should therefore declare the filtering model that maps histories into belief states and the lower-validity functional that turns those beliefs into a safe action set, as formalized in theorems 156 to 158. If the dynamic model is ambiguous, the report should also state whether the ambiguity is genuinely stagewise rectangular or only available as a nonrectangular family. The former supports the exact robust Bellman recursion of theorem 160; the latter should be accompanied by a reported rectangular outer hull and treated through the conservative lower-value reduction of theorem 164, not smuggled into an "exact" Bellman story. Without that declaration, talk of "state-aware" benchmark control is usually just a deterministic fairy tale pasted over partial observability.

If the control action includes portfolio redesign, the estimation layer should additionally report either certified lower coverage/survival summaries for each candidate action or a certified lower coverage frontier $v(\rho; T)$. This lets the controller prune frontier-dominated actions using theorem 115 before solving the dynamic program and prevents high-dimensional portfolio weights from leaking silently into a low-transparency control layer.

9.22 Estimation-to-control handoff

The final output of the estimation layer should not be a pile of unrelated intervals. It should be a control-ready certificate in the sense of theorem 177: an anytime-valid validity envelope, a declared ambiguity set for the relevant transition law, and a solver certificate for the chosen planner. This is the minimal interface needed for theorems 178 and 179 to convert statistical uncertainty into an operational safety-and-value guarantee. A point estimate for validity plus a point estimate for transitions is not an interface; it is wishful compression.

9.23 Exploration budgets for regeneration learning

When regeneration actions are learned online rather than fixed in advance, the protocol should report the confidence system and optimism rule that govern exploration inside the safe action set. In practice this

means specifying the belief-action abstraction, the confidence bonuses, and the resulting regret certificate from theorems 167 to 169. Exploration that is not tied to an explicit width budget is not a learning strategy; it is unpriced risk.

9.24 Retirement, regeneration, and scope restriction

The theory suggests a fail-closed rule:

> *Retire, regenerate, or scope-restrict a benchmark when the optimistic upper bound on current validity falls below the governance threshold for the intended operational model class.*

Operationally this is best exposed as the status map from theorem 145: *active*, *watchlist*, *regenerate-or-replace*, or *retire-or-scope-restrict*. That labeling is not cosmetic. It forces the report to distinguish epistemic ambiguity from demonstrated invalidity and to state whether a declared repair action actually exists. Fix a threshold $\theta \in (0,1)$. If the upper envelope in (72) satisfies

$$\mathcal{V}_B(0)\,\overline{J}_B(t)\exp\Big(-\int_0^t \underline{q}_B(u)\,\mathrm{d}u\Big) \le \theta \mathcal{V}_B(0), \tag{314}$$

then even the optimistic estimate is below threshold, and continued use for that operational class is no longer justified. If the failure is frontier-specific rather than universal, scope restriction is preferable to binary retirement.

10 Toy Regimes and Phase Structure

A full empirical calibration is outside the scope of the present paper, but the state-variable theory already implies a useful phase portrait. Consider the passive linear proxy

$$\dot{v}_t = -\Big[\lambda s_t + \rho j_t + \gamma k_t + \phi(1-f_t) + \psi(1-u_t) - \eta n_t\Big]_+ v_t, \qquad v_t := \mathcal{V}_B(t), \tag{315}$$

with state variables evolving more slowly than v_t between declared regime times. Upward validity movements are represented by declared regeneration jumps, not by negative passive hazard.

Regime I: Static-decay regime. If n_t is near zero while s_t, j_t, k_t, or support erosion remain positive, the net decay rate stays positive and the benchmark has a short finite half-life. This approximates long-lived public benchmarks with limited refresh and increasing recursive exposure.

Regime II: Regenerated-stable regime. If verified novelty inflow is sufficiently strong and incoming items remain genuinely distinct and verifiable, the passive hazard can be driven close to zero and scheduled regeneration jumps can restore validity after auditing. This is the design target of dynamic benchmarks.

Regime III: False-stability regime. The benchmark appears alive because scores still move, but the movement is driven by evaluator coupling or benchmark-specific exploitability rather than the intended capability. Here discriminative power may remain nontrivial while construct validity deteriorates. Raw leaderboard motion becomes epistemically misleading.

Regime IV: Class-relative saturation regime. A benchmark can lose frontier discriminative power while retaining narrower diagnostic or pedagogical value for weaker systems. In this case the benchmark is not universally dead; it has a shorter half-life for one operational class than for another, as formalized in theorem 37. This is why Π_t matters.

These regimes suggest a phase diagram with horizontal axis N_t and vertical axis effective decay pressure $S_t + \lambda J_t + \mu K_t + \nu(1 - F_t) + \xi(1 - U_t)$. Regeneration moves left-to-right; uncontrolled recursive exposure moves bottom-to-top.

Regime V: Portfolio-fragile regime. A benchmark family appears diversified at the item level, but most of its components share the same judge family, exposure channels, or construction pipeline. Then the portfolio behaves as though it were a single instrument with a strong common-mode hazard, and the apparent breadth does little to extend the family half-life. This is the portfolio analogue of hidden evaluator coupling.

Known-best versus robustly-diversified portfolios. If one benchmark is known to dominate all others at the target horizon, then deterministic optimization puts all weight on that benchmark. A diversified portfolio becomes rational only when the hazard profile is uncertain, when common-mode exposure must be capped, or when governance requires breadth across task manifolds. This sounds obvious once stated, but scoreboards often behave as though diversification were self-justifying. It is not.

Analytic Example 1: Constant hazard with a certified jump envelope. Suppose $q_B^{\mathrm{lin}}(t) \equiv q > 0$ on $[0, T]$ and the declared cumulative jump lower envelope is $\underline{J}_B^{\mathrm{safe}}(t) \equiv e^{-\Lambda}$ for all $t \le T$. Then theorem 34 yields

$$\mathcal{V}_B(t) \ge \mathcal{V}_B(0) e^{-(qt+\Lambda)}.$$

Hence $\tau_{1/2}(B) > T$ whenever $qT + \Lambda < \log 2$, and the certified threshold-invalidity time at level α satisfies $T_\alpha(B) > T$ whenever $qT + \Lambda < \log(1/\alpha)$. This is the simplest closed-form illustration of how passive decay and jump risk add on the log scale.

Analytic Example 2: Two-benchmark robust design. Consider two candidate benchmarks with horizon survivals $(\sigma_1^{(1)}, \sigma_2^{(1)})$ and $(\sigma_1^{(2)}, \sigma_2^{(2)})$ under two scenarios. The robust design problem is

$$\max_{w \in [0,1]} \ \min\{w\sigma_1^{(1)} + (1-w)\sigma_2^{(1)},\ w\sigma_1^{(2)} + (1-w)\sigma_2^{(2)}\}.$$

When the two affine scenario-survival lines cross inside $[0, 1]$, the optimal portfolio equalizes them; otherwise the optimum is a corner. This is the exact two-benchmark analogue of theorems 104 and 107: diversification appears only when scenario uncertainty makes equalization valuable.

Analytic Example 3: Anchor-error widening. Under the scalar anchor bound from theorem 83, if $\Gamma_A(t) = 0.82$ and the admissible validation bound is $\bar{e}_A(t) = 0.07$, then

$$\mathcal{O}_B(t) \in [0.75, 0.89].$$

If a benchmark-dependent violation budget $\bar{\Delta}_{\mathrm{nd}}(t) = 0.03$ is also declared in the score-resolved model, the affine recovery interval from theorem 88 acquires an additional half-width of

$$\frac{\bar{\Delta}_{\mathrm{nd}}(t)}{1 - e_A^+(t) - e_A^-(t)}.$$

This makes visible how anchor dependence and benchmark dependence enter the construct-validity certificate separately rather than being buried in one generic uncertainty term.

11 Implications for Benchmark Governance

11.1 From scoreboards to validity budgets

If benchmark validity is time-varying and hazard-driven, then a leaderboard score without a validity budget is incomplete. A benchmark report should ideally include estimated leakage, regeneration rate, verification fidelity, evaluator coupling, anchor-based construct-validity intervals, a current validity interval, and, where portfolio redesign is live, a certified coverage frontier or certified lower coverage/survival summaries for the action class. This is closer to instrument reporting in mature empirical sciences than to static scoreboards.

11.2 Portfolio governance and common-mode audits

Benchmark families should be governed as portfolios, not as piles of unrelated scoreboards. Reporting only per-benchmark validity can hide the more important question: whether the family is exposed to a dominant common-mode decay channel. A useful portfolio report therefore includes the weighted survival curve, the portfolio half-life, and a qualitative or quantitative audit of common hazards such as shared judges, shared construction pipelines, shared retrieval corpora, or shared benchmark-aware communities.

11.3 Portfolio design should state its scenario set

A benchmark portfolio should disclose why its weights are what they are. If the weights follow a single deterministic survival estimate, then the implied design should usually be sparse or even concentrated by theorem 104. If the design is diversified, the report should specify the scenario set, uncertainty model, exposure-cap logic, and coverage targets that justify diversification, ideally in the explicit forms of theorems 106 and 109. Otherwise portfolio weighting becomes decorative rather than epistemically meaningful.

11.4 Coverage claims should carry uncertainty budgets

Coverage matrices are often treated as if they were ground truth. Usually they are not. They are outputs of taxonomies, audits, or expert judgments and can be wrong in structured ways. If a portfolio's breadth claim depends on those matrices, the report should state whether coverage is nominal, fixed-robust, or learned-robust in the senses of theorems 117 to 119, 121 and 122. Otherwise "coverage" becomes another scoreboard number with hidden fragility.

11.5 Dependence-aware audit sets should match the sampling reality

If coverage matrices are learned from clustered or time-dependent audits, then the uncertainty set should be valid under that dependence pattern rather than under a fictional i.i.d. story. A learned set with poor dependence handling is just a brittle confidence costume. The dependence-aware transfer theorem theorem 126 makes the governance requirement explicit: adaptive monitoring is fine, but only after the uncertainty set itself has a uniform validity guarantee under the actual audit process.

11.6 Half-life is not the only lifetime scalar that matters

A half-life is one threshold at one loss level. Operational governance often needs a stricter question: by when does the benchmark become invalid under the declared threshold and uncertainty law? Reporting threshold invalidity times and lower safe quantiles makes the failure mode legible in ways a single $\tau_{1/2}$ cannot. When tail failures matter, shortfall-tail summaries are often the honest object.

11.7 Sequential retirement should be anytime-valid

Benchmark programs are usually monitored repeatedly, not inspected once and sealed in amber. A retirement or continuation decision therefore should be tied to an anytime-valid envelope, not to a one-shot interval that quietly assumes nobody peeked early. The safe default is simple: retire only when the upper envelope crosses the governance threshold, continue only while the lower envelope remains above it, and state the error budget attached to that online rule.

11.8 Retirement and regeneration should be optimized inside a safe action set

A benchmark program should not treat retirement and regeneration as ad hoc committee rituals. Once an anytime-valid envelope exists, the admissible action set should be declared up front and the retirement–regeneration trade-off should be optimized inside that safe region as in theorems 148 and 150. Otherwise the program is doing dynamic control while pretending it is merely "monitoring."

11.9 Public versus private is not enough

Hiding test sets can reduce some exposure channels, but saturation evidence suggests that privacy alone does not robustly prevent lifetime erosion [20]. In the present framework, privacy affects leakage only indirectly; if semantic derivatives, benchmark-aware communities, or evaluator coupling remain high, durability still degrades.

11.10 Dynamic evaluation must be metrologically constrained

A badly designed dynamic benchmark can create novelty without validity. Regeneration that lacks provenance, verification, or construct alignment simply replaces one failure mode with another. Regeneration is therefore not a synonym for quality; it is a control input whose value depends on F_t, L_t, and K_t as well as on N_t.

11.11 Progress claims should be temporally indexed

A benchmark score is a claim about a model relative to a benchmark at a particular validity state. In recursive corpora, scores are not timeless properties. They are conditional on benchmark age, exposure history, maintenance quality, and the operational model class under consideration.

12 Limitations

This is a theory paper and therefore has the usual vices if left unbounded by discipline. First, the state variables are abstractions, and real estimators for them can be noisy, model-dependent, or incomplete. In particular, a small construct-validity component can come from a small resolved-pair mass, poor order fidelity on resolved pairs, or both. Those cases should not be collapsed into one vague scalar story; the resolved-mass estimate and resolution-status label must be reported alongside the scalar. Second, the target-claim functional G_t is partly conceptual and partly domain-specific; construct validity is therefore only as good as the declared claim and the external anchors used to audit it. Third, the shared state-reduced exact law is a modeling assumption rather than a theorem of nature; only the envelope results avoid that stronger commitment, and the interval-propagation theorem is conditional on a justified hazard map. Fourth, the factor-specific exact law is stronger still, and factor-specific hazard identification requires separate access to $\mathcal{D}_B$ and $\mathcal{C}_B$ together with calibrated ecological-state recovery and sufficiently rich support; aggregate validity alone does not suffice. Fifth, the linear proxy is a governance convenience, not a universal law. Sixth, although the theory deliberately avoids treating origin labels as primitives, provenance metadata can still be useful auxiliary evidence in broader estimators. The claim here is only that origin labels are not sufficient primitives for benchmark governance. Seventh, the interval results inherit whatever errors or misspecification exist in the calibration correspondence or state-bracketing procedure. Eighth, even in the strengthened empirical protocol, many settings will remain set identified rather than point identified, and matched interventions generally recover effect signs more readily than effect magnitudes. Ninth, hazard-map identification is inherently local to calibrated and sufficiently excited portions of state space; off-support extrapolations remain structural assumptions even when they look visually smooth. The new extrapolation results do not repeal that fact: they require declared shape restrictions such as a justified Lipschitz modulus or monotone order, and those restrictions may themselves be wrong. Tenth, anchor-panel aggregation can be much less informative than its panel size suggests when common blind spots or positive dependence are strong, and latent-factor bounds can still be loose when the factor itself is badly measured. Eleventh, factor-specific intervention design helps most under genuinely justified exclusive regimes; shared-coordinate interventions still need separate factor traces, credible exclusion restrictions, or a full-rank local design matrix. Under imperfect exclusion or invalid-IV style direct effects, the theory honestly widens the answer into a sensitivity region, but that region is only as meaningful as the declared violation budgets. Twelfth, coverage matrices are often uncertain objects in their own right, and robust coverage frontiers are only as meaningful as the auditing process that produced the uncertainty set. Learned uncertainty sets improve honesty but do not abolish sampling bias, taxonomy drift, or audit misspecification; their guarantees are only as good as the finite-sample coverage statement they actually satisfy. Thirteenth, time-to-invalidity risk summaries inherit whatever scenario law, threshold choice, and tail-risk functional one declares; a fancy CVaR number can still be nonsense if the scenario set is decorative. Fourteenth, sequential monitoring gains validity only if the online envelope is genuinely anytime valid for the estimator and dependence structure in use; otherwise continuous peeking

turns the stopping rule back into theater. Fifteenth, the new sharp imperfect-exclusion results are only as informative as the declared violation class and its attainability logic; a narrow identified set based on fantasy exclusions is still fantasy. Sixteenth, dependence-aware learned uncertainty sets still take a valid uniform set-coverage guarantee as an input; they do not conjure that guarantee from arbitrary dependence without additional empirical work. Seventeenth, the dynamic retirement–regeneration control layer is only as good as the declared transition kernel and reward model. Bellman equations do not rescue bad operational assumptions. Eighteenth, the new structural-sharpness results for imperfect exclusion contract identified sets only if the structural violation class is actually credible; sign, monotonicity, sparsity, or linear-coupling restrictions that are merely wished into existence can make the theory look sharper than the evidence warrants. Nineteenth, the mixing and martingale audit regimes have now been extended to observed-regime and posterior-mixture nonstationary audits, but long-memory or adversarially drifting audit processes can still defeat the declared confidence construction. Twentieth, safe POMDP control is only as good as the declared filtering model and observation kernel, while robust POMDP value guarantees additionally depend on the truth lying inside the reported ambiguity set; a polished belief state built from a wrong hidden-state model is still a wrong state. Twenty-first, nondifferential directional anchor error is a real sharpening but not a free lunch: if anchor mistakes depend directly on the benchmark comparator even after conditioning on the latent target order, the affine correction in theorem 85 can be wrong and the honest fallback is the looser partial-identification logic from theorems 83 and 93. Twenty-second, rectangular-hull planning under nonrectangular ambiguity is conservative by design; its value lower bound can be materially looser than the true nonrectangular robust value when impossible kernel combinations make the hull much larger than the actually admissible family. Twenty-third, optimistic regeneration learning controls regret only on the confidence event and only on the chosen safe belief-action abstraction; hidden confounding, unpriced drift, or undercoverage in the bonuses can make the reported exploration budget fiction. Twenty-fourth, approximate safe control bounds certify solver error only relative to the declared model class and reachable information states; hidden model misspecification can dominate the reported regret bound. Twenty-fifth, the grid-based approximation theorem assumes Lipschitz regularity of rewards and transitions in a chosen metric, and those assumptions can fail badly when benchmark ecology changes discontinuously. Twenty-sixth, portfolio-level lifetime gains vanish under large common-mode hazards, and portfolio diversification is only as meaningful as the declared uncertainty, exposure, and coverage model. The framework is therefore fail-closed only to the extent that its uncertainty envelopes, dependence budgets, shape restrictions, exclusion logic, learned uncertainty sets, scenario declarations, and control models are themselves honestly constructed.

13 Conclusion

Benchmarks now live inside recursive corpora. In that environment they are not static mirrors of capability but decaying instruments whose usefulness depends on semantic leakage, evaluator coupling, construct drift, support erosion, verification fidelity, and regeneration. The right primitive is therefore not nominal data origin but benchmark validity, defined here as the product of discriminative power and a construct-validity component that incorporates both resolved-pair mass and thresholded order fidelity, and governed by a state-dependent hazard. That hazard induces a benchmark half-life.

The resulting picture is neither fatalistic nor naive. Benchmarks can and often do decay under persistent exposure pressure. But regeneration, verifiability, provenance-aware auditing, and deliberate metrological

design can extend or stabilize their useful life. The scientific implication is direct: benchmark results should be interpreted together with a life-cycle state, not as timeless capability facts. The identification implication is equally direct: aggregate validity, factor-specific hazards, anchor-based order-fidelity and construct-validity intervals, and portfolio survival do not become visible under the same information regime, so reports should state which regime they inhabit. The engineering implication is equally direct: benchmark governance must move from static scoreboards to dynamic validity management, with identification status, dependence budgets, exclusion logic, coverage uncertainty budgets, estimation-to-control certificates, and scenario declarations reported explicitly rather than smuggled in as false precision. Half-life remains useful, but it should sit inside a wider family of threshold invalidity times and tail-risk summaries whenever operational failure thresholds differ across use cases. Once monitoring becomes live, the governance problem is no longer only inferential; it is also sequential and control-theoretic. Nonstationary audits may require regime-indexed or posterior-mixture uncertainty sets rather than one frozen coverage matrix, and partially observed benchmark ecologies require belief-state rather than fairy-tale fully observed control. Safe benchmark programs therefore need three things at once: honest uncertainty sets, honest action sets, and honest learning budgets for regeneration. When those action rules are implemented approximately or learned online, the approximation and exploration error should itself be metrologized through Bellman residuals, reachable-state error bounds, confidence widths, or explicit regret certificates rather than hidden behind solver mystique.

14 Suggested Empirical Program

A concrete empirical follow-up could proceed in eleven stages.

1. Select benchmark families with long public exposure histories and enough leaderboard data to estimate changing discriminative power.
2. Reconstruct approximate recursive corpora from web snapshots, tutorials, solution repositories, public evaluation harnesses, model-generated discussions, and benchmark-aware prompting traces.
3. Estimate exact, lexical, semantic, and variant leakage side by side to quantify the gap between exact-match and semantic hazard proxies.
4. When feasible, separately track discriminative-power and construct-validity traces so that factor-specific hazards can be identified or bounded rather than conflated into a single composite hazard.
5. Build time-indexed validity envelopes by combining score-separation statistics, construct-audit scores, judge-coupling proxies, and provenance observability indicators.
6. Maintain at least one dependence-robust anchor panel for each major benchmark family and document whether the dependence budget is framed by conditional independence, shared blind spots, covariance bounds, or a latent-factor model.
7. For shared-coordinate interventions, elicit asymmetric or sign-restricted spillover classes and report sharp identified sets rather than only symmetric sensitivity intervals.
8. Learn coverage uncertainty sets under the actual audit dependence structure and verify stopping-time validity before using those sets for adaptive portfolio updates.

9. For benchmark families, publish the scenario set and exposure caps that justify portfolio weights, rather than presenting diversification as self-justifying.

10. Estimate or otherwise conservatively bracket regeneration transition kernels and solve the safe Bellman control problem for retirement versus regeneration under declared utility and cost models.

11. Compare static and dynamic benchmarks by estimated half-life, threshold invalidity times, and life-cycle state rather than by instantaneous score spread alone.

This program would not merely test the theory. It would produce the first benchmark-life-cycle dashboards suitable for real governance and safe action planning.

A Additional Proofs and Useful Lemmas

A.1 General envelope lemma

Lemma 180 (Envelope integration lemma)**.** *If an absolutely continuous function $v : [0,T] \to (0,\infty)$ satisfies*

$$-a(t) \le \frac{\mathrm{d}}{\mathrm{d}t} \log v(t) \le -b(t) \tag{316}$$

for almost every $t \in [0,T]$, then

$$v(0) \exp\Big(-\int_0^t a(u)\,\mathrm{d}u\Big) \le v(t) \le v(0) \exp\Big(-\int_0^t b(u)\,\mathrm{d}u\Big). \tag{317}$$

Proof. Integrate the lower and upper inequalities separately and exponentiate. □

A.2 Half-life under constant hazard

If $q_B^{\mathrm{lin}}(t) \equiv h > 0$ and the validity dynamics satisfy equality with the linear proxy, then

$$\mathcal{V}_B(t) = \mathcal{V}_B(0) e^{-ht} \qquad \text{and} \qquad \tau_{1/2}(B) = \frac{\log 2}{h}. \tag{318}$$

For $h = 0$, the half-life does not occur on any finite horizon under the equality model. Negative passive hazards are excluded by construction because regeneration gains are represented by declared jumps rather than by a negative passive decay rate.

A.3 A conservative retirement lemma

Let $\theta \in (0,1)$ be a governance threshold and define the optimistic validity envelope

$$\overline{V}_B(t) := \mathcal{V}_B(0) \exp\Big(-\int_0^t \underline{q}_B(u)\,\mathrm{d}u\Big). \tag{319}$$

If $\overline{V}_B(t) \le \theta \mathcal{V}_B(0)$, then even the optimistic estimate places the benchmark below threshold, so a fail-closed governance rule retires, regenerates, or scope-restricts the benchmark at time t.

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